6.0 Data Sufficiency

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Data sufficiency questions appear in the Quantitative section of the GMAT® exam. Multiple-choice data sufficiency questions are intermingled with problem solving questions throughout the section. You will have 62 minutes to complete the Quantitative section of the GMAT exam, or about 2 minutes to answer each question. These questions require knowledge of the following topics:

- Arithmetic
- Elementary algebra
- Commonly known concepts of geometry

Data sufficiency questions are designed to measure your ability to analyze a quantitative problem, recognize which given information is relevant, and determine at what point there is sufficient information to solve a problem. In these questions, you are to classify each problem according to the five fixed answer choices, rather than find a solution to the problem.

Each data sufficiency question consists of a question, often accompanied by some initial information, and two statements, labeled (1) and (2), which contain additional information. You must decide whether the information in each statement is sufficient to answer the question or—if neither statement provides enough information—whether the information in the two statements together is sufficient. It is also possible that the statements, in combination, do not give enough information to answer the question.

Begin by reading the initial information and the question carefully. Next, consider the first statement. Does the information provided by the first statement enable you to answer the question? Go on to the second statement. Try to ignore the information given in the first statement when you consider whether the second statement provides information that, by itself, allows you to answer the question. Now you should be able to say, for each statement, whether it is sufficient to determine the answer.

Next, consider the two statements in tandem. Do they, together, enable you to answer the question?

Look again at your answer choices. Select the one that most accurately reflects whether the statements provide the information required to answer the question.

6.1 Test-Taking Strategies

1. Do not waste valuable time solving a problem.

You only need to determine whether sufficient information is given to solve it.

2. Consider each statement separately.

First, decide whether each statement alone gives sufficient information to solve the problem. Be sure to disregard the information given in statement (1) when you evaluate the information given in statement (2). If either, or both, of the statements give(s) sufficient information to solve the problem, select the answer corresponding to the description of which statement(s) give(s) sufficient information to solve the problem.

3. Judge the statements in tandem if neither statement is sufficient by itself.

It is possible that the two statements together do not provide sufficient information. Once you decide, select the answer corresponding to the description of whether the statements together give sufficient information to solve the problem.

4. Answer the question asked.

For example, if the question asks, "What is the value of y?" for an answer statement to be sufficient, you must be able to find one and only one value for y. Being able to determine minimum or maximum values for an answer (e.g., $y = x + 2$) is not sufficient, because such answers constitute a range of values rather than the specific value of y.

5. Be very careful not to make unwarranted assumptions based on the images represented.

Figures are not necessarily drawn to scale; they are generalized figures showing little more than intersecting line segments and the relationships of points, angles, and regions. For example, if a figure described as a rectangle looks like a square, do *not* conclude that it is actually a square just by looking at the figure.

If statement 1 is sufficient, then the answer must be **A or D.**

If statement 2 is not sufficient, then the answer must be **A.**

If statement 2 is sufficient, then the answer must be **D.**

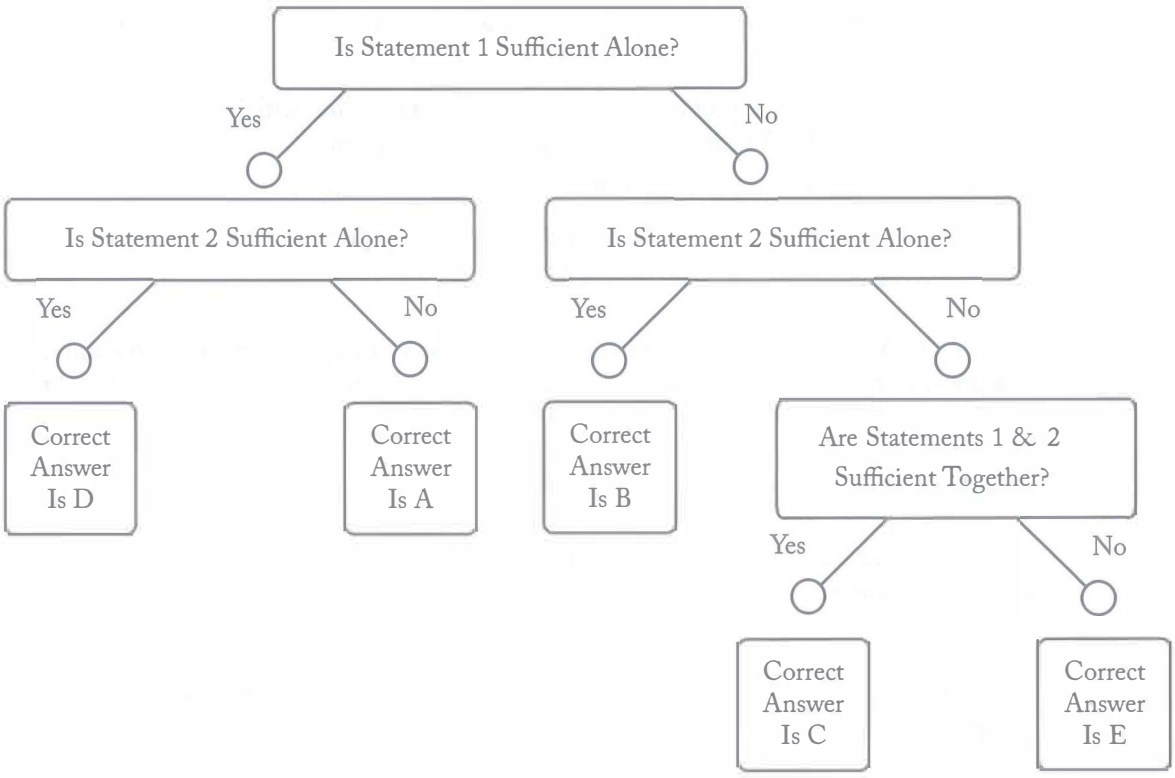
If statement 1 is not sufficient, then the answer must be **B, C, or E.**

If statement 2 is sufficient, then the answer must be **B.**

If statement 2 is not sufficient, then the answer must be **C or E.**

If both statements together are sufficient, then the answer must be **C.**

If both statements together are still not sufficient, then the answer must be **E.**



6.2 The Directions

These directions are similar to those you will see for data sufficiency questions when you take the GMAT exam. If you read the directions carefully and understand them clearly before going to sit for the test, you will not need to spend much time reviewing them when you take the GMAT exam.

Each data sufficiency problem consists of a question and two statements, labeled (1) and (2), that give data. You have to decide whether the data given in the statements are *sufficient* for answering the question. Using the data given in the statements *plus* your knowledge of mathematics and everyday facts (such as the number of days in July or the meaning of *counterclockwise*), you must indicate whether the data given in the statements are sufficient for answering the questions and then indicate one of the following answer choices:

(A) Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient to answer the question asked;

(B) Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient to answer the question asked;

(C) BOTH statements (1) and (2) TOGETHER are sufficient to answer the question asked, but NEITHER statement ALONE is sufficient;

(D) EACH statement ALONE is sufficient to answer the question asked;

(E) Statements (1) and (2) TOGETHER are NOT sufficient to answer the question asked, and additional data are needed.

NOTE: In data sufficiency problems that ask for the value of a quantity, the data given in the statements are sufficient only when it is possible to determine exactly one numerical value for the quantity.

Numbers: All numbers used are real numbers.

Figures: A figure accompanying a data sufficiency problem will conform to the information given in the question but will not necessarily conform to the additional information given in statements (1) and (2).

Lines shown as straight can be assumed to be straight and lines that appear jagged can also be assumed to be straight.

You may assume that the positions of points, angles, regions, and so forth exist in the order shown and that angle measures are greater than zero degrees.

All figures lie in a plane unless otherwise indicated.

To register for the GMAT exam go to www.mba.com

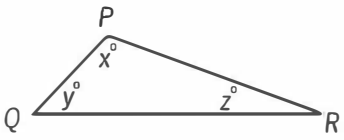
6.3 Practice Questions

Each data sufficiency problem consists of a question and two statements, labeled (1) and (2), which contain certain data. Using these data and your knowledge of mathematics and everyday facts (such as the number of days in July or the meaning of the word *counterclockwise*), decide whether the data given are sufficient for answering the question and then indicate one of the following answer choices:

A Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.
B Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.
C BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
D EACH statement ALONE is sufficient.
E Statements (1) and (2) TOGETHER are NOT sufficient.

Note: In data sufficiency problems that ask for the value of a quantity, the data given in the statements are sufficient only when it is possible to determine exactly one numerical value for the quantity.

Example:



In, ΔPQR what is the value of x?

(1) $PQ = PR$

(2) $y = 40$

Explanation: According to statement (1) $PQ = PR$; therefore, ΔPQR is isosceles and $y = z$. Since $x + y + z = 180$, it follows that $x + 2y = 180$. Since statement (1) does not give a value for y, you cannot answer the question using statement (1) alone. According to statement (2), $y = 40$; therefore, $x + z = 140$. Since statement (2) does not give a value for z, you cannot answer the question using statement (2) alone. Using both statements together, since $x + 2y = 180$ and the value of y is given, you can find the value of x. Therefore, BOTH statements (1) and (2) TOGETHER are sufficient to answer the questions, but NEITHER statement ALONE is sufficient.

Numbers: All numbers used are real numbers.

Figures:

- Figures conform to the information given in the question, but will not necessarily conform to the additional information given in statements (1) and (2).
- Lines shown as straight are straight, and lines that appear jagged are also straight.
- The positions of points, angles, regions, etc., exist in the order shown, and angle measures are greater than zero.
- All figures lie in a plane unless otherwise indicated.

*DS02562

231. What is the number of pages of a certain journal article?

(1) The size of each page is $5\frac{1}{2}$ inches by 8 inches.

(2) The average (arithmetic mean) number of words per page is 250.

DS10471

232. If a certain vase contains only roses and tulips, how many tulips are there in the vase?

(1) The number of roses in the vase is 4 times the number of tulips in the vase.

(2) There is a total of 20 flowers in the vase.

DS03802

233. The cost of 10 pounds of apples and 2 pounds of grapes was \$12. What was the cost per pound of apples?

(1) The cost per pound of grapes was \$2.

(2) The cost of 2 pounds of apples was less than the cost of 1 pound of grapes.

DS05863

234. What was the median annual salary for the employees at Company X last year?

(1) Last year there were 29 employees at Company X.

(2) Last year 12 employees at Company X had an annual salary of \$24,000.

DS03422

235. How many basic units of currency X are equivalent to 250 basic units of currency Y ?

(1) 100 basic units of currency X are equivalent to 625 basic units of currency Y.

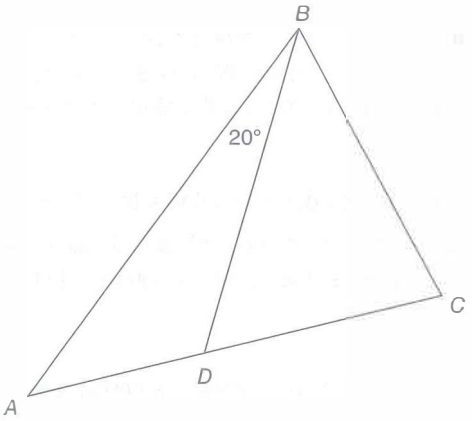
(2) 2,000 basic units of currency X are equivalent to 12,500 basic units of currency Y.

DS12265

236. A company bought 3 printers and 1 scanner. What was the price of the scanner?

(1) The total price of the printers and the scanner was \$1,300.

(2) The price of each printer was 4 times the price of the scanner.



DS17639

237. In the figure above, point D is on $\overline{AC}$. What is the degree measure of $\angle BAC$?

(1) The measure of $\angle BDC$ is 60°.

(2) The degree measure of $\angle BAC$ is less than the degree measure of $\angle BCD$.

DS07822

238. Each of the 256 solid-colored marbles in a box is either blue, green, or purple. What is the ratio of the number of blue marbles to the number of purple marbles in the box?

(1) The number of green marbles in the box is 4 times the number of blue marbles in the box.

(2) There are 192 green marbles in the box.

DS15940

239. A certain mixture of paint requires blue, yellow, and red paints in ratios of 2:3:1, respectively, and no other ingredients. If there are ample quantities of the blue and red paints available, is there enough of the yellow paint available to make the desired amount of the mixture?

(1) Exactly 20 quarts of the mixture are needed.

(2) Exactly 10 quarts of the yellow paint are available.

DS05338

240. The research funds of a certain company were divided among three departments, X, Y, and Z. Which one of the three departments received the greatest proportion of the research funds?

(1) The research funds received by departments X and Y were in the ratio 3 to 5, respectively.

(2) The research funds received by departments X and Z were in the ratio 2 to 1, respectively.

*These numbers correlate with the online test bank question number. See the GMAT Official Online Index in the back of this book.

DS03138

241. In a certain class, some students donated cans of food to a local food bank. What was the average (arithmetic mean) number of cans donated per student in the class?

(1) The students donated a total of 56 cans of food.

(2) The total number of cans donated was 40 greater than the total number of students in the class.

DS00254

242. Each of the n employees at a certain company has a different annual salary. What is the median of the annual salaries of the n employees?

(1) When the annual salaries of the n employees are listed in increasing order, the median is the 15th salary.

(2) The sum of the annual salaries of the n employees is $913,500.

DS10687

243. In a recent town election, what was the ratio of the number of votes in favor of a certain proposal to the number of votes against the proposal?

(1) There were 60 more votes in favor of the proposal than against the proposal.

(2) There were 240 votes in favor of the proposal.

DS02541

244. How many men are in a certain company's vanpool program?

(1) The ratio of men to women in the program is 3 to 2.

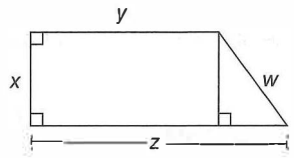
(2) The men and women in the program fill 6 vans.

DS08054

245. Each of the marbles in a jar is either red or white or blue. If one marble is to be selected at random from the jar, what is the probability that the marble will be blue?

(1) There are a total of 24 marbles in the jar, 8 of which are red.

(2) The probability that the marble selected will be white is $\frac{1}{2}$.



DS04594

246. In the figure above, what is the value of z?

(1) $x = y = 1$

(2) $w = 2$

DS04630

247. What is the value of 10 percent of y?

(1) 5 percent of y is 60.

(2) y is 80 percent of 1,500.

DS12062

248. Last semester, Professor K taught two classes, A and B. Each student in class A handed in 7 assignments, and each student in class B handed in 5 assignments. How many students were in class A?

(1) The students in both classes combined handed in a total of 85 assignments.

(2) There were 10 students in class B.

DS06802

249. Was the amount of John's heating bill for February greater than it was for January?

(1) The ratio of the amount of John's heating bill for February to that for January was $\frac{26}{25}$.

(2) The sum of the amounts of John's heating bills for January and February was $183.60.

DS06662

250. If sequence S has 120 terms, what is the 105th term of S?

(1) The first term of S is –8.

(2) Each term of S after the first term is 10 more than the preceding term.

DS00858

251. Machine R and machine S work at their respective constant rates. How much time does it take machine R, working alone, to complete a certain job?

(1) The amount of time that it takes machine S, working alone, to complete the job is $\frac{3}{4}$ the amount of time that it takes machine R, working alone, to complete the job.

(2) Machine R and machine S, working together, take 12 minutes to complete the job.

DS06065

252. If $u > 0$ and $v > 0$, which is greater, u^v or v^u ?

(1) $u = 1$

(2) $v > 2$

DS00660

253. What was the range of the selling prices of the 30 wallets sold by a certain store yesterday?

(1) $\frac{1}{3}$ of the wallets had a selling price of \$24 each.

(2) The lowest selling price of the wallets was $\frac{1}{3}$ the highest selling price of the wallets.

DS08723

254. Three houses are being sold through a real estate agent. What is the asking price for the house with the second-largest asking price?

(1) The difference between the greatest and the least asking price is \$130,000.

(2) The difference between the two greater asking prices is \$85,000.

DS04605

255. If $a + b + c = 12$, what is the value of b ?

(1) $a + b = 8$

(2) $b + c = 6$

DS11254

256. Is $rw = 0$?

(1) $-6 < r < 5$

(2) $6 < w < 10$

DS06633

257. Is $x = \frac{1}{y}$?

(1) $xy = 1$

(2) $\frac{1}{xy} = 1$

DS07949

258. How many people in a group of 50 own neither a fax machine nor a laser printer?

(1) The total number of people in the group who own a fax machine or a laser printer or both is less than 50.

(2) The total number of people in the group who own both a fax machine and a laser printer is 15.

DS06475

259. What is the value of w^{-2} ?

(1) $w^{-1} = \frac{1}{2}$

(2) $w^3 = 8$

DS07839

260. A certain investment earned a fixed rate of 4 percent interest per year, compounded annually, for five years. The interest earned for the third year of the investment was how many dollars greater than that for the first year?

(1) The amount of the investment at the beginning of the second year was \$4,160.00.

(2) The amount of the investment at the beginning of the third year was \$4,326.40.

DS06397

261. What is the circumference of circle C ?

(1) The radius of circle C is 2π.

(2) The center of circle C is located at point (7,8) in the xy-plane.

DS07813

262. What is the value of t ?

(1) $s + t = 6 + s$

(2) $t^3 = 216$

DS11109

263. For a certain car repair, the total charge consisted of a charge for parts, a charge for labor, and a 6 percent sales tax on both the charge for parts and the charge for labor. If the charge for parts, excluding sales tax, was \$50.00, what was the total charge for the repair?

(1) The sales tax on the charge for labor was \$9.60.

(2) The total sales tax was \$12.60.

DS06905

264. George has a total of B books in his library, 25 of which are hardcover fiction books. What is the value of B ?

(1) 40 of the B books are fiction and the rest are nonfiction.

(2) 60 of the B books are hardcovers and the rest are paperbacks.

DS12031

265. Is $w + h^4$ positive?

(1) h is positive.

(2) w is positive.

DS15377

266. If a is a 3-digit integer and b is a 3-digit integer, is the units digit of the product of a and b greater than 5 ?

(1) The units digit of a is 4.

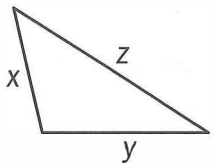
(2) The units digit of b is 7.

DS02450

267. In each of the last five years, Company K donated p percent of its annual profits to a certain scholarship fund. Did Company K donate more than \$10,000 to the scholarship fund last year?

(1) Two years ago, Company K had annual profits of \$3 million and donated \$15,000 to the scholarship fund.

(2) Last year, Company K had annual profits of \$2.5 million.



DS06901

268. Is the area of the triangular region above less than 20?

(1) $x^2 + y^2 \neq z^2$

(2) $x + y < 13$

DS04366

269. A, B, C, and D are points on a line. If C is the midpoint of line segment AB and if D is the midpoint of line segment CB, is the length of line segment DB greater than 5 ?

(1) The length of line segment AC is greater than 8.

(2) The length of line segment CD is greater than 6.

DS11805

270. The people in a line waiting to buy tickets to a show are standing one behind the other. Adam and Beth are among the people in the line, and Beth is standing behind Adam with a number of people between them. If the number of people in front of Adam plus the number of people behind Beth is 18, how many people in the line are behind Beth?

(1) There are a total of 32 people in the line.

(2) 23 people in the line are behind Adam.

DS08730

271. Square $ABCD$ is inscribed in circle O. What is the area of square region $ABCD$?

(1) The area of circular region O is 64π.

(2) The circumference of circle O is 16π.

DS12533

272. Lines k and m are parallel to each other. Is the slope of line k positive?

(1) Line k passes through the point (3,2).

(2) Line m passes through the point (–3,2).

DS19520

273. In cross section, a tunnel that carries one lane of one-way traffic is a semicircle with radius 4.2 m. Is the tunnel large enough to accommodate the truck that is approaching the entrance to the tunnel?

(1) The maximum width of the truck is 2.4 m.

(2) The maximum height of the truck is 4 m.

DS13122

274. In a certain group of 50 people, how many are doctors who have a law degree?

(1) In the group, 36 people are doctors.

(2) In the group, 18 people have a law degree.

DS01544

275. Of a group of 50 households, how many have at least one cat or at least one dog, but not both?

(1) The number of households that have at least one cat and at least one dog is 4.

(2) The number of households that have no cats and no dogs is 14.

DS02441

276. Robin invested a total of \$12,000 in two investments, X and Y, so that the investments earned the same amount of simple annual interest. How many dollars did Robin invest in investment Y ?

(1) Investment X paid 3 percent simple annual interest, and investment Y paid 6 percent simple annual interest.

(2) Robin invested more than \$1,000 in investment X.

DS03999

277. In a real estate office that employs n salespeople, f of them are females and x of the females are new employees. What is the value of n?

(1) If an employee were randomly selected from the n employees, the probability of selecting a female would be $\frac{2}{3}$.

(2) If an employee were randomly selected from the f female employees, the probability of selecting a new employee would be $\frac{1}{2}$.

DS09315

278. Is $\frac{x+1}{y+1} > \frac{x}{y}$?

(1) $0 < x < y$

(2) $xy > 0$

DS01216

279. Do at least 60 percent of the students in Pat's class walk to school?

(1) At least 60 percent of the female students in Pat's class walk to school.

(2) The number of students in Pat's class who walk to school is twice the number of students who do not walk to school.

DS03628

280. A certain plumber charges \$92 for each job completed in 4 hours or less and \$23 per hour for each job completed in more than 4 hours. If it took the plumber a total of 7 hours to complete two separate jobs, what was the total amount charged by the plumber for the two jobs?

(1) The plumber charged \$92 for one of the two jobs.

(2) The plumber charged \$138 for one of the two jobs.

DS02585

281. If x and y are positive numbers, is $\frac{x+1}{y+1} > \frac{x}{y}$?

(1) $x > 1$

(2) $x < y$

DS01619

282. If a and b are positive integers, is $\frac{a}{b} < \frac{9}{11}$?

(1) $\frac{a}{b} < 0.818$

(2) $\frac{b}{a} > 1.223$

DS04536

283. Every object in a box is either a sphere or a cube, and every object in the box is either red or green. How many objects are in the box?

(1) There are six cubes and five green objects in the box.

(2) There are two red spheres in the box.

DS01425

284. If x and y are positive integers, is xy even?

(1) $x^2 + y^2 - 1$ is divisible by 4.

(2) $x + y$ is odd.

DS14502

285. If a and b are integers, is $a + b + 3$ an odd integer?

(1) ab is an odd integer.

(2) $a - b$ is an even integer.

DS08308

286. If x and y are positive integers, what is the value of $\sqrt{x} + \sqrt{y}$?

(1) $x + y = 15$

(2) $\sqrt{xy} = 6$

DS05312

287. A certain truck uses $\frac{1}{12} + kv^2$ gallons of fuel per mile when its speed is v miles per hour, where k is a constant. At what speed should the truck travel so that it uses $\frac{5}{12}$ gallon of fuel per mile?

(1) The value of k is $\frac{1}{10{,}800}$.

(2) When the truck travels at 30 miles per hour, it uses $\frac{1}{6}$ gallon of fuel per mile.

DS08420

288. On June 1, Mary paid Omar $360 for rent and utilities for the month of June. Mary moved out early, and Omar refunded the money she paid for utilities, but not for rent, for the days in June after she moved out. How many dollars did Omar refund to Mary?

(1) Mary moved out on June 24.

(2) The amount Mary paid for utilities was less than $\frac{1}{5}$ the amount Mary paid for rent.

DS04057

289. If $x = 2t$ and $y = \frac{t}{3}$, what is the value of $x^2 - y^2$?

(1) $t^2 - 3 = 6$

(2) $t^3 = -27$

DS02939

290. The 10 students in a history class recently took an examination. What was the maximum score on the examination?

(1) The mean of the scores was 75.

(2) The standard deviation of the scores was 5.

DS01341

291. Last school year, each of the 200 students at a certain high school attended the school for the entire year. If there were 8 cultural performances at the school during the last school year, what was the average (arithmetic mean) number of students attending each cultural performance?

(1) Last school year, each student attended at least one cultural performance.

(2) Last school year, the average number of cultural performances attended per student was 4.

DS14569

292. A clothing manufacturer makes jackets that are wool or cotton or a combination of wool and cotton. The manufacturer has 3,000 pounds of wool and 2,000 pounds of cotton on hand. Is this enough wool and cotton to make at least 1,000 jackets?

(1) Each wool jacket requires 4 pounds of wool, and no cotton.

(2) Each cotton jacket requires 6 pounds of cotton, and no wool.

DS05377

293. If n is an integer, what is the greatest common divisor of 12 and n?

(1) The product of 12 and n is 432.

(2) The greatest common divisor of 24 and n is 12.

DS11287

294. Each month, Jim receives a base salary plus a 10 percent commission on the price of each car he sells that month. If Jim sold 15 cars last month, what was the total amount of base salary and commissions that Jim received that month?

(1) Last month, Jim's base salary was $3,000.

(2) Last month, Jim sold 3 cars whose prices totaled $60,000 and 5 cars whose prices totaled $120,000.

DS17615

295. If x is a positive integer greater than 1, what is the value of x?

(1) $2x$ is a common factor of 18 and 24.

(2) x is a factor of 6.

DS13408

296. By what percentage was the price of a certain television set discounted for a sale?

(1) The price of the television set before it was discounted for the sale was 25 percent greater than the discounted price.

(2) The price of the television set was discounted by $60 for the sale.

DS05049

297. Jack wants to use a circular rug on his rectangular office floor to cover two small circular stains, each less than $\frac{\pi}{100}$ square feet in area and each more than 3 feet from the nearest wall. Can the rug be placed to cover both stains?

(1) Jack's rug covers an area of 9π square feet.

(2) The centers of the stains are less than 4 feet apart.

×	a	b	c
a	d	e	f
b	e	g	h
c	f	h	j

DS05772

298. In the multiplication table above, each letter represents an integer. What is the value of c ?

(1) $c = f$

(2) $h \neq 0$

DS09379

299. If n is an integer, is $(0.1)^n$ greater than $(10)^n$?

(1) $n > -10$

(2) $n < 10$

DS19199

300. For a basic monthly fee of F yen (¥F), Naoko's first cell phone plan allowed him to use a maximum of 420 minutes on calls during the month. Then, for each of x additional minutes he used on calls, he was charged ¥M, making his total charge for the month ¥T, where $T = F + xM$. What is the value of F ?

(1) Naoko used 450 minutes on calls the first month and the total charge for the month was ¥13,755.

(2) Naoko used 400 minutes on calls the second month and the total charge for the month was ¥13,125.

DS13949

301. Is the sum of the prices of the 3 books that Shana bought less than \$48 ?

(1) The price of the most expensive of the 3 books that Shana bought is less than \$17.

(2) The price of the least expensive of the 3 books that Shana bought is exactly \$3 less than the price of the second most expensive book.

DS12943

302. If r and t are three-digit positive integers, is r greater than t ?

(1) The tens digit of r is greater than each of the three digits of t.

(2) The tens digit of r is less than either of the other two digits of r.

DS14788

303. Is the product of two positive integers x and y divisible by the sum of x and y ?

(1) $x = y$

(2) $x = 2$

DS05330

304. A company makes and sells two products, P and Q. The costs per unit of making and selling P and Q are \$8.00 and \$9.50, respectively, and the selling prices per unit of P and Q are \$10.00 and \$13.00, respectively. In one month the company sold a total of 834 units of these products. Was the total profit on these items more than \$2,000.00 ?

(1) During the month, more units of P than units of Q were sold.

(2) During the month, at least 100 units of Q were sold.

DS03045

305. Jill has applied for a job with each of two different companies. What is the probability that she will get job offers from both companies?

(1) The probability that she will get a job offer from neither company is 0.3.

(2) The probability that she will get a job offer from exactly one of the two companies is 0.5.

DS01257

306. A conveyor belt moves bottles at a constant speed of 120 centimeters per second. If the conveyor belt moves a bottle from a loading dock to an unloading dock, is the distance that the conveyor belt moves the bottle less than 90 meters? (1 meter = 100 centimeters)

(1) It takes the conveyor belt less than 1.2 minutes to move the bottle from the loading dock to the unloading dock.

(2) It takes the conveyor belt more than 1.1 minutes to move the bottle from the loading dock to the unloading dock.

DS02706

307. If x, y, and z are positive numbers, what is the value of the average (arithmetic mean) of x and z ?

(1) $x - y = y - z$

(2) $x^2 - y^2 = z$



DS04428

308. The rectangular rug shown in the figure above has an accent border. What is the area of the portion of the rug that excludes the border?

(1) The perimeter of the rug is 44 feet.

(2) The width of the border on all sides is 1 foot.

DS07227

309. Terry holds 12 cards, each of which is red, white, green, or blue. If a person is to select a card randomly from the cards Terry is holding, is the probability less than $\frac{1}{2}$ that the card selected will be either red or white?

(1) The probability that the person will select a blue card is $\frac{1}{3}$.

(2) The probability that the person will select a red card is $\frac{1}{6}$.

DS06537

310. If $y \neq 2xz$, what is the value of $\frac{2xz + yz}{2xz - y}$?

(1) $2x + y = 3$

(2) $z = 2$



DS04852

311. In the parallelogram shown, what is the value of x?

(1) $y = 2x$

(2) $x + z = 120$

DS06096

312. In a product test of a common cold remedy, x percent of the patients tested experienced side effects from the use of the drug and y percent experienced relief of cold symptoms. What percent of the patients tested experienced both side effects and relief of cold symptoms?

(1) Of the 1,000 patients tested, 15 percent experienced neither side effects nor relief of cold symptoms.

(2) Of the patients tested, 30 percent experienced relief of cold symptoms without side effects.

DS13588

313. Is $x < 5$?

(1) $x^2 > 5$

(2) $x^2 + x < 5$

DS11257

314. Is zp negative?

(1) $pz^4 < 0$

(2) $p + z^4 = 14$

DS04157

315. In each game of a certain tournament, a contestant either loses 3 points or gains 2 points. If Pat had 100 points at the beginning of the tournament, how many games did Pat play in the tournament?

(1) At the end of the tournament, Pat had 104 points.

(2) Pat played fewer than 10 games.

DS05631

316. At the beginning of the year, the Finance Committee and the Planning Committee of a certain company each had n members, and no one was a member of both committees. At the end of the year, 5 members left the Finance Committee and 3 members left the Planning Committee. How many members did the Finance Committee have at the beginning of the year?

(1) The ratio of the total number of members who left at the end of the year to the total number of members at the beginning of the year was 1:6.

(2) At the end of the year, 21 members remained on the Planning Committee.

DS03268

317. Can a certain rectangular sheet of glass be positioned on a rectangular tabletop so that it covers the entire tabletop and its edges are parallel to the edges of the tabletop?

(1) The tabletop is 36 inches wide by 60 inches long.

(2) The area of one side of the sheet of glass is 2,400 square inches.

DS15561

318. If $xy \neq 0$, is $x^3 + y^3 > 0$?

(1) $x + y > 0$

(2) $xy > 0$

DS13541

319. Max purchased a guitar for a total of $624, which consisted of the price of the guitar and the sales tax. Was the sales tax rate greater than 3 percent?

(1) The price of the guitar that Max purchased was less than $602.

(2) The sales tax for the guitar that Max purchased was less than $30.

DS06027

320. What is the sum of a certain pair of consecutive odd integers?

(1) At least one of the integers is negative.

(2) At least one of the integers is positive.

DS08197

321. The sum of 4 different odd integers is 64. What is the value of the greatest of these integers?

(1) The integers are consecutive odd numbers.

(2) Of these integers, the greatest is 6 more than the least.

DS13130

322. Was the number of books sold at Bookstore X last week greater than the number of books sold at Bookstore Y last week?

(1) Last week, more than 1,000 books were sold at Bookstore X on Saturday and fewer than 1,000 books were sold at Bookstore Y on Saturday.

(2) Last week, less than 20 percent of the books sold at Bookstore X were sold on Saturday and more than 20 percent of the books sold at Bookstore Y were sold on Saturday.

DS04540

323. From May 1 to May 30 in the same year, the balance in a checking account increased. What was the balance in the checking account on May 30 ?

(1) If, during this period of time, the increase in the balance in the checking account had been 12 percent, then the balance in the account on May 30 would have been $504.

(2) During this period of time, the increase in the balance in the checking account was 8 percent.

DS08365

324. A merchant discounted the sale price of a coat and the sale price of a sweater. Which of the two articles of clothing was discounted by the greater dollar amount?

(1) The percent discount on the coat was 2 percentage points greater than the percent discount on the sweater.

(2) Before the discounts, the sale price of the coat was $10 less than the sale price of the sweater.

DS01168

325. If the positive integer n is added to each of the integers 69, 94, and 121, what is the value of n ?

(1) $69 + n$ and $94 + n$ are the squares of two consecutive integers.

(2) $94 + n$ and $121 + n$ are the squares of two consecutive integers.

DS05269

326. Last year, in a certain housing development, the average (arithmetic mean) price of 20 new houses was $160,000. Did more than 9 of the 20 houses have prices that were less than the average price last year?

(1) Last year the greatest price of one of the 20 houses was $219,000.

(2) Last year the median of the prices of the 20 houses was $150,000.

DS14527

327. For a certain city's library, the average cost of purchasing each new book is $28. The library receives $15,000 from the city each year; the library also receives a bonus of $2,000 if the total number of items checked out over the course of the year exceeds 5,000. Did the library receive the bonus last year?

(1) The library purchased an average of 50 new books each month last year and received enough money from the city to cover this cost.

(2) The lowest number of items checked out in one month was 459.

DS00395

328. Each gift certificate sold yesterday by a certain bookstore cost either \$10 or \$50. If yesterday the bookstore sold more than 5 gift certificates that cost \$50 each, what was the total number of gift certificates sold yesterday by the bookstore?

(1) Yesterday the bookstore sold fewer than 10 gift certificates that cost \$10 each.

(2) The total cost of gift certificates sold yesterday by the bookstore was \$460.

DS15045

329. Three dice, each of which has its 6 sides numbered 1 through 6, are tossed. The sum of the 3 numbers that are facing up is 12. Is at least 1 of these numbers 5 ?

(1) None of the 3 numbers that are facing up is divisible by 3.

(2) Of the numbers that are facing up, 2, but not all 3, are equal.

DS01324

330. On the number line, point R has coordinate r and point T has coordinate t. Is $t < 0$?

(1) $-1 < r < 0$

(2) The distance between R and T is equal to r^2.

DS06659

331. S is a set of points in the plane. How many distinct triangles can be drawn that have three of the points in S as vertices?

(1) The number of distinct points in S is 5.

(2) No three of the points in S are collinear.

DS16078

332. Stores L and M each sell a certain product at a different regular price. If both stores discount their regular price of the product, is the discount price at Store M less than the discount price at Store L ?

(1) At Store L the discount price is 10 percent less than the regular price; at Store M the discount price is 15 percent less than the regular price.

(2) At Store L the discount price is \$5 less than the regular store price; at Store M the discount price is \$6 less than the regular price.

DS16529

333. If d denotes a decimal, is $d \geq 0.5$?

(1) When d is rounded to the nearest tenth, the result is 0.5.

(2) When d is rounded to the nearest integer, the result is 1.

DS08231

334. In the two-digit integers 3■ and 2▲, the symbols ■ and ▲ represent different digits, and the product (3■)(2▲) is equal to 864. What digit does ■ represent?

(1) The sum of ■ and ▲ is 10.

(2) The product of ■ and ▲ is 24.



DS07262

335. Two points, N and Q (not shown), lie to the right of point M on line ℓ. What is the ratio of the length of QN to the length of MQ ?

(1) Twice the length of MN is 3 times the length of MQ.

(2) Point Q is between points M and N.

DS05639

336. Did the sum of the prices of three shirts exceed \$60 ?

(1) The price of the most expensive of the shirts exceeded \$30.

(2) The price of the least expensive of the shirts exceeded \$20.

DS03057

337. What is the total number of coins that Bert and Claire have?

(1) Bert has 50 percent more coins than Claire.

(2) The total number of coins that Bert and Claire have is between 21 and 28.

DS05668

338. A telephone station has x processors, each of which can process a maximum of y calls at any particular time, where x and y are positive integers. If 500 calls are sent to the station at a particular time, can the station process all of the calls?

(1) $x = 600$

(2) $100 < y < 200$

	Price per Flower
Roses	$1.00
Daisies	$0.50

DS07953

339. Kim and Sue each bought some roses and some daisies at the prices shown above. If Kim bought the same total number of roses and daisies as Sue, was the price of Kim's purchase of roses and daisies higher than the price of Sue's purchase of roses and daisies?

(1) Kim bought twice as many daisies as roses.

(2) Kim bought 4 more roses than Sue bought.

DS14406

340. Jazz and blues recordings accounted for 6 percent of the $840 million revenue from the sales of recordings in Country Y in 2000. What was the revenue from the sales of jazz and blues recordings in Country Y in 1998?

(1) Jazz and blues recordings accounted for 5 percent of the revenue from the sales of recordings in Country Y in 1998.

(2) The revenue from the sales of jazz and blues recordings in Country Y increased by 40 percent from 1998 to 2000.

DS06315

341. On a certain nonstop trip, Marta averaged x miles per hour for 2 hours and y miles per hour for the remaining 3 hours. What was her average speed, in miles per hour, for the entire trip?

(1) $2x + 3y = 280$

(2) $y = x + 10$

DS12730

342. If x is a positive integer, what is the value of $\sqrt{x+24} - \sqrt{x}$?

(1) $\sqrt{x}$ is an integer.

(2) $\sqrt{x+24}$ is an integer.

DS13982

343. A tank is filled with gasoline to a depth of exactly 2 feet. The tank is a cylinder resting horizontally on its side, with its circular ends oriented vertically. The inside of the tank is exactly 6 feet long. What is the volume of the gasoline in the tank?

(1) The inside of the tank is exactly 4 feet in diameter.

(2) The top surface of the gasoline forms a rectangle that has an area of 24 square feet.



DS16197

344. Of the four numbers represented on the number line above, is r closest to zero?

(1) $q = -s$

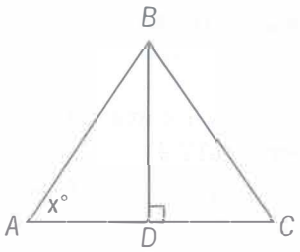
(2) $-t < q$

DS18414

345. A group consisting of several families visited an amusement park where the regular admission fees were ¥5,500 for each adult and ¥4,800 for each child. Because there were at least 10 people in the group, each paid an admission fee that was 10% less than the regular admission fee. How many children were in the group?

(1) The total of the admission fees paid for the adults in the group was ¥29,700.

(2) The total of the admission fees paid for the children in the group was ¥4,860 more than the total of the admission fees paid for the adults in the group.



DS16536

346. What is the area of triangular region ABC above?

(1) The product of BD and AC is 20.

(2) $x = 45$

DS05265

347. In the xy-coordinate plane, is point R equidistant from points $(-3,-3)$ and $(1,-3)$?

(1) The x-coordinate of point R is -1.

(2) Point R lies on the line $y = -3$.

DS09603

348. What is the ratio of the average (arithmetic mean) height of students in class X to the average height of students in class Y?

(1) The average height of the students in class X is 120 centimeters.

(2) The average height of the students in class X and class Y combined is 126 centimeters.

DS04631

349. Is the positive two-digit integer N less than 40 ?

(1) The units digit of N is 6 more than the tens digit.

(2) N is 4 less than 4 times the units digit.

DS06318

350. If $2^{x+y} = 4^8$, what is the value of y ?

(1) $x^2 = 81$

(2) $x - y = 2$

DS03680

351. Each week a certain salesman is paid a fixed amount equal to \$300, plus a commission equal to 5 percent of the amount of his sales that week over \$1,000. What is the total amount the salesman was paid last week?

(1) The total amount the salesman was paid last week is equal to 10 percent of the amount of his sales last week.

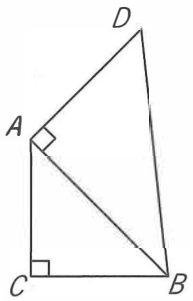
(2) The salesman's sales last week totaled \$5,000.

DS01383

352. At a bakery, all donuts are priced equally and all bagels are priced equally. What is the total price of 5 donuts and 3 bagels at the bakery?

(1) At the bakery, the total price of 10 donuts and 6 bagels is \$12.90.

(2) At the bakery, the price of a donut is \$0.15 less than the price of a bagel.



DS06869

353. In the figure above, is the area of triangular region ABC equal to the area of triangular region DBA ?

(1) $(AC)^2 = 2(AD)^2$

(2) ΔABC is isosceles.

DS08105

354. If r and s are positive integers, can the fraction $\frac{r}{s}$ be expressed as a decimal with only a finite number of nonzero digits?

(1) s is a factor of 100.

(2) r is a factor of 100.

DS16384

355. If $r > 0$ and $s > 0$, is $\frac{r}{s} < \frac{s}{r}$?

(1) $\frac{r}{3s} = \frac{1}{4}$

(2) $s = r + 4$

DS06789

356. If k is an integer such that $56 < k < 66$, what is the value of k ?

(1) If k were divided by 2, the remainder would be 1.

(2) If $k + 1$ were divided by 3, the remainder would be 0.

DS13965

357. If x is a positive integer, then is x prime?

(1) $3x + 1$ is prime.

(2) $5x + 1$ is prime.

$k, n, 12, 6, 17$

DS00172

358. What is the value of n in the list above?

(1) $k < n$

(2) The median of the numbers in the list is 10.

DS07508

359. If x and y are integers, what is the value of $x + y$?

(1) $3 < \frac{x+y}{2} < 4$

(2) $2 < x < y < 5$

DS00764

360. Last year, if Arturo spent a total of \$12,000 on his mortgage payments, real estate taxes, and home insurance, how much did he spend on his real estate taxes?

(1) Last year, the total amount that Arturo spent on his real estate taxes and home insurance was $33\frac{1}{3}$ percent of the amount that he spent on his mortgage payments.

(2) Last year, the amount that Arturo spent on his real estate taxes was 20 percent of the total amount he spent on his mortgage payments and home insurance.

DS06038

361. If a, b, c, and d are positive numbers, is $\frac{a}{b} < \frac{c}{d}$?

(1) $0 < \frac{c-a}{d-b}$

(2) $\left(\frac{ad}{bc}\right)^2 < \frac{ad}{bc}$

DS12008

362. Is the number of members of Club X greater than the number of members of Club Y?

(1) Of the members of Club X, 20 percent are also members of Club Y.

(2) Of the members of Club Y, 30 percent are also members of Club X.



DS16361

363. On the number line above, p, q, r, s, and t are five consecutive even integers in increasing order. What is the average (arithmetic mean) of these five integers?

(1) $q + s = 24$

(2) The average (arithmetic mean) of q and r is 11.

DS06657

364. If $\lceil x \rceil$ denotes the least integer greater than or equal to x, is $\lceil x \rceil = 0$?

(1) $-1 < x < 1$

(2) $x < 0$

DS12718

365. If x and y are integers, is $x > y$?

(1) $x + y > 0$

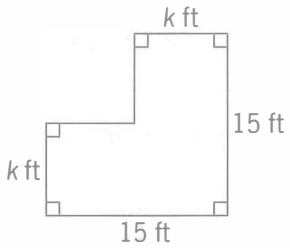
(2) $y^x < 0$

DS03046

366. If r and s are the roots of the equation $x^2 + bx + c = 0$, where b and c are constants, is $rs < 0$?

(1) $b < 0$

(2) $c < 0$



DS02888

367. The figure above represents an L-shaped garden. What is the value of k?

(1) The area of the garden is 189 square feet.

(2) The perimeter of the garden is 60 feet.

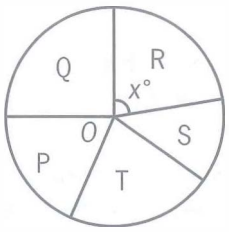
DS01049

368. The only articles of clothing in a certain closet are shirts, dresses, and jackets. The ratio of the number of shirts to the number of dresses to the number of jackets in the closet is 9:4:5, respectively. If there are more than 7 dresses in the closet, what is the total number of articles of clothing in the closet?

(1) The total number of shirts and jackets in the closet is less than 30.

(2) The total number of shirts and dresses in the closet is 26.

TOTAL EXPENSES FOR THE FIVE DIVISIONS OF COMPANY H



DS16542

369. The figure above represents a circle graph of Company H's total expenses broken down by the expenses for each of its five divisions. If O is the center of the circle and if Company H's total expenses are $5,400,000, what are the expenses for Division R?

(1) $x = 94$

(2) The total expenses for Divisions S and T are twice as much as the expenses for Division R.

DS13641

370. If x is negative, is $x < -3$?

(1) $x^2 > 9$

(2) $x^3 < -9$

DS04897

371. What is the number of cans that can be packed in a certain carton?

(1) The interior volume of this carton is 2,304 cubic inches.

(2) The exterior of each can is 6 inches high and has a diameter of 4 inches.

r	s	t
u	v	w
x	y	z

DS08301

372. Each of the letters in the table above represents one of the numbers 1, 2, or 3, and each of these numbers occurs exactly once in each row and exactly once in each column. What is the value of r?

(1) $v + z = 6$

(2) $s + t + u + x = 6$

DS00328

373. Material A costs \$3 per kilogram, and Material B costs \$5 per kilogram. If 10 kilograms of Material K consists of x kilograms of Material A and y kilograms of Material B, is $x > y$?

(1) $y > 4$

(2) The cost of the 10 kilograms of Material K is less than \$40.

DS16164

374. At what speed was a train traveling on a trip when it had completed half of the total distance of the trip?

(1) The trip was 460 miles long and took 4 hours to complete.

(2) The train traveled at an average rate of 115 miles per hour on the trip.

DS12047

375. Tom, Jane, and Sue each purchased a new house. The average (arithmetic mean) price of the three houses was \$120,000. What was the median price of the three houses?

(1) The price of Tom's house was \$110,000.

(2) The price of Jane's house was \$120,000.

DS13958

376. What is the value of x if $x^3 < x^2$?

(1) $-2 < x < 2$

(2) x is an integer greater than -2.

DS08451

377. For any integers x and y, min(x, y) and max(x, y) denote the minimum and the maximum of x and y, respectively. For example, min(5, 2) = 2 and max(5, 2) = 5. For the integer w, what is the value of min(10, w)?

(1) w = max(20, z) for some integer z

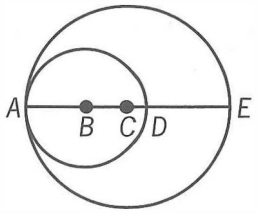
(2) w = max(10, w)

DS01473

378. A certain bookcase has 2 shelves of books. On the upper shelf, the book with the greatest number of pages has 400 pages. On the lower shelf, the book with the least number of pages has 475 pages. What is the median number of pages for all of the books on the 2 shelves?

(1) There are 25 books on the upper shelf.

(2) There are 24 books on the lower shelf.



DS12070

379. In the figure above, points A, B, C, D, and E lie on a line. A is on both circles, B is the center of the smaller circle, C is the center of the larger circle, D is on the smaller circle, and E is on the larger circle. What is the area of the region inside the larger circle and outside the smaller circle?

(1) $AB = 3$ and $BC = 2$

(2) $CD = 1$ and $DE = 4$

DS08995

380. In planning for a car trip, Joan estimated both the distance of the trip, in miles, and her average speed, in miles per hour. She accurately divided her estimated distance by her estimated average speed to obtain an estimate for the time, in hours, that the trip would take. Was her estimate within 0.5 hour of the actual time that the trip took?

(1) Joan's estimate for the distance was within 5 miles of the actual distance.

(2) Joan's estimate for her average speed was within 10 miles per hour of her actual average speed.

DS12239

381. A certain list consists of 3 different numbers. Does the median of the 3 numbers equal the average (arithmetic mean) of the 3 numbers?

(1) The range of the 3 numbers is equal to twice the difference between the greatest number and the median.

(2) The sum of the 3 numbers is equal to 3 times one of the numbers.

DS12806

382. Line ℓ lies in the xy-plane and does not pass through the origin. What is the slope of line ℓ?

(1) The x-intercept of line ℓ is twice the y-intercept of line ℓ.

(2) The x- and y-intercepts of line ℓ are both positive.

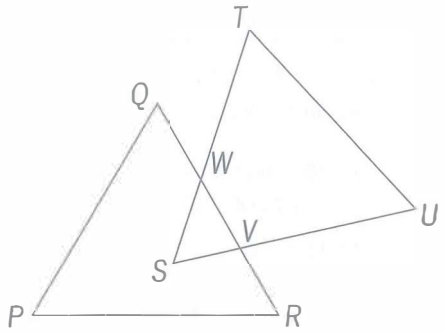
$$y = ax - 5$$
$$y = x + 6$$
$$y = 3x + b$$

DS07713

383. In the xy-plane, the straight-line graphs of the three equations above each contain the point (p,r). If a and b are constants, what is the value of b?

(1) $a = 2$

(2) $r = 17$



DS06861

384. In the figure above, PQR and STU are identical equilateral triangles, and $PQ = 6$. What is the perimeter of polygon $PQWTUVR$?

(1) Triangle SWV has perimeter 9.

(2) VW has length 3.5.

DS09973

385. The range of the numbers in set S is x, and the range of the numbers in set T is y. If all of the numbers in set T are also in set S, is x greater than y?

(1) Set S consists of 7 numbers.

(2) Set T consists of 6 numbers.

DS13857

386. The hypotenuse of a right triangle is 10 cm. What is the perimeter, in centimeters, of the triangle?

(1) The area of the triangle is 25 square centimeters.

(2) The 2 legs of the triangle are of equal length.

Shipment	S1	S2	S3	S4	S5	S6
Fraction of the Total Value of the Six Shipments	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{3}{20}$	$\frac{2}{15}$	$\frac{1}{10}$

DS01427

387. Six shipments of machine parts were shipped from a factory on two trucks, with each shipment entirely on one of the trucks. Each shipment was labeled either S1, S2, S3, S4, S5, or S6. The table shows the value of each shipment as a fraction of the total value of the six shipments. If the shipments on the first truck had a value greater than $\frac{1}{2}$ of the total value of the six shipments, was S3 shipped on the first truck?

(1) S2 and S4 were shipped on the first truck.

(2) S1 and S6 were shipped on the second truck.

DS11723

388. If x, y, and z are three-digit positive integers and if $x = y + z$, is the hundreds digit of x equal to the sum of the hundreds digits of y and z?

(1) The tens digit of x is equal to the sum of the tens digits of y and z.

(2) The units digit of x is equal to the sum of the units digits of y and z.

	Favorable	Unfavorable	Not Sure
Candidate M	40	20	40
Candidate N	30	35	35

DS05162

389. The table above shows the results of a survey of 100 voters who each responded "Favorable" or "Unfavorable" or "Not Sure" when asked about their impressions of Candidate M and of Candidate N. What was the number of voters who responded "Favorable" for both candidates?

(1) The number of voters who did not respond "Favorable" for either candidate was 40.

(2) The number of voters who responded "Unfavorable" for both candidates was 10.

DS00340

390. A school administrator will assign each student in a group of n students to one of m classrooms. If $3 < m < 13 < n$, is it possible to assign each of the n students to one of the m classrooms so that each classroom has the same number of students assigned to it?

(1) It is possible to assign each of $3n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

(2) It is possible to assign each of $13n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

DS07441

391. If q, s, and t are all different numbers, is $q < s < t$?

(1) $t - q = |t - s| + |s - q|$

(2) $t > q$

DS11538

392. What is the median number of employees assigned per project for the projects at Company Z?

(1) 25 percent of the projects at Company Z have 4 or more employees assigned to each project.

(2) 35 percent of the projects at Company Z have 2 or fewer employees assigned to each project.

DS04409

393. Last year, a certain company began manufacturing product X and sold every unit of product X that it produced. Last year the company's total expenses for manufacturing product X were equal to $100,000 plus 5 percent of the company's total revenue from all units of product X sold. If the company made a profit on product X last year, did the company sell more than 21,000 units of product X last year?

(1) The company's total revenue from the sale of product X last year was greater than $110,000.

(2) For each unit of product X sold last year, the company's revenue was $5.

DS01641

394. Beginning in January of last year, Carl made deposits of $120 into his account on the 15th of each month for several consecutive months and then made withdrawals of $50 from the account on the 15th of each of the remaining months of last year. There were no other transactions in the account last year. If the closing balance of Carl's account for May of last year was $2,600, what was the range of the monthly closing balances of Carl's account last year?

(1) Last year the closing balance of Carl's account for April was less than $2,625.

(2) Last year the closing balance of Carl's account for June was less than $2,675.

DS16368

395. Are all of the numbers in a certain list of 15 numbers equal?

(1) The sum of all the numbers in the list is 60.

(2) The sum of any 3 numbers in the list is 12.

DS16565

396. If the average (arithmetic mean) of six numbers is 75, how many of the numbers are equal to 75?

(1) None of the six numbers is less than 75.

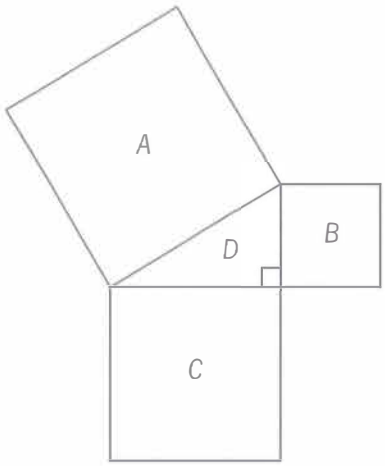
(2) None of the six numbers is greater than 75.

DS16188

397. What amount did Jean earn from the commission on her sales in the first half of 1988 ?

(1) In 1988 Jean's commission was 5 percent of the total amount of her sales.

(2) The amount of Jean's sales in the second half of 1988 averaged \$10,000 per month more than in the first half.



DS16572

398. In the figure above, if the area of triangular region D is 4, what is the length of a side of square region A ?

(1) The area of square region B is 9.

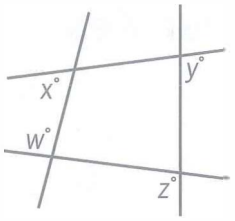
(2) The area of square region C is $\frac{64}{9}$.

DS16168

399. If n is a positive integer and $k = 5.1 \times 10^n$, what is the value of k ?

(1) $6{,}000 < k < 500{,}000$

(2) $k^2 = 2.601 \times 10^9$



DS06875

400. What is the value of $x + y$ in the figure above?

(1) $w = 95$

(2) $z = 125$

DS16370

401. If n and k are positive integers, is $\sqrt{n+k} > 2\sqrt{n}$?

(1) $k > 3n$

(2) $n + k > 3n$

DS16589

402. In a certain business, production index p is directly proportional to efficiency index e, which is in turn directly proportional to investment index i. What is p if $i = 70$?

(1) e = 0.5 whenever $i = 60$.

(2) $p = 2.0$ whenever $i = 50$.

DS16085

403. If n is a positive integer, what is the tens digit of n ?

(1) The hundreds digit of $10n$ is 6.

(2) The tens digit of $n + 1$ is 7.

DS16204

404. What is the value of $\frac{2t + t - x}{t - x}$?

(1) $\frac{2t}{t - x} = 3$

(2) $t - x = 5$

6.4 Answer Key

231. E
232. C
233. A
234. E
235. D
236. C
237. A
238. C
239. C
240. C
241. C
242. E
243. C
244. E
245. C
246. C
247. D
248. C
249. A
250. C
251. C
252. C
253. E
254. E
255. C
256. E
257. D
258. E
259. D
260. D
261. A
262. D
263. D
264. E
265. B
266. C
267. C
268. E
269. B
270. C
271. D
272. E
273. C
274. E
275. C
276. A
277. E
278. A
279. B
280. B
281. B
282. D
283. E
284. D
285. D
286. C
287. D
288. E
289. D
290. E
291. B
292. C
293. D
294. E
295. A
296. A
297. C
298. E
299. E
300. B
301. C
302. C
303. C
304. E
305. C
306. A
307. E
308. E
309. E
310. E
311. D
312. E
313. B
314. E
315. E
316. D
317. E
318. A
319. A
320. C
321. D
322. C
323. C
324. E
325. D
326. B
327. D
328. E
329. C
330. C
331. C
332. C
333. B
334. D
335. A
336. B
337. C
338. A
339. B
340. B
341. A
342. E
343. D
344. A
345. C
346. A
347. A
348. E
349. D
350. B
351. D
352. A
353. C
354. A
355. D
356. E
357. E
358. C
359. D
360. B
361. B
362. C
363. D
364. C
365. C
366. B
367. A
368. D
369. A
370. A
371. E
372. D
373. B
374. E
375. B
376. B
377. D
378. C
379. D
380. E
381. D
382. A
383. D
384. A
385. E
386. D
387. B
388. A
389. A
390. B
391. A
392. C
393. B
394. C
395. B
396. D
397. E
398. D
399. D
400. C
401. A
402. B
403. A
404. A

6.5 Answer Explanations

The following discussion of data sufficiency is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to data sufficiency. The particular questions in this chapter are generally representative of the kinds of data sufficiency questions you will encounter on the GMAT exam. Remember that it is the problem solving strategy that is important, not the specific details of a particular question.

DS02562

231. What is the number of pages of a certain journal article?

(1) The size of each page is $5\frac{1}{2}$ inches by 8 inches.

(2) The average (arithmetic mean) number of words per page is 250.

Arithmetic Applied problems

(1) Given that the size of each page is 5½ inches by 8 inches, it is clearly not possible to determine the number of pages; NOT sufficient.

(2) Given that the average number of words per page is 250, it is not possible to determine the number of pages because nothing is known about the number of words other than the number of words is a positive integer multiple of 250; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the number of pages, because the total number of words is not known.

The correct answer is E;
both statements together are still not sufficient.

DS10471

232. If a certain vase contains only roses and tulips, how many tulips are there in the vase?

(1) The number of roses in the vase is 4 times the number of tulips in the vase.

(2) There is a total of 20 flowers in the vase.

Algebra Simultaneous equations

Let R and T, respectively, be the number of roses and tulips in the vase. What is the value of T?

(1) Given that $R = 4T$, it is not possible to determine the value of T. For example, $R = 4$ and $T = 1$ is possible. However, $R = 8$ and $T = 2$ is also possible; NOT sufficient.

(2) Given that $R + T = 20$, it is not possible to determine the value of T. For example, $R = 10$ and $T = 10$ is possible. However, $R = 15$ and $T = 5$ is also possible; NOT sufficient.

Taking (1) and (2) together, substituting $R = 4T$ into $R + T = 20$ gives $4T + T = 20$, or $5T = 20$, or $T = 4$.

The correct answer is C;
both statements together are sufficient.

DS03802

233. The cost of 10 pounds of apples and 2 pounds of grapes was \$12. What was the cost per pound of apples?

(1) The cost per pound of grapes was \$2.

(2) The cost of 2 pounds of apples was less than the cost of 1 pound of grapes.

Algebra Simultaneous equations

Let A be the cost, in dollars per pound, of the apples and let G be the cost, in dollars per pound, of the grapes. It is given that $10A + 2G = 12$. What is the value of A?

(1) Given that $G = 2$, it follows that $10A + 2(2) = 12$, or $10A = 8$, or $A = 0.8$; SUFFICIENT.

(2) Given that $2A < G$, different possible values of A can be found by using $10A + 2G = 12$, or $2G = 12 - 10A$, or $G = 6 - 5A$, and picking values for A, determining the corresponding values for G, and then determining whether the inequality $2A < G$ is true. Thus, the values of A and B could be, respectively, 0.8 and 2, since both

$10(0.8) + 2(2) = 8 + 4 = 12$ and $2(0.8) < 2$ are true. On the other hand, the values of A and B could be, respectively, 0.6 and 3, since both $10(0.6) + 2(3) = 6 + 6 = 12$ and $2(0.6) < 3$ are true; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS05863

234. What was the median annual salary for the employees at Company X last year?

(1) Last year there were 29 employees at Company X.

(2) Last year 12 employees at Company X had an annual salary of $24,000.

Arithmetic Statistics

(1) Given that there were 29 employees, the median salary cannot be determined because no information about the salaries is given; NOT sufficient.

(2) Given that there were 12 salaries of $24,000, the median salary cannot be determined. For example, if there were 12 salaries of $24,000 and 17 salaries of $20,000, then the median salary would be $20,000. On the other hand, if there were 12 salaries of $24,000 and 17 salaries of $30,000, then the median salary would be $30,000; NOT sufficient.

Taking (1) and (2) together is still not sufficient because the same examples used in (2) also satisfy (1).

The correct answer is E;
both statements together are still not sufficient.

DS03422

235. How many basic units of currency X are equivalent to 250 basic units of currency Y ?

(1) 100 basic units of currency X are equivalent to 625 basic units of currency Y.

(2) 2,000 basic units of currency X are equivalent to 12,500 basic units of currency Y.

Arithmetic Ratio and proportion

Let x be the number of basic units of currency X that are equivalent to 1 basic unit of currency Y. What is the value of $250x$?

(1) Given that 100 basic units of currency X are equivalent to 625 basic units of currency Y, it follows that $\frac{100}{625}$ basic units of currency X are equivalent to 1 basic unit of currency Y. Therefore, $x = \frac{100}{625}$ and $250x = 250\left(\frac{100}{625}\right)$; SUFFICIENT.

(2) Given that 2,000 basic units of currency X are equivalent to 12,500 basic units of currency Y, it follows that $\frac{2,000}{12,500}$ basic units of currency X are equivalent to 1 basic unit of currency Y. Therefore, $x = \frac{2,000}{12,500}$ and $250x = 250\left(\frac{2,000}{12,500}\right)$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS12265

236. A company bought 3 printers and 1 scanner. What was the price of the scanner?

(1) The total price of the printers and the scanner was $1,300.

(2) The price of each printer was 4 times the price of the scanner.

Algebra Simultaneous equations

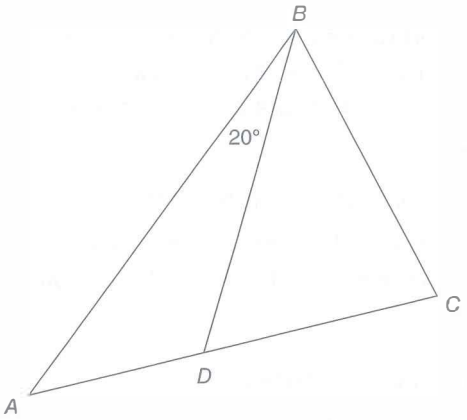
Let P_1, P_2, and P_3 be the prices, in dollars, of the 3 printers and let S be the price, in dollars, of the scanner. What is the value of S ?

(1) Given that $P_1 + P_2 + P_3 + S = 1,300$, it is clear that the value of S cannot be determined; NOT sufficient.

(2) Given that $P_1 = P_2 = P_3 = 4S$, it is possible that $P_1 = P_2 = P_3 = 400$ and $S = 100$. On the other hand, it is also possible that $P_1 = P_2 = P_3 = 800$ and $S = 200$; NOT sufficient.

Taking (1) and (2) together, substituting $P_1 = 4S$, $P_2 = 4S$, and $P_3 = 4S$ into $P_1 + P_2 + P_3 + S = 1,300$ gives $4S + 4S + 4S + S = 1,300$, or $13S = 1,300$, and therefore $S = 100$.

The correct answer is C;
both statements together are sufficient.



DS17639

237. In the figure above, point D is on $\overline{AC}$. What is the degree measure of $\angle BAC$?

(1) The measure of $\angle BDC$ is 60°.

(2) The degree measure of $\angle BAC$ is less than the degree measure of $\angle BCD$.

Geometry Angles

(1) Given that the measure of $\angle BDC$ is 60°, it follows that the measure of $\angle BDA$ is $180° - 60° = 120°$. Therefore, since the measures of the angles in a triangle add to 180°, the measure of $\angle BAC$ is $180 - (20° + 120°) = 40°$; SUFFICIENT.

(2) Given that the degree measure of $\angle BAC$ is less than the degree measure of $\angle BCD$, when the line segment $\overline{AC}$ rotates a few degrees counterclockwise about point A and point D stays on $\overline{AC}$, then the degree measure of $\angle BAC$ decreases (and the degree measure of $\angle BCD$ increases, and the degree measure of $\angle ABD$ remains 20°). Therefore, the degree measure of $\angle BAC$ can vary; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS07822

238. Each of the 256 solid-colored marbles in a box is either blue, green, or purple. What is the ratio of the number of blue marbles to the number of purple marbles in the box?

(1) The number of green marbles in the box is 4 times the number of blue marbles in the box.

(2) There are 192 green marbles in the box.

Arithmetic Ratio and Proportion

Let B, G, and P be the numbers of marbles, respectively, that are blue, green, and purple. From the given information it follows that $B + G + P = 256$. What is the value of $\frac{B}{P}$?

(1) Given that $G = 4B$, it is possible that $\frac{B}{P} = \frac{1}{251}$ (choose $B = 1$, $G = 4$, and $P = 251$) and it is possible that $\frac{B}{P} = \frac{1}{123}$ (choose $B = 2$, $G = 8$, and $P = 246$); NOT sufficient.

(2) Given that $G = 192$, it is possible that $\frac{B}{P} = \frac{1}{63}$ (choose $B = 1$, $G = 192$, and $P = 63$) and it is possible that $\frac{B}{P} = \frac{1}{31}$ (choose $B = 2$, $G = 192$, and $P = 62$); NOT sufficient.

Taking (1) and (2) together, it follows from $G = 4B$ and $G = 192$ that $192 = 4B$, or $B = 48$. Thus, $B + G + P = 256$ becomes $48 + 192 + P = 256$, and so $P = 256 - 48 - 192 = 16$. Therefore, $\frac{B}{P} = \frac{48}{16}$.

The correct answer is C; both statements together are sufficient.

DS15940

239. A certain mixture of paint requires blue, yellow, and red paints in ratios of 2:3:1, respectively, and no other ingredients. If there are ample quantities of the blue and red paints available, is there enough of the yellow paint available to make the desired amount of the mixture?

(1) Exactly 20 quarts of the mixture are needed.

(2) Exactly 10 quarts of the yellow paint are available.

Arithmetic Ratios

Given that the mixture requires blue paint, yellow paint, and red paint in the ratios 2:3:1, it follows that $\frac{3}{2 + 3 + 1} = \frac{1}{2}$ of the mixture will be yellow paint. Determining whether there is enough yellow paint available depends on how much of the mixture is needed and how much yellow paint is available.

(1) This indicates that exactly 20 quarts of the paint mixture are needed, so $\frac{1}{2}(20) = 10$ quarts of yellow paint are needed. However, there is no information about how much yellow paint is available; NOT sufficient.

(2) This indicates that exactly 10 quarts of yellow paint are available, but there is no information about how much of the mixture or how much yellow paint is needed; NOT sufficient.

Taking (1) and (2) together, 10 quarts of yellow paint are needed and 10 quarts are available.

The correct answer is C;
both statements together are sufficient.

DS05338

240. The research funds of a certain company were divided among three departments, X, Y, and Z. Which one of the three departments received the greatest proportion of the research funds?

(1) The research funds received by departments X and Y were in the ratio 3 to 5, respectively.

(2) The research funds received by departments X and Z were in the ratio 2 to 1, respectively.

Algebra Order; Ratio

Let x, y, and z be the research funds, respectively, of departments X, Y, and Z. Which of $\frac{x}{x+y+z}$, $\frac{y}{x+y+z}$, or $\frac{z}{x+y+z}$ is the greatest, or respectively, which of x, y, or z is the greatest?

(1) Given that $\frac{x}{y}=\frac{3}{5}$ where x and y are positive, it follows that $x<y$. However, nothing is known about z other than z is positive, so it cannot be determined which of x, y, or z is the greatest; NOT sufficient.

(2) Given that $\frac{x}{z}=\frac{2}{1}$ where x and z are positive, it follows that $z<x$. However, nothing is known about y other than y is positive, so it cannot be determined which of x, y, or z is the greatest; NOT sufficient.

Taking (1) and (2) together, it follows from (1) that $x<y$ and it follows from (2) that $z<x$. Therefore, $z<x<y$ and y is the greatest.

The correct answer is C;
both statements together are sufficient.

DS03138

241. In a certain class, some students donated cans of food to a local food bank. What was the average (arithmetic mean) number of cans donated per student in the class?

(1) The students donated a total of 56 cans of food.

(2) The total number of cans donated was 40 greater than the total number of students in the class.

Arithmetic Statistics

Let s be the number of students and let c be the number of cans. What is the value of $\frac{c}{s}$?

(1) Given that $c=56$, it is not possible to determine the value of $\frac{c}{s}$ because nothing is known about the value of s other than s is a positive integer; NOT sufficient.

(2) Given that $c=40+s$, it is not possible to determine the value of $\frac{c}{s}$. For example, if $c=40$ and $s=40$, then $\frac{c}{s}=1$. However, if $c=80$ and $s=40$, then $\frac{c}{s}=2$; NOT sufficient.

Taking (1) and (2) together, it follows that $40+s=c=56$, or $s=56-40=16$. Therefore, $c=56$, $s=16$, and $\frac{c}{s}=\frac{56}{16}$.

The correct answer is C;
both statements together are sufficient.

DS00254

242. Each of the n employees at a certain company has a different annual salary. What is the median of the annual salaries of the n employees?

(1) When the annual salaries of the n employees are listed in increasing order, the median is the 15th salary.

(2) The sum of the annual salaries of the n employees is $913,500.

Arithmetic Statistics

(1) Given that the median is the 15th salary when the n salaries, which are all different from each other, are listed in increasing order, it follows that n is an odd number. This is because if there were an even number of salaries, where S_k and S_{k+1} are the two

middle salaries, then the median salary would be midway between the different numbers S_k and S_{k+1}, and thus the median salary could not be any of the salaries. Also, since the 15th salary is the middle salary, there must be a total of 29 salaries, as indicated below.

first 14 salaries	15th salary	last 14 salaries
$S_1, S_2, \ldots, S_{14}$	S_{15}	$S_{16}, S_{17}, \ldots, S_{29}$

However, no explicit values are given for any of the salaries, and therefore the median of the 29 salaries cannot be determined; NOT sufficient.

(2) Given that the sum of the annual salaries is \$913,500, the median salary cannot be determined. For example, since \$913,500 = 21 × \$43,500, there could be 21 different salaries each greater than \$40,000, and so the median could be greater than \$40,000. However, since \$913,500 = 29 × \$31,500, there could be 29 different salaries each less than \$35,000, and so the median could be less than \$35,000; NOT sufficient.

Taking (1) and (2) together, it follows from (1) that there are 29 salaries and it follows from (2) that the sum of the salaries is \$913,500. However, the median salary still cannot be determined. For example, let the first 13 salaries be \$31,500 – \$1,400, \$31,500 – \$1,300, . . ., and \$31,500 – \$200; and let the last 13 salaries be \$31,500 + \$200, \$31,500 + \$300, . . ., and \$31,500 + \$1,400. Then the 3 middle salaries (the 14th, 15th, and 16th salaries) could be \$31,500 – \$100, \$31,500, and \$31,500 + \$100, and thus the median salary could be \$31,500. However, the 3 middle salaries (the 14th, 15th, and 16th salaries) could be \$31,500 – \$100, \$31,500 + \$40, and \$31,500 + \$60, and thus the median salary could be \$31,540.

The correct answer is E;
both statements together are still not sufficient.

DS10687

243. In a recent town election, what was the ratio of the number of votes in favor of a certain proposal to the number of votes against the proposal?

(1) There were 60 more votes in favor of the proposal than against the proposal.

(2) There were 240 votes in favor of the proposal.

Arithmetic Ratio and Proportion

Let F be the number of votes in favor and let A be the number of votes against. What is the value of $\frac{F}{A}$?

(1) Given that $F = 60 + A$, it is possible that $\frac{F}{A} = 2$ (choose $F = 120$ and $A = 60$) and it is possible that $\frac{F}{A} = 3$ (choose $F = 90$ and $A = 30$); NOT sufficient.

(2) Given that $F = 240$, it is possible that $\frac{F}{A} = 1$ (choose $F = 240$ and $A = 240$) and it is possible that $\frac{F}{A} = 2$ (choose $F = 240$ and $A = 120$); NOT sufficient.

Taking (1) and (2) together, substitute $F = 240$ into $F = 60 + A$ to get $240 = 60 + A$, or $A = 180$. Therefore, $\frac{F}{A} = \frac{240}{180}$.

The correct answer is C;
both statements together are sufficient.

DS02541

244. How many men are in a certain company's vanpool program?

(1) The ratio of men to women in the program is 3 to 2.

(2) The men and women in the program fill 6 vans.

Arithmetic Applied problems

Let m be the number of men in the program and let w be the number of women in the program. What is the value of m?

(1) Given that $\frac{m}{w} = \frac{3}{2}$, it is not possible to determine the value of m. For example, if $m = 3$ and $w = 2$, then $\frac{m}{w} = \frac{3}{2}$ is true; and if $m = 6$ and $w = 4$, then $\frac{m}{w} = \frac{3}{2}$ is true; NOT sufficient.

(2) Given that the men and women fill 6 vans, it is clearly not possible to determine the value of m; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the value of m. Even if it is assumed that each van has the same maximum capacity, it is possible that $m = 18$ and $w = 12$ (a total of 30 people, which would fill 6 vans each having a capacity of 5 people) and it is possible that $m = 36$ and $w = 24$ (a total of 60 people, which would fill 6 vans each having a capacity of 10 people).

The correct answer is E;
both statements together are still not sufficient.

DS08054

245. Each of the marbles in a jar is either red or white or blue. If one marble is to be selected at random from the jar, what is the probability that the marble will be blue?

(1) There are a total of 24 marbles in the jar, 8 of which are red.

(2) The probability that the marble selected will be white is $\frac{1}{2}$.

Arithmetic Probability

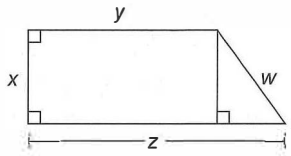
This problem can be solved by determining the number of marbles in the jar and the number that are blue.

(1) This indicates that there are 24 marbles in the jar and 8 of them are red. The number of marbles that are blue is not known; NOT sufficient.

(2) This indicates neither the number of marbles in the jar nor the number that are blue; NOT sufficient.

Taking (1) and (2) together, there are 24 marbles in the jar, 8 of which are red and $\frac{1}{2}(24) = 12$ of which are white. Then, the number of blue marbles is $24 - 8 - 12 = 4$, and the probability that one marble selected at random will be blue is $\frac{4}{24} = \frac{1}{6}$.

The correct answer is C;
both statements together are sufficient.



DS04594

246. In the figure above, what is the value of z?

(1) $x = y = 1$

(2) $w = 2$

Geometry Triangles; Pythagorean theorem

(1) Given that $x = y = 1$, the length of the base of the triangle to the right, which equals $z - y = z - 1$, can vary, and thus the value of z can vary; NOT sufficient.

(2) Given that $w = 2$, all dimensions of the rectangle to the left can vary, and thus the value of z can vary; NOT sufficient.

Taking (1) and (2) together, then $z = y + (z - y) = 1 + (z - 1)$, and the value of $z - 1$ can be determined by applying the Pythagorean theorem to the triangle that has hypotenuse of length $w = 2$ and sides of lengths $x = 1$ and $z - 1$. Specifically, the Pythagorean theorem gives $1^2 + (z - 1)^2 = 2^2$, or $1 + (z - 1)^2 = 4$, and solving this equation gives $(z - 1)^2 = 3$, or $z - 1 = \sqrt{3}$, or $z = 1 + \sqrt{3}$.

The correct answer is C;
both statements together are sufficient.

DS04630

247. What is the value of 10 percent of y?

(1) 5 percent of y is 60.

(2) y is 80 percent of 1,500.

Arithmetic Percents

This problem can be solved by determining the value of y.

(1) If 5 percent of y is 60, then $0.05y = 60$ and $y = \frac{60}{0.05} = 1{,}200$; SUFFICIENT.

(2) If y is 80 percent of 1,500, then $y = (0.80)(1{,}500) = 1{,}200$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS12062

248. Last semester, Professor K taught two classes, A and B. Each student in class A handed in 7 assignments, and each student in class B handed in 5 assignments. How many students were in class A?

(1) The students in both classes combined handed in a total of 85 assignments.

(2) There were 10 students in class B.

Algebra Simultaneous equations

Let a be the number of students in class A and let b be the number of students in class B. Then the students in class A handed in a total of $7a$ assignments and the students in class B handed in a total of $5b$ assignments. What is the value of a?

(1) Given that $7a + 5b = 85$, it is not possible to determine the value of a. For example, it is possible that $a = 5$ and $b = 10$, since $7(5) + 5(10) = 85$. On the other hand, it is also possible that $a = 10$ and $b = 3$, since $7(10) + 5(3) = 85$; NOT sufficient.

(2) Given that $b = 10$, it is not possible to determine the value of a. For example, it is possible that $a = b = 10$; and it is also possible that $a = 5$ and $b = 10$; NOT sufficient.

Taking (1) and (2) together, substituting $b = 10$ into $7a + 5b = 85$ gives $7a + 5(10) = 85$, or $7a = 35$, or $a = 5$.

The correct answer is C;
both statements together are sufficient.

DS06802

249. Was the amount of John's heating bill for February greater than it was for January?

(1) The ratio of the amount of John's heating bill for February to that for January was $\frac{26}{25}$.

(2) The sum of the amounts of John's heating bills for January and February was $183.60.

Arithmetic Applied problems

Let J and F be the amounts, respectively and in dollars, of the heating bills for January and February. Is $F > J$?

(1) Given that $\frac{F}{J} = \frac{26}{25}$, it follows that $\frac{F}{J} > 1$. Multiplying both sides of the inequality $\frac{F}{J} > 1$ by the positive quantity J gives $F > J$; SUFFICIENT.

(2) Given that $J + F = 183.60$, it is not possible to determine whether $F > J$. For example, if $J = 83.60$ and $F = 100.00$, then $J + F = 183.60$ and $F > J$. On the other hand, if $J = 100.00$ and $F = 83.60$, then $J + F = 183.60$ and $F < J$; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS06662

250. If sequence S has 120 terms, what is the 105th term of S?

(1) The first term of S is -8.

(2) Each term of S after the first term is 10 more than the preceding term.

Arithmetic Sequences

This problem can be solved by determining at least one term of the sequence and how each term is derived from the preceding term(s).

(1) This indicates that the first term of the sequence is -8, but does not indicate how subsequent terms are derived; NOT sufficient.

(2) This indicates how each term is derived from the preceding term, but does not indicate at least one term of the sequence; NOT sufficient.

Taking (1) and (2) together, the first term is -8 and each subsequent term is 10 more than the preceding term from which the 105th term can be determined.

The correct answer is C;
both statements together are sufficient.

DS00858

251. Machine R and machine S work at their respective constant rates. How much time does it take machine R, working alone, to complete a certain job?

(1) The amount of time that it takes machine S, working alone, to complete the job is $\frac{3}{4}$ the amount of time that it takes machine R, working alone, to complete the job.

(2) Machine R and machine S, working together, take 12 minutes to complete the job.

Algebra Applied problems

For machines R and S, let r_R and r_S be their respective rates, in jobs per minute; and let t_R and t_S be their respective times, in minutes, to complete 1 job. Then $1 = r_R t_R$ (i.e. 1 job is completed by machine R working at rate r_R for t_R minutes) and $1 = r_S t_S$ (i.e. 1 job is completed by machine S working at rate r_S for t_S minutes). What is the value of t_R?

(1) Given that $t_S = \frac{3}{4} t_R$, the value of t_R cannot be determined because nothing else is known about t_R and t_S other than they are positive real numbers; NOT sufficient.

(2) Given that 1 job is completed by machines R and S working together at a combined rate $r_R + r_S$ for 12 minutes, or $1 = (r_R + r_S)(12)$, the value of t_R cannot be determined. For example, if $r_R = \frac{1}{24}$ and $r_S = \frac{1}{24}$, then $1 = (r_R + r_S)(12)$ is true and from $1 = r_R t_R$, or $1 = \left(\frac{1}{24}\right) t_R$, it follows that $t_R = 24$. However, if $r_R = \frac{1}{48}$ and $r_S = \frac{3}{48}$, then $1 = (r_R + r_S)(12)$ is true and from $1 = r_R t_R$, or $1 = \left(\frac{1}{48}\right) t_R$, it follows that $t_R = 48$; NOT sufficient.

Taking (1) and (2) together, from (1) it follows that $t_S = \frac{3}{4} t_R$ and from (2) it follows that $(r_R + r_S)(12) = 1$, or $r_R + r_S = \frac{1}{12}$. Using $r_R = \frac{1}{t_R}$ and $r_S = \frac{1}{t_S}$ (obtained from $1 = r_R t_R$ and $1 = r_S t_S$), the last equality in the previous sentence becomes $\frac{1}{t_R} + \frac{1}{t_S} = \frac{1}{12}$. Substituting $t_S = \frac{3}{4} t_R$ into $\frac{1}{t_R} + \frac{1}{t_S} = \frac{1}{12}$ gives $\frac{1}{t_R} + \frac{1}{(3/4)t_R} = \frac{1}{12}$, or $\frac{1}{t_R} + \frac{4}{3} \cdot \frac{1}{t_R} = \frac{1}{12}$, or $\frac{7}{3} \cdot \frac{1}{t_R} = \frac{1}{12}$. Therefore, $\frac{1}{t_R} = \frac{3}{7} \cdot \frac{1}{12} = \frac{1}{28}$, or $t_R = 28$.

The correct answer is C;
both statements together are sufficient.

DS06065

252. If $u > 0$ and $v > 0$, which is greater, u^v or v^u?

(1) $u = 1$

(2) $v > 2$

Arithmetic Exponents

(1) Given that $u = 1$, then $u^v = 1^v = 1$ and $v^u = v^1 = v$. If $v = 2$, then $u^v = 1$ is less than $v^u = 2$. However, if $v = 0.5$, then $u^v = 1$ is greater than $v^u = 0.5$; NOT sufficient.

(2) Given that $v > 2$, it is possible that u^v is less than v^u (for example, if $u = 1$ and $v = 4$, then $u^v = 1^4 = 1$ and $v^u = 4^1 = 4$), and it is possible that u^v is greater than v^u (for example, if $u = 3$ and $v = 4$, then $u^v = 3^4 = 81$ and $v^u = 4^3 = 64$); NOT sufficient.

Taking (1) and (2) together, it follows from (1) that $u^v = 1$ and $v^u = v$, and it follows from (2) that $v > 2$. Therefore, $u^v = 1 < 2 < v = v^u$, and so v^u is greater than u^v.

The correct answer is C;
both statements together are sufficient.

DS00660

253. What was the range of the selling prices of the 30 wallets sold by a certain store yesterday?

(1) $\frac{1}{3}$ of the wallets had a selling price of \$24 each.

(2) The lowest selling price of the wallets was $\frac{1}{3}$ the highest selling price of the wallets.

Arithmetic Statistics

Since the range of a data set is the greatest value in the data set minus the least value in the data set, this problem can be solved if the least selling price and the greatest selling price of the wallets sold by the store yesterday can be determined.

(1) This indicates that 10 of the wallets had a selling price of \$24 each, but does not indicate the least selling price or the greatest selling price of the wallets sold by the store yesterday; NOT sufficient.

(2) This indicates that $L = \frac{1}{3} G$, where L represents the least selling price and G represents the greatest selling price, but does

not give enough information to determine $G - L$; NOT sufficient.

Taking (1) and (2) together, the least and greatest selling prices of the wallets sold by the store yesterday could be \$10 and \$30 for a range of \$30 – \$10 = \$20, or the least and greatest selling prices of the wallets sold by the store yesterday could be \$20 and \$60 for a range of \$60 – \$20 = \$40.

The correct answer is E;
both statements together are still not sufficient.

DS08723

254. Three houses are being sold through a real estate agent. What is the asking price for the house with the second-largest asking price?

(1) The difference between the greatest and the least asking price is \$130,000.

(2) The difference between the two greater asking prices is \$85,000.

Algebra Simultaneous equations

Let x, y, and z, where $x \leq y \leq z$, be the asking prices of the three houses. This problem can be solved by determining the value of y.

(1) This indicates that $z - x = \$130{,}000$, but does not give the value of y; NOT sufficient.

(2) This indicates that $z - y = \$85{,}000$, but does not give the value of y; NOT sufficient.

Taking (1) and (2) together, x, y, and z could be \$100,000, \$145,000, and \$230,000, respectively, in which case the second highest selling price is \$145,000, or they could be \$200,000, \$245,000, and \$330,000, respectively, in which case the second highest selling price is \$245,000.

The correct answer is E;
both statements together are still not sufficient.

DS04605

255. If $a + b + c = 12$, what is the value of b?

(1) $a + b = 8$

(2) $b + c = 6$

Algebra Simultaneous equations

(1) Given that $a + b = 8$, it follows from $(a + b) + c = 12$ that $8 + c = 12$, or $c = 4$. However, the value of b still cannot be determined. For example, if $a = 6$, $b = 2$, and $c = 4$, then $a + b + c = 12$ and $a + b = 8$. On the other hand, if $a = 4$, $b = 4$, and $c = 4$, then $a + b + c = 12$ and $a + b = 8$; NOT sufficient.

(2) Given that $b + c = 6$, it follows from $a + (b + c) = 12$ that $a + 6 = 12$, or $a = 6$. However, the value of b still cannot be determined. For example, if $a = 6$, $b = 3$, and $c = 3$, then $a + b + c = 12$ and $b + c = 6$. On the other hand, if $a = 6$, $b = 2$, and $c = 4$, then $a + b + c = 12$ and $b + c = 6$; NOT sufficient.

Taking (1) and (2) together, it follows from (1) that $c = 4$ and it follows from (2) that $a = 6$. Substituting these values in $a + b + c = 12$ gives $6 + b + 4 = 12$, or $b = 2$.

The correct answer is C;
both statements together are sufficient.

DS11254

256. Is $rw = 0$?

(1) $-6 < r < 5$

(2) $6 < w < 10$

Algebra Inequalities

This problem can be solved if it can be determined that either or both of r and w is zero or that neither r nor w is zero.

(1) This indicates that r can be, but does not have to be, zero and gives no indication of the value of w; NOT sufficient.

(2) This indicates that w cannot be zero, but gives no indication of the value of r; NOT sufficient.

Taking (1) and (2) together, r could be zero, in which case $rw = 0$ **or** r could be nonzero, in which case $rw \neq 0$.

The correct answer is E;
both statements together are still not sufficient.

DS06633

257. Is $x = \frac{1}{y}$?

(1) $xy = 1$

(2) $\frac{1}{xy} = 1$

Algebra Equations

(1) This indicates that neither x nor y can be zero because if $x = 0$ or $y = 0$, then $xy = 0$. Since $xy = 1$ and $y \neq 0$, then dividing both sides by y gives $x = \frac{1}{y}$; SUFFICIENT.

(2) This indicates that neither x nor y can be zero because if $x = 0$ or $y = 0$, then $xy = 0$, and division by zero is undefined. Multiplying both sides of the equation $\frac{1}{xy} = 1$ by x gives $\frac{1}{y} = x$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS07949

258. How many people in a group of 50 own neither a fax machine nor a laser printer?

(1) The total number of people in the group who own a fax machine or a laser printer or both is less than 50.

(2) The total number of people in the group who own both a fax machine and a laser printer is 15.

Algebra Sets

(1) This indicates that the total number who own either a fax machine or a laser printer or both is less than 50, but does not indicate how much less than 50 the total number is. Thus, the number of people in the group who own neither a fax machine nor a laser printer cannot be determined; NOT sufficient.

(2) This indicates that, of the 50 people, 15 own both a fax machine and a laser printer, but does not indicate how many own one or the other. Thus, the number of people in the group who own neither a fax machine nor a laser printer cannot be determined; NOT sufficient.

Taking (1) and (2) together, it is known that 15 people own both a fax machine and a laser printer and that the total number who own either a fax machine or a laser printer or both is less than 50, but the exact number who own neither still cannot be determined. For example, if 20 people own only a fax machine and 10 people own only a laser printer, then both (1) and (2) are true and the number of people who own neither a fax machine nor a laser printer is $50 - 20 - 10 - 15 = 5$. However, if 10 people own only a fax machine and 10 people own only a laser printer, then both (1) and (2) are true and the number of people who own neither a fax machine nor a laser printer is $50 - 10 - 10 - 15 = 15$.

The correct answer is E;
both statements together are still not sufficient.

DS06475

259. What is the value of w^{-2}?

(1) $w^{-1} = \frac{1}{2}$

(2) $w^3 = 8$

Algebra Exponents

(1) Since $w^{-1} = \frac{1}{2}$, then $(w^{-1})^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ or $w^{-2} = \frac{1}{4}$; SUFFICIENT.

(2) Since $w^3 = 8$, then $w = \sqrt[3]{8} = 2$ and $w^{-2} = 2^{-2} = \frac{1}{4}$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS07839

260. A certain investment earned a fixed rate of 4 percent interest per year, compounded annually, for five years. The interest earned for the third year of the investment was how many dollars greater than that for the first year?

(1) The amount of the investment at the beginning of the second year was $4,160.00.

(2) The amount of the investment at the beginning of the third year was $4,326.40.

Arithmetic Applied problems; Percents

This problem can be solved by determining the dollar amount of the interest earned for the third year minus the dollar amount of interest earned in the first year, or, in symbols, $[A(1.04)^3 - A(1.04)^2] - [A(1.04)^1 - A]$, where A represents the dollar amount of the investment.

This can be determined if the value of A can be determined.

(1) This indicates that $A(1.04)^1 = \$4{,}160.00$, since the amount of the investment at the beginning of the second year is the same as the amount of the investment at the end of the first year. Thus, the value of A can be determined by dividing 4,160.00 by 1.04; SUFFICIENT.

(2) This indicates that $A(1.04)^2 = \$4{,}326.40$, since the amount of the investment at the beginning of the third year is the same as the amount of the investment at the end of the second year. Thus, the value of A can be determined by dividing 4,326.40 by $(1.04)^2$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06397

261. What is the circumference of circle C ?

(1) The radius of circle C is 2π.

(2) The center of circle C is located at point (7,8) in the xy-plane.

Geometry Circles

What is the value of $2\pi r$, where r is the radius of circle C?

(1) Given that $r = 2\pi$, the value of $2\pi r$ is $2\pi \cdot 2\pi$; SUFFICIENT.

(2) Given that the center of circle C is located at (7,8), the radius of circle C can be any positive real number, and therefore the value of $2\pi r$ cannot be determined; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS07813

262. What is the value of t ?

(1) $s + t = 6 + s$

(2) $t^3 = 216$

Algebra Equations

(1) Given that $s + t = 6 + s$, subtracting s from both sides gives $t = 6$; SUFFICIENT.

(2) Given that $t^3 = 216$, taking the cube root of both sides gives $t = 6$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS11109

263. For a certain car repair, the total charge consisted of a charge for parts, a charge for labor, and a 6 percent sales tax on both the charge for parts and the charge for labor. If the charge for parts, excluding sales tax, was \$50.00, what was the total charge for the repair?

(1) The sales tax on the charge for labor was \$9.60.

(2) The total sales tax was \$12.60.

Arithmetic Applied problems; Percents

To find the total charge, T, for the repair, requires determining the value, in dollars, of $T = (1.06)(50 + L) = 53 + 1.06L$, where L represents the charge, in dollars, for labor. This value can be determined if the value of L can be determined.

(1) This indicates that $0.06L = 9.60$, from which L can be determined by dividing 9.60 by 0.06; SUFFICIENT.

(2) This indicates that $(0.06)(50 + L) = 12.60$. So, $3.00 + 0.06L = 12.60$ or $0.06L = 9.60$, which is the same as the equation in (1); SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06905

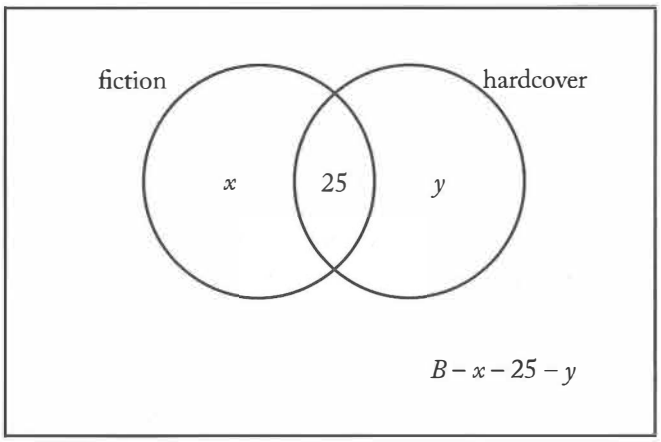
264. George has a total of B books in his library, 25 of which are hardcover fiction books. What is the value of B ?

(1) 40 of the B books are fiction and the rest are nonfiction.

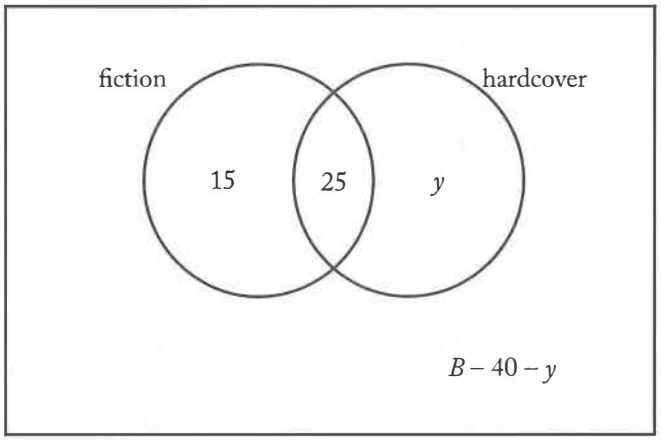
(2) 60 of the B books are hardcovers and the rest are paperbacks.

Arithmetic Applied problems

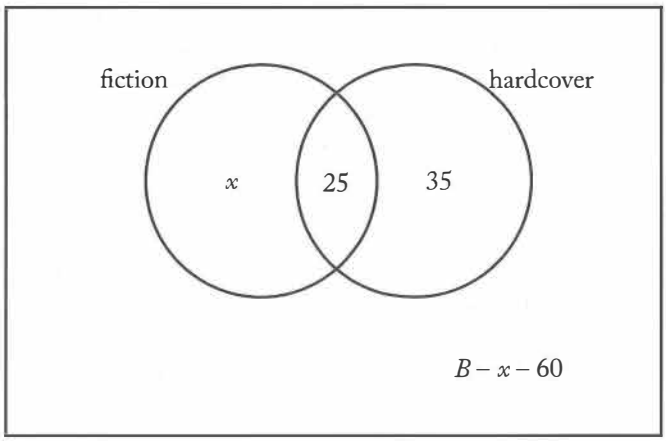
The diagram below incorporates the given information that George has a total of B books, of which 25 are both fiction and hardcover. In the diagram, x represents the number of books that are fiction and not hardcover, and y represents the number of books that are hardcover and not fiction.



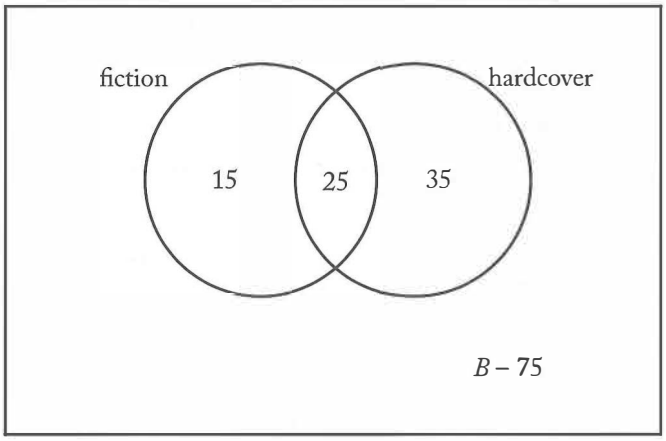
(1) Given that 40 of the books are fiction, it follows that $x + 25 = 40$, or $x = 15$. Using this additional information, the diagram (shown below) clearly shows that the value of B cannot be determined. For example, $y = 35$ and $B = 80$ is possible (note that $B - 40 - y = 5$), and $y = 35$ and $B = 100$ is possible (note that $B - 40 - y = 25$); NOT sufficient.



(2) Given that 60 of the books are hardcover, it follows that $25 + y = 60$, or $y = 35$. Using this additional information, the diagram (shown below) clearly shows that the value of B cannot be determined. For example, $x = 15$ and $B = 80$ is possible (note that $B - x - 60 = 5$), and $x = 15$ and $B = 100$ is possible (note that $B - x - 60 = 25$); NOT sufficient.



Taking (1) and (2) together, there is still not enough information to determine the value of B, since in the diagram below $B = 80$ is possible and $B = 100$ is possible.



The correct answer is E;
both statements together are still not sufficient.

DS12031

265. Is $w + h^4$ positive?

(1) h is positive.

(2) w is positive.

Arithmetic Properties of numbers

(1) Given that h is positive, then $w + h^4$ can be positive (for example, when $w = 1$ and $h = 1$) and $w + h^4$ can be negative (for example, when $w = -2$ and $h = 1$); NOT sufficient.

(2) Given that w is positive, then because $b^4 \geq 0$, it follows that $w + b^4 \geq w + 0 = w$ and thus $w + b^4 > 0$ since $w > 0$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

DS15377

266. If a is a 3-digit integer and b is a 3-digit integer, is the units digit of the product of a and b greater than 5 ?

(1) The units digit of a is 4.

(2) The units digit of b is 7.

Arithmetic Operations with integers

Determine whether the units digit of the product of the two 3-digit integers a and b is greater than 5.

(1) It is given that the units digit of a is 4. If the units digit of b is 0 or 1, then the units digit of ab is not greater than 5. However, if the units digit of b is 2, then the units digit of ab is greater than 5; NOT sufficient.

(2) It is given that the units digit of b is 7. If the units digit of a is 0, then the units digit of ab is not greater than 5. However, if the units digit of a is 1, then the units digit of ab is greater than 5; NOT sufficient.

Taking (1) and (2) together, the units digit of a is 4 and the units digit of b is 7. Therefore, the units digit of ab is 8, which is greater than 5.

The correct answer is C; both statements together are sufficient.

DS02450

267. In each of the last five years, Company K donated p percent of its annual profits to a certain scholarship fund. Did Company K donate more than \$10,000 to the scholarship fund last year?

(1) Two years ago, Company K had annual profits of \$3 million and donated \$15,000 to the scholarship fund.

(2) Last year, Company K had annual profits of \$2.5 million.

Arithmetic Applied problems; Percents

Let $\$k$ be Company K's profit last year.

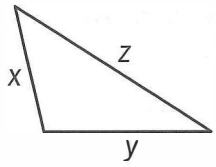
Is $\left(\frac{p}{100}\right)k > 10{,}000$?

(1) Given that $\left(\frac{p}{100}\right)(3 \text{ million}) = 15{,}000$, it follows that $p = \frac{(15{,}000)(100)}{3 \text{ million}} = \frac{1}{2}$. However, the question cannot be answered since no information is available about the value of k. For example, if $k = 3$ million, then $\left(\frac{p}{100}\right)k = 15{,}000$, and the answer to the question is yes. However, if $k = 1$ million, then $\left(\frac{p}{100}\right)k = 5{,}000$, and the answer to the question is no; NOT sufficient.

(2) Given that $k = 2.5$ million, it follows that $\left(\frac{p}{100}\right)k = \left(\frac{p}{100}\right)(2.5 \text{ million})$. However, the question cannot be answered since no information is available about the value of p. For example, if $p = 1$, then $\left(\frac{p}{100}\right)k = 25{,}000$, and the answer to the question is yes. However, if $p = 0.1$, then $\left(\frac{p}{100}\right)k = 2{,}500$, and the answer to the question is no; NOT sufficient.

Taking (1) and (2) together, the values of both p and k can be determined. Therefore, the value of $\left(\frac{p}{100}\right)k$ can be determined, and thus the answer to the question can be determined.

The correct answer is C; both statements together are sufficient.



DS06901

268. Is the area of the triangular region above less than 20 ?

(1) $x^2 + y^2 \neq z^2$

(2) $x + y < 13$

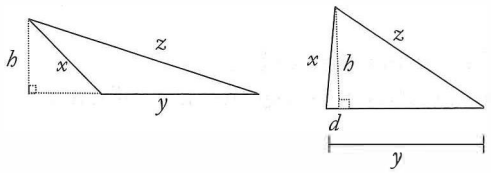
Geometry Triangles; Area

Determine whether the area of the given triangle is less than 20.

(1) This indicates the triangle is not a right triangle but gives no information about the base and height or how to determine them in order to find the area; NOT sufficient.

(2) This indicates that two sides of the triangle have lengths x and y such that $x + y < 13$. If, for example, the triangle is a right triangle with $x = 3$ and $y = 4$, then $x + y = 3 + 4 = 7 < 13$ and the area of the triangle is $\frac{1}{2}(3)(4) = 6$, which is less than 20. But if the triangle is a right triangle with $x = y = \sqrt{42}$, then $x + y = \sqrt{42} + \sqrt{42} = 2\sqrt{42} = \sqrt{168} < \sqrt{169} = 13$ and the area is $\frac{1}{2}(\sqrt{42})(\sqrt{42}) = 21$, which is greater than 20; NOT sufficient.

Taking (1) and (2) together, since the triangle is not a right triangle from (1), the diagrams show two other possibilities: an obtuse triangle and an acute triangle.



For the obtuse triangle, $h < x$ since because h is a leg of a right triangle with hypotenuse x. The area A is given by $A = \frac{1}{2}yh < \frac{1}{2}xy$. If $x = y = 6$, then $x + y = 6 + 6 = 12 < 13$ and $A < \frac{1}{2}(6^2) = 18$. The area of the triangular region is less than 20 for the obtuse triangle. For the acute triangle, if $x = y = \sqrt{42}$, then $x + y = \sqrt{42} + \sqrt{42} = 2\sqrt{42} = \sqrt{168} < \sqrt{169} = 13$. Let $d = 1$. Then, $h^2 = (\sqrt{42})^2 - 1 = 42 - 1 = 41$ and $h = \sqrt{41}$. Thus, for the acute triangle, the area of the triangular region is $\frac{1}{2}yh = \frac{1}{2}(\sqrt{42})(\sqrt{41})$ and $\frac{1}{2}(\sqrt{42})(\sqrt{41}) > \frac{1}{2}(\sqrt{40})(\sqrt{40})$ or $\frac{1}{2}(\sqrt{42})(\sqrt{41}) > 20$. Therefore, it is possible that the area of the triangular region can be less than 20 (in the case of the obtuse triangle) and it is possible that the area of the triangular region can be greater than 20 (in the case of the acute triangle).

The correct answer is E;
both statements together are still not sufficient.

DS04366

269. A, B, C, and D are points on a line. If C is the midpoint of line segment AB and if D is the midpoint of line segment CB, is the length of line segment DB greater than 5 ?

(1) The length of line segment AC is greater than 8.

(2) The length of line segment CD is greater than 6.

Geometry Lines and segments

It is given that C is the midpoint of $\overline{AB}$, so $AC = CB = \frac{1}{2}(AB)$. It is given that D is the midpoint of $\overline{CB}$, so $CD = DB = \frac{1}{2}(CB)$. Determine if $DB > 5$.

(1) Given that $AC > 8$, it follows that $CB > 8$ since $AC = CB$, and so $DB = \frac{1}{2}(CB) > \frac{1}{2}(8)$ or $DB > 4$. However, this means that DB could be 6 and $6 > 5$ or DB could be 4.1 and $4.1 < 5$; NOT sufficient.

(2) Given that $CD > 6$, it follows that $DB > 6$ since $CD = DB$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS11805

270. The people in a line waiting to buy tickets to a show are standing one behind the other. Adam and Beth are among the people in the line, and Beth is standing behind Adam with a number of people between them. If the number of people in front of Adam plus the number of people behind Beth is 18, how many people in the line are behind Beth?

(1) There are a total of 32 people in the line.

(2) 23 people in the line are behind Adam.

Arithmetic Order

Beth is standing in line behind Adam with a number of people between them. Let x be the number of people ahead of Adam, let y be the number of people between Adam and Beth, and let z be the number of people behind Beth. It is given that $x + z = 18$. Determine z.

(1) This indicates that there are 32 people in the line. Two of these people are Adam and Beth, so there are 30 other people in line besides Adam and Beth. Therefore, $x + y + z = 30$ and $x + z = 18$. From this, $y = 12$, but z cannot be determined uniquely. For example, if $x = 5$, then $z = 13$, but if $x = 10$, then $z = 8$; NOT sufficient.

(2) This indicates that $y + 1 + z = 23$ because the people behind Adam consist of the people between Adam and Beth, Beth herself, and the people behind Beth. If, for example, $y = 4$, then $z = 18$, but if $y = 9$, then $z = 13$; NOT sufficient.

Taking (1) and (2) together, $y = 12$ from (1) and $y + z = 22$ from (2). Therefore, $z = 10$, and there are 10 people in line behind Beth.

The correct answer is C; both statements together are sufficient.

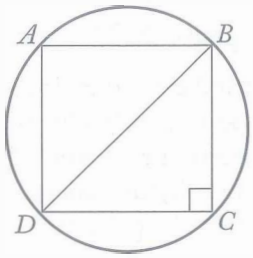
DS08730

271. Square *ABCD* is inscribed in circle *O*. What is the area of square region *ABCD*?

(1) The area of circular region *O* is 64π.

(2) The circumference of circle *O* is 16π.

Geometry Circles; Area



Since the area of square *ABCD* is the square of its side length, to solve this problem it is necessary to determine the side length of *ABCD*. From the figure, a diameter of the circle coincides with a diagonal of the square. If the length of a diameter of the circle can be determined, then the length of a diagonal of the square can be determined, from which the side length of the square can be determined using the Pythagorean theorem.

(1) This indicates that the area of the circle is 64π, so letting r represent the radius of the circle, $\pi r^2 = 64\pi$. Thus, $r^2 = 64$, $r = 8$, and $d = 2(8) = 16$, where d represents the length of the diameter of the circle and also the diagonal of the square; SUFFICIENT.

(2) This indicates that the circumference of the circle is 16π, so letting d represent the length of the diameter of the circle as well as the diagonal of the square, $\pi d = 16\pi$. Thus, $d = 16$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

DS12533

272. Lines *k* and *m* are parallel to each other. Is the slope of line *k* positive?

(1) Line *k* passes through the point (3,2).

(2) Line *m* passes through the point (–3,2).

Geometry Coordinate geometry

(1) This indicates that line *k* passes through the point (3,2). One point is not enough to determine whether the slope of line *k* is positive. If, for example, line *k* also passes through the point (0,0), then the slope of line *k* is $\frac{2-0}{3-0} = \frac{2}{3}$, which is positive. But if line *k* also passes through the point (5,0),

then the slope of line k is $\frac{2-0}{3-5}=\frac{2}{-2}$, which is negative; NOT sufficient.

(2) This indicates that line m passes through the point (–3,2). One point is not enough to determine whether the slope of line m is positive. If, for example, line m also passes through the point (0,0), then the slope of line m is $\frac{2-0}{-3-0}=\frac{2}{-3}$, which is negative. In this case, the slope of line k is also negative since parallel lines k and m have the same slope. But if line m also passes through the point (0,5), then the slope of line m is $\frac{2-5}{-3-0}=\frac{-3}{-3}$, which is positive. In this case, the slope of line k is also positive since parallel lines k and m have the same slope; NOT sufficient.

Taking (1) and (2) together gives no more information than (1) or (2) alone and so whether the slope of line k is positive cannot be determined.

The correct answer is E;
both statements together are still not sufficient.

DS19520

273. In cross section, a tunnel that carries one lane of one-way traffic is a semicircle with radius 4.2 m. Is the tunnel large enough to accommodate the truck that is approaching the entrance to the tunnel?

(1) The maximum width of the truck is 2.4 m.

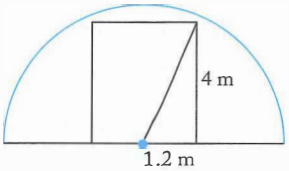
(2) The maximum height of the truck is 4 m.

Geometry Circles; Pythagorean theorem

(1) Given that the maximum width of the truck is 2.4 m, it is possible that the tunnel is large enough (for example, if the maximum height of the truck is sufficiently small) and it is possible that the tunnel is not large enough (for example, if the maximum height of the truck is sufficiently large); NOT sufficient.

(2) Given that the maximum height of the truck is 4 m, it is possible that the tunnel is large enough (for example, if the maximum width is sufficiently small) and it is possible that the tunnel is not large enough (for example, if the maximum width of the truck is sufficiently large); NOT sufficient.

Taking (1) and (2) together, it is possible for the truck to pass through the tunnel, because even if the truck were rectangular with width 2.4 m and height 4 m, the truck could pass through the tunnel if it were centered in the tunnel as shown in the diagram. In this position, it follows from the Pythagorean theorem that every point on the truck is located at most a distance of $\sqrt{(1.2)^2+4^2}$ meters from the center axis of the tunnel, and this distance is less than 4.2 meters, the radius of the tunnel, since $(1.2)^2 + 4^2 = 17.44$ is less than $(4.2)^2 = 17.64$.



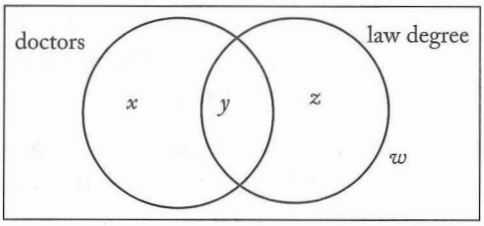
The correct answer is C;
both statements together are sufficient.

DS13122

274. In a certain group of 50 people, how many are doctors who have a law degree?

(1) In the group, 36 people are doctors.

(2) In the group, 18 people have a law degree.



Arithmetic Sets

Using the labels in the Venn diagram, $w + x + y + z = 50$. Determine the value of y.

(1) This indicates that $x + y = 36$ but does not give a unique value for y. For example, if $x = 18$, then $y = 18$; but if $x = 10$, then $y = 26$; NOT sufficient.

(2) This indicates that $y + z = 18$ but does not give a unique value for y. For example, if $z = 8$, then $y = 10$; but if $z = 10$, then $y = 8$; NOT sufficient.

Taking (1) and (2) together, if $w = 6$, $x = 26$, $y = 10$, and $z = 8$, then $w + x + y + z = 50$, $x + y = 36$, and $y + z = 18$. In this case, $y = 10$. However, if $w = 2$, $x = 30$, $y = 6$, and $z = 12$, then $w + x + y + z = 50$, $x + y = 36$, and $y + z = 18$. In this case, $y = 6$.

Alternatively, this problem can be solved by means of a contingency table. Letting n represent the number of doctors with a law degree, the table with the given information is shown below.

	doctor	not a doctor	total
law degree	n		
no law degree			
total			50

(1) This indicates that there are 36 doctors, so the number of people who are not doctors is $50 - 36 = 14$. The table with this information is shown below.

	doctor	not a doctor	total
law degree	n		
no law degree			
total	36	14	50

It is obvious from the table that there is not enough information to determine the value of n, the number of doctors with a law degree; NOT sufficient.

(2) This indicates that there are 18 people with a law degree, so the number of people without a law degree is $50 - 18 = 32$. The table with this information is shown below.

	doctor	not a doctor	total
law degree	n		18
no law degree			32
total			50

It is obvious from the table that there is not enough information to determine the value of n, the number of doctors with a law degree; NOT sufficient.

The table with the information from both (1) and (2) is shown below.

	doctor	not a doctor	total
law degree	n		18
no law degree			32
total	36	14	50

It is obvious from the table that there is still not enough information to determine the value of n, the number of doctors with a law degree.

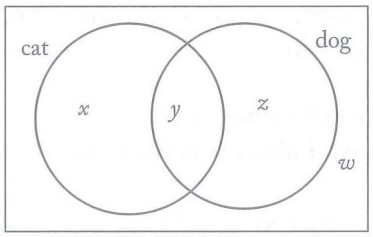
The correct answer is E;
both statements together are still not sufficient.

DS01544

275. Of a group of 50 households, how many have at least one cat or at least one dog, but not both?

(1) The number of households that have at least one cat and at least one dog is 4.

(2) The number of households that have no cats and no dogs is 14.



Arithmetic Sets

Using the labels on the Venn diagram, $w + x + y + z = 50$. Determine the value of $x + z$.

(1) This indicates that $y = 4$, so $w + x + z = 46$, but the value of $x + z$ cannot be uniquely determined; NOT sufficient.

(2) This indicates that $w = 14$, so $x + y + z = 36$, but the value of $x + z$ cannot be uniquely determined; NOT sufficient.

Taking (1) and (2) together, $14 + x + 4 + z = 50$, so $x + z = 50 - 14 - 4 = 32$.

The correct answer is C;
both statements together are sufficient.

DS02441

276. Robin invested a total of $12,000 in two investments, X and Y, so that the investments earned the same amount of simple annual interest. How many dollars did Robin invest in investment Y ?

(1) Investment X paid 3 percent simple annual interest, and investment Y paid 6 percent simple annual interest.

(2) Robin invested more than $1,000 in investment X.

Algebra Applied problems

For investments X and Y, let r_X and r_Y be the annual percentage interest rates, respectively, and let x and y be the investment amounts, respectively and in dollars. Then $x + y = 12{,}000$ and $r_X \cdot x = r_Y \cdot y$. What is the value of y?

(1) Given that $r_X = 3$ and $r_Y = 6$, then from $r_X \cdot x = r_Y \cdot y$ it follows that $3x = 6y$, or $x = 2y$. Therefore, $x + y = 12{,}000$ becomes $2y + y = 12{,}000$, or $3y = 12{,}000$, or $y = 4{,}000$; SUFFICIENT.

(2) Given that $x > 1{,}000$, it is not possible to determine the value of y. For example, $y = 4{,}000$ is possible (choose $r_X = 3$, $r_Y = 6$, and $x = 8{,}000$) and $y = 6{,}000$ is possible (choose $r_X = 3$, $r_Y = 3$, and $x = 6{,}000$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS03999

277. In a real estate office that employs n salespeople, f of them are females and x of the females are new employees. What is the value of n?

(1) If an employee were randomly selected from the n employees, the probability of selecting a female would be $\frac{2}{3}$.

(2) If an employee were randomly selected from the f female employees, the probability of selecting a new employee would be $\frac{1}{2}$.

Arithmetic Probability

(1) Given that $\frac{f}{n} = \frac{2}{3}$, it is possible that $n = 3$ (choose $f = 2$ and $x = 1$) and it is possible that $n = 6$ (choose $f = 4$ and $x = 2$); NOT sufficient.

(2) Given that $\frac{x}{f} = \frac{1}{2}$, the value of n cannot be determined because $\frac{x}{f} = \frac{1}{2}$ is true for each of the two choices of values of n, f, and x that were used in (1); NOT sufficient.

Taking (1) and (2) together is still not sufficient because both (1) and (2) hold for each of the two choices of values of n, f, and x that were used in (1).

The correct answer is E;
both statements together are still not sufficient.

DS09315

278. Is $\frac{x+1}{y+1} > \frac{x}{y}$?

(1) $0 < x < y$

(2) $xy > 0$

Algebra Inequalities

(1) Given that $0 < x < y$, the following steps show how to obtain $\frac{x+1}{y+1} > \frac{x}{y}$.

y	$>$	x	given
$xy + y$	$>$	$xy + x$	add xy to both sides
$y(x+1)$	$>$	$x(y+1)$	factor
$\frac{x+1}{y+1}$	$>$	$\frac{x}{y}$	divide both sides by $y(y+1)$

In the last step, the direction of the inequality is not changed because both y and $y + 1$ are positive, and hence the product $y(y + 1)$ is positive. These steps can be discovered by performing standard algebraic manipulations that transform $\frac{x+1}{y+1} > \frac{x}{y}$ into $y > x$, and then verifying that it is mathematically valid to reverse the steps; SUFFICIENT.

(2) Given that $xy > 0$, it is possible that $\frac{x+1}{y+1} > \frac{x}{y}$ can be true (choosing $x = 1$ and $y = 2$, the inequality becomes $\frac{2}{3} > \frac{1}{2}$) and it is possible that $\frac{x+1}{y+1} > \frac{x}{y}$ can be false (choosing $x = -1$ and $y = -2$, the inequality becomes $\frac{0}{-1} > \frac{-1}{-2}$ or $0 > \frac{1}{2}$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS01216

279. Do at least 60 percent of the students in Pat's class walk to school?

(1) At least 60 percent of the female students in Pat's class walk to school.

(2) The number of students in Pat's class who walk to school is twice the number of students who do not walk to school.

Arithmetic Percents

This problem can be solved by determining the total number of students in Pat's class and the number who walk to school.

(1) This indicates that at least 60% of the female students in Pat's class walk to school. However, it does not give any information about the other students in Pat's class; NOT sufficient.

(2) Letting x represent the number of students in Pat's class who do not walk to school, this indicates that $2x$ students in Pat's class walk to school and that the total number of students in Pat's class is $x + 2x = 3x$. From this, the percent of students in Pat's class who walk to school is $\left[\frac{2x}{3x}100\right]\% = \left[\frac{2}{3}(100)\right]\%$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS03628

280. A certain plumber charges \$92 for each job completed in 4 hours or less and \$23 per hour for each job completed in more than 4 hours. If it took the plumber a total of 7 hours to complete two separate jobs, what was the total amount charged by the plumber for the two jobs?

(1) The plumber charged \$92 for one of the two jobs.

(2) The plumber charged \$138 for one of the two jobs.

Arithmetic Applied problems

Find the total amount charged for two jobs lasting a total of 7 hours if the plumber charges \$92 for a job lasting 4 hours or less and \$23 per hour for a job lasting more than 4 hours.

(1) This indicates that one of the two jobs lasted 4 hours or less. If that job lasted 2 hours, then the other job lasted 5 hours, and the charge for that job would be (\$23)(5) = \$115, making the total for the two jobs \$92 + \$115 = \$207. However, if the jobs lasted 1 hour and 6 hours, respectively, then the total charge for the two jobs would be \$92 + (6)(\$23) = \$92 + \$138 = \$230; NOT sufficient.

(2) This indicates that one job lasted $\frac{\$138}{\$23} = 6$ hours. Therefore, the other job lasted for 1 hour and the charge for that job was \$92, making the total charge for the two jobs \$92 + \$138 = \$230; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS02585

281. If x and y are positive numbers, is $\frac{x+1}{y+1} > \frac{x}{y}$?

(1) $x > 1$

(2) $x < y$

Algebra Inequalities

Since y is positive, multiplying both sides of the inequality $\frac{x+1}{y+1} > \frac{x}{y}$ by $y(y+1)$ gives $y(x+1) > x(y+1)$ or $xy + y > xy + x$, which is equivalent to $y > x$. So, determining whether the inequality $\frac{x+1}{y+1} > \frac{x}{y}$ is true is equivalent to determining whether the inequality $y > x$ is true.

(1) This indicates that $x > 1$. If, for example, $x = 3$ and $y = 4$, then $y > x$ is true. However, if $x = 4$ and $y = 3$, then $y > x$ is not true; NOT sufficient.

(2) This indicates that $x < y$, so $y > x$ is true; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS01619

282. If a and b are positive integers, is $\frac{a}{b} < \frac{9}{11}$?

(1) $\frac{a}{b} < 0.818$

(2) $\frac{b}{a} > 1.223$.

Arithmetic Inequalities

If $11a < 9b$ and b is positive, then $\frac{11a}{b} < 9$ and $\frac{a}{b} < \frac{9}{11}$. Therefore, this problem can be solved by determining if $11a < 9b$.

(1) From this,

$\frac{a}{b} < 0.818$ given

$a < 0.818b$ multiply both sides by b, which is positive

$11a < 8.998b$ multiply both sides by 11

Then, since $8.998b < 9b$, it follows that $11a < 9b$; SUFFICIENT.

(2) From this,

$\frac{b}{a} > 1.223$ given

$b > 1.223a$ multiply both sides by a, which is positive

$9b > 11.007a$ multiply both sides by 9

Then, since $11.007a > 11a$, it follows that $9b > 11a$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS04536

283. Every object in a box is either a sphere or a cube, and every object in the box is either red or green. How many objects are in the box?

(1) There are six cubes and five green objects in the box.

(2) There are two red spheres in the box.

Arithmetic Sets

This problem can be solved using a contingency table set up as shown below, where T represents the number to be determined.

	cube	sphere	total
red			
green			
total			T

(1) The following table displays the information that there are 6 cubes and 5 green objects:

	cube	sphere	total
red			
green			5
total	6		T

It is obvious that there is not enough information to determine a unique value for T; NOT sufficient.

(2) The following table displays the information that there are 2 red spheres:

	cube	sphere	total
red		2	
green			
total			T

It is obvious that there is not enough information to determine a unique value for T; NOT sufficient.

The following table displays the information from (1) and (2) taken together:

	cube	sphere	total
red		2	
green			5
total	6		T

It is obvious that there is still not enough information to determine a unique value for T.

The correct answer is E;
both statements together are still not sufficient.

DS01425

284. If x and y are positive integers, is xy even?

(1) $x^2 + y^2 - 1$ is divisible by 4.

(2) $x + y$ is odd.

Arithmetic Properties of numbers

Determine whether the product of two positive integers, x and y, is even.

(1) This indicates that $x^2 + y^2 - 1$ is divisible by 4, so $x^2 + y^2 - 1 = 4q$ for some integer q. Then $x^2 + y^2 = 4q + 1$, which means $x^2 + y^2$ is odd. Both x^2 and y^2 cannot be even because, in that case, their sum would be even, and both cannot be odd because, in that case, their sum would also be even. Therefore, one of x^2 and y^2 is even, and the

other is odd. It follows that one of x or y is even and the other is odd, so xy is even; SUFFICIENT.

(2) If $x + y$ is odd, then one of x and y is even and the other is odd because if both were even or both were odd, the sum would be even. It follows that xy is even; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS14502

285. If a and b are integers, is $a + b + 3$ an odd integer?

(1) ab is an odd integer.

(2) $a - b$ is an even integer.

Arithmetic Properties of numbers

Determine whether $a + b + 3$ is odd for integers a and b.

(1) This indicates that, since ab is odd, both a and b are odd because if one of a or b is even or both a and b are even, then ab is even. Therefore, $a + b$ is even and $a + b + 3$ is odd; SUFFICIENT.

(2) This indicates that, since $a - b$ is even, both a and b are even or both are odd because if one of them is even and the other is odd, then $a - b$ is odd. Therefore, $a + b$ is even, and $a + b + 3$ is odd; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS08308

286. If x and y are positive integers, what is the value of $\sqrt{x} + \sqrt{y}$?

(1) $x + y = 15$

(2) $\sqrt{xy} = 6$

Algebra Operations with radicals

(1) From this, if $x = 1$ and $y = 14$, then $x + y = 15$ and $\sqrt{x} + \sqrt{y} = \sqrt{1} + \sqrt{14} = 1 + \sqrt{14}$. But $\sqrt{14} < \sqrt{16} = 4$, so $1 + \sqrt{14} < 1 + 4 = 5$. However, if $x = 9$ and $y = 6$, then $x + y = 15$ and $\sqrt{x} + \sqrt{y} = \sqrt{9} + \sqrt{6} = 3 + \sqrt{6}$. But $\sqrt{6} > \sqrt{4} = 2$ so, $3 + \sqrt{6} > 3 + 2 = 5$; NOT sufficient.

(2) From this, if $x = 1$ and $y = 36$, then $\sqrt{xy} = 6$ and $\sqrt{x} + \sqrt{y} = \sqrt{1} + \sqrt{36} = 1 + 6 = 7$. However, if $x = 4$ and $y = 9$, then $\sqrt{xy} = 6$ and $\sqrt{x} + \sqrt{y} = \sqrt{4} + \sqrt{9} = 2 + 3 = 5$; NOT sufficient.

Taking (1) and (2) together, $x + y = 15$ and $\sqrt{xy} = 6$, so

$$\begin{aligned} \sqrt{x(15-x)} &= 6 && \text{substitute } 15 - x \text{ for } y \\ x(15-x) &= 36 && \text{square both sides} \\ 15x - x^2 &= 36 && \text{use distributive property} \\ x^2 - 15x + 36 &= 0 && \text{collect like terms} \\ (x-12)(x-3) &= 0 && \text{factor} \end{aligned}$$

Thus, $x = 12$ or $x = 3$. If $x = 12$, then $y = 3$ and $\sqrt{x} + \sqrt{y} = \sqrt{12} + \sqrt{3}$. If $x = 3$, then $y = 12$ and $\sqrt{x} + \sqrt{y} = \sqrt{3} + \sqrt{12}$.

Algebraically, if the value of $\left(\sqrt{x} + \sqrt{y}\right)^2 = \left(\sqrt{x}\right)^2 + 2\sqrt{xy} + \left(\sqrt{y}\right)^2 = x + y + 2\sqrt{xy}$ can be determined, then the value of $\sqrt{x} + \sqrt{y}$ can be determined, and conversely.

(1) From this, $\left(\sqrt{x} + \sqrt{y}\right)^2 = x + y + 2\sqrt{xy} = 15 + 2\sqrt{x(15-x)}$, but the value of x is unknown, and the value of $\left(\sqrt{x} + \sqrt{y}\right)^2$ cannot be determined; NOT sufficient.

(2) From this, $\left(\sqrt{x} + \sqrt{y}\right)^2 = x + y + 2\sqrt{xy} = x + \frac{36}{x} + 2(6)$, but the value of x is unknown, and the value of $\left(\sqrt{x} + \sqrt{y}\right)^2$ cannot be determined; NOT sufficient.

Taking (1) and (2) together, $\left(\sqrt{x} + \sqrt{y}\right)^2 = x + y + 2\sqrt{xy} = 15 + (2)(6) = 27$ and so $\sqrt{x} + \sqrt{y} = 27 = 3\sqrt{3}$. Note that in the nonalgebraic solution $\sqrt{x} + \sqrt{y} = \sqrt{3} + \sqrt{12}$ and $\sqrt{3} + \sqrt{12} = \sqrt{3} + 2\sqrt{3} = 3\sqrt{3}$.

The correct answer is C;
both statements together are sufficient.

DS05312

287. A certain truck uses $\frac{1}{12} + kv^2$ gallons of fuel per mile when its speed is v miles per hour, where k is a constant. At what speed should the truck travel so that it uses $\frac{5}{12}$ gallon of fuel per mile?

(1) The value of k is $\frac{1}{10,800}$.

(2) When the truck travels at 30 miles per hour, it uses $\frac{1}{6}$ gallon of fuel per mile.

Algebra Applied problems

This problem can be solved by determining the positive value of v so that $\frac{1}{12} + kv^2 = \frac{5}{12}$.

(1) This indicates the value of k and so the value of v can be determined by solving $\frac{1}{12} + \frac{1}{10,800}v^2 = \frac{5}{12}$ for v; SUFFICIENT.

(2) This indicates that $\frac{1}{12} + k(30)^2 = \frac{1}{6}$. The value of k can be determined by solving this equation for k. Then that value of k can be substituted into $\frac{1}{12} + kv^2 = \frac{5}{12}$, which can then be solved for the value of v; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS08420

288. On June 1, Mary paid Omar $360 for rent and utilities for the month of June. Mary moved out early, and Omar refunded the money she paid for utilities, but not for rent, for the days in June after she moved out. How many dollars did Omar refund to Mary?

(1) Mary moved out on June 24.

(2) The amount Mary paid for utilities was less than $\frac{1}{5}$ the amount Mary paid for rent.

Algebra Applied problems; Proportions

Let R and U be the amounts, respectively and in dollars, that Mary paid for rent and utilities for the month of June. Given that $R + U = 360$, find the value of $\left(\frac{n}{30}\right)U$, where n is the number of days in June—which has 30 days—after Mary moved out.

(1) Given that $n = 6$, it follows that $\left(\frac{n}{30}\right)U = \frac{1}{5}U$. However, it is clear that more than one value of U is possible, and thus the value of $\left(\frac{n}{30}\right)U$ cannot be determined; NOT sufficient.

(2) Given that $U < \frac{1}{5}R$, it is clear that more than one value of U is possible, and thus the value of $\left(\frac{n}{30}\right)U$ cannot be determined; NOT sufficient.

Taking (1) and (2) together, the problem reduces to whether the value of $\frac{1}{5}U$, or equivalently the value of U, can be determined when $R + U = 360$ and $U < \frac{1}{5}R$. However, it is still the case that the value of U cannot be determined. For example, if $U = \frac{1}{10}(360) = 36$ and $R = \frac{9}{10}(360) = 324$, then both $R + U = 360$ and $U < \frac{1}{5}R$ are true. On the other hand, if $U = \frac{1}{36}(360) = 10$ and $R = \frac{35}{36}(360) = 350$, then both $R + U = 360$ and $U < \frac{1}{5}R$ are true.

The correct answer is E;
both statements together are still not sufficient.

DS04057

289. If $x = 2t$ and $y = \frac{t}{3}$, what is the value of $x^2 - y^2$?

(1) $t^2 - 3 = 6$

(2) $t^3 = -27$

Algebra Simplifying algebraic expressions

This problem can be solved by determining the value of $x^2 - y^2 = (2t)^2 - \left(\frac{t}{3}\right)^2 = 4t^2 - \frac{t^2}{9}$, and this can be determined if the value of t^2 or the value of t can be determined.

(1) This indicates that $t^2 - 3 = 6$ and so $t^2 = 6 + 3 = 9$; SUFFICIENT.

(2) This indicates that $t^3 = -27$ and so $t = -3$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS02939

290. The 10 students in a history class recently took an examination. What was the maximum score on the examination?

(1) The mean of the scores was 75.

(2) The standard deviation of the scores was 5.

Arithmetic Statistics

(1) Given that the mean of the scores was 75, then all 10 scores could have been 75 and the maximum score would be 75. However, it is also possible that the scores were 74, 76, and 8 scores of 75, and the maximum would be 76; NOT sufficient.

(2) Given that the standard deviation was 5, it will be helpful to use a specific value for the mean, such as 75. From the definition of standard deviation, it follows that $\sqrt{\frac{\sum(S-75)^2}{10}} = 5$, where $\sum(S-75)^2$ denotes the sum of the squares of the differences of the scores from 75. It will be useful to solve for $\sum(S-75)^2$. Doing this gives $\frac{\sum(S-75)^2}{10} = 25$, and so the condition $\sum(S-75)^2 = 250$ will ensure that the standard deviation of the scores is 5. One list of 10 scores with standard deviation 5 and mean 75 is five occurrences of 70 and five occurrences of 80, since for these scores we have $\sum(S-75)^2 = 5(70-75)^2 + 5(80-75)^2$, which equals $5(25) + 5(25) = 250$; and the mean is 75 because the sum of the differences of the scores from 75 is equal to 0. Another list of 10 scores with standard deviation 5 and mean 75 is 70, 70, 70, 70, 75, 75, 75, 80, 80, and 85, since for these scores we have $\sum(S-75)^2 = 4(70-75)^2 + 3(75-75)^2 + 2(80-75)^2 + (85-75)^2$, which equals $4(25) + 2(25) + 100 = 250$; and the mean is 75 because the sum of the differences of the scores from 75 is equal to 0. The maximum for the first list is 80, and the maximum for the second list is 85; NOT sufficient.

Taking (1) and (2) together is still not sufficient because both (1) and (2) hold for the lists used in (2).

The correct answer is E;
both statements together are still not sufficient.

DS01341

291. Last school year, each of the 200 students at a certain high school attended the school for the entire year. If there were 8 cultural performances at the school during the last school year, what was the average (arithmetic mean) number of students attending each cultural performance?

(1) Last school year, each student attended at least one cultural performance.

(2) Last school year, the average number of cultural performances attended per student was 4.

Arithmetic Statistics

Determine the average number of students attending each of 8 cultural performances at a school that has 200 students.

(1) This indicates that each student attended at least one cultural performance, but the average number of students attending each performance cannot be uniquely determined. It could be as low as $\frac{200}{8} = 25$ students if each student attended exactly one performance, or it could be as high as $\frac{200}{1} = 200$ if each student attended every performance; NOT sufficient.

(2) This indicates that, since the average number of performances attended per student was 4, $\frac{\text{total attendance at performances}}{\text{total number of students}} = 4$, so the total attendance at the performances is $(4)(200) = 800$ students. It follows that the average number of students attending each performance is $\frac{800}{8} = 100$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS14569

292. A clothing manufacturer makes jackets that are wool or cotton or a combination of wool and cotton. The manufacturer has 3,000 pounds of wool and 2,000 pounds of cotton on hand. Is this enough wool and cotton to make at least 1,000 jackets?

(1) Each wool jacket requires 4 pounds of wool, and no cotton.

(2) Each cotton jacket requires 6 pounds of cotton, and no wool.

Arithmetic Applied problems

(1) Given that each wool jacket requires 4 pounds of wool and no cotton, then at most a total of 750 wool jackets can be made (because $\frac{3,000}{4} = 750$), and possibly no other jackets. Therefore, it is possible that there is enough wool and cotton to make at least 1,000 jackets (for example, if there is enough cotton to make 250 cotton jackets) and it is possible that there is not enough wool and cotton to make at least 1,000 jackets (for example, if there is less than the amount of cotton needed to make 250 cotton jackets); NOT sufficient.

(2) Given that each cotton jacket requires 6 pounds of cotton and no wool, then at most a total of 333 cotton jackets can be made (because rounding $\frac{2,000}{6}$ down to the nearest integer gives 333), and possibly no other jackets. Therefore, it is possible that there is enough wool and cotton to make at least 1,000 jackets (for example, if there is enough wool to make 667 wool jackets), and it is possible that there is not enough wool and cotton to make at least 1,000 jackets (for example, if there is less than the amount of wool needed to make 667 wool jackets); NOT sufficient.

Taking (1) and (2) together, there is enough wool and cotton to make 750 wool jackets and 333 cotton jackets for a total of 1,083 jackets.

The correct answer is C;
both statements together are sufficient.

DS05377

293. If n is an integer, what is the greatest common divisor of 12 and n?

(1) The product of 12 and n is 432.

(2) The greatest common divisor of 24 and n is 12.

Arithmetic Properties of integers

(1) Given that $12n = 432$, it follows that $n = 36$ and the greatest common divisor of 12 and 36 is 12. However, it is not necessary to **solve** this equation and then **calculate** the greatest common divisor of two numbers to determine whether it is possible to answer the question asked. It is sufficient to observe that the value of n can be determined by solving this equation and then observe that the greatest common divisor of 12 and the value of n can be determined; SUFFICIENT.

(2) Given that 12 is the greatest common divisor of 24 and n, it follows that 12 is a common divisor of 24 and n, and hence 12 is a divisor of n. Thus, 12 is a common divisor of 12 and n because 12 is a divisor of 12 and 12 is a divisor of n. Also, no number greater than 12 can be a common divisor of 12 and n because no number greater than 12 can be a divisor of 12. Therefore, 12 is the greatest common divisor of 12 and n; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS11287

294. Each month, Jim receives a base salary plus a 10 percent commission on the price of each car he sells that month. If Jim sold 15 cars last month, what was the total amount of base salary and commissions that Jim received that month?

(1) Last month, Jim's base salary was $3,000.

(2) Last month, Jim sold 3 cars whose prices totaled $60,000 and 5 cars whose prices totaled $120,000.

Arithmetic Applied problems

Determine the total of Jim's base salary plus a 10% commission on the price of each of the 15 cars he sold last month.

(1) This indicates that Jim's base salary was $3,000 but gives no information about the prices of the 15 cars he sold last month; NOT sufficient.

(2) This indicates that Jim's commission on 8 cars that he sold last month can be determined but gives no information about his base salary or about the prices of the other 7 cars he sold last month; NOT sufficient.

Taking (1) and (2) together gives Jim's base salary and information about the prices of 8 of the 15 cars he sold last month, but does not provide

information about the selling prices of the other 7 cars he sold.

The correct answer is E;
both statements together are still not sufficient.

DS17615

295. If x is a positive integer greater than 1, what is the value of x?

(1) $2x$ is a common factor of 18 and 24.

(2) x is a factor of 6.

Arithmetic Properties of numbers

(1) Given that $2x$, where x is an integer greater than 1, is a common factor of 18 and 24, then $2x$ is an even integer greater than or equal to 4 that is a factor of 18 and 24. The factors of 18 are 1, 2, 3, 6, 9, and 18. The factors of 24 are 1, 2, 3, 4, 6, 8, 12, and 24. Since the only even integer greater than or equal to 4 that belongs to both lists is 6, it follows that $2x = 6$, or $x = 3$; SUFFICIENT.

(2) Given that x, an integer greater than 1, is a factor of 6, then x could be 2, 3, or 6; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS13408

296. By what percent was the price of a certain television set discounted for a sale?

(1) The price of the television set before it was discounted for the sale was 25 percent greater than the discounted price.

(2) The price of the television set was discounted by \$60 for the sale.

Arithmetic Percents

Let B represent the price, in dollars, for the television before the sale and let D represent the discounted price, in dollars. The percent by which B was discounted to obtain D can be determined if the value of $\frac{B-D}{B}$ can be determined.

(1) This indicates that B is 25% greater than D or that $B = 1.25D$. Then, $\frac{B-D}{B} = \frac{1.25D-D}{1.25D} = \frac{0.25}{1.25} = 0.2$ or 20%; SUFFICIENT.

(2) This indicates that $D = B - 60$, but B is unknown. For example, B could be 100 and the value of $\frac{B-D}{B}$ would be $\frac{60}{100}$, or B could be 500 and the value of $\frac{B-D}{B}$ would be $\frac{60}{500}$; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS05049

297. Jack wants to use a circular rug on his rectangular office floor to cover two small circular stains, each less than $\frac{\pi}{100}$ square feet in area and each more than 3 feet from the nearest wall. Can the rug be placed to cover both stains?

(1) Jack's rug covers an area of 9π square feet.

(2) The centers of the stains are less than 4 feet apart.

Geometry Applied problems; Circles

If a circle with radius r has area less than $\frac{\pi}{100}$, then $\pi r^2 < \frac{\pi}{100}$, or $r^2 < \frac{1}{100}$. Hence, $r < \frac{1}{10}$, so the circle's radius is less than $\frac{1}{10}$ ft.

(1) Given that the circular rug has area 9π ft^2, the radius of the rug, R ft, satisfies the equation $\pi R^2 = 9\pi$, and so $R^2 = 9$, or $R = 3$. If the centers of the circular stains are 4 ft apart, then the midpoint of the centers will be a distance of at most $\left(\frac{1}{10} + 2 + \frac{1}{10}\right)$ ft $= 2.2$ ft from any point on the stains.

Thus, if the circular rug (which has radius 3 ft) were placed so that its center lies on that midpoint, then the rug would cover both stains. However, if the centers of the circular stains are 7 ft apart, then the circular rug (which has diameter 6 ft) cannot be placed to cover both stains; NOT sufficient.

(2) Given that the centers of the circular stains are less than 4 ft apart, it is not possible to determine whether the circular rug can be placed to cover both stains since nothing is given about the size of the circular rug; NOT sufficient.

Taking (1) and (2) together, the centers of the circular stains are less than 4 ft apart and it follows from the discussion in (1) that the circular rug can be placed to cover both stains.

The correct answer is C;
both statements together are sufficient.

×	a	b	c
a	d	e	f
b	e	g	h
c	f	h	j

DS05772

298. In the multiplication table above, each letter represents an integer. What is the value of c?

(1) $c = f$

(2) $h \neq 0$

Arithmetic Properties of numbers

(1) Given that $c = f$, the examples below, in which $a = b = 1$, show that the value of c could be 1 and the value of c could be 2; NOT sufficient.

×	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

×	1	1	2
1	1	1	2
1	1	1	2
2	2	2	4

(2) Given that $h \neq 0$, the examples in (1) show that the value of c could be 1 and the value of c could be 2; NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS09379

299. If n is an integer, is $(0.1)^n$ greater than $(10)^n$?

(1) $n > -10$

(2) $n < 10$

Arithmetic Exponents

$(0.1)^n = \frac{1}{10^n}$ will be greater than 10^n if 1 is greater than 10^{2n}, and this will be the case if and only if $n < 0$. Determine if $n < 0$.

(1) This indicates that n is greater than -10. This includes nonnegative values of n as well as negative values of n; NOT sufficient.

(2) This indicates that n is less than 10, but this includes negative values of n as well as nonnegative values of n; NOT sufficient.

Taking (1) and (2) together, n can have any value between -10 and 10 and can therefore be negative or nonnegative.

The correct answer is E;
both statements together are still not sufficient.

DS19199

300. For a basic monthly fee of F yen (¥F), Naoko's first cell phone plan allowed him to use a maximum of 420 minutes on calls during the month. Then, for each of x additional minutes he used on calls, he was charged ¥M, making his total charge for the month ¥T, where $T = F + xM$. What is the value of F?

(1) Naoko used 450 minutes on calls the first month and the total charge for the month was ¥13,755.

(2) Naoko used 400 minutes on calls the second month and the total charge for the month was ¥13,125.

Algebra Simultaneous equations

(1) Given that Naoko used 450 minutes and was charged ¥13,125, it follows that $F + (450 - 420)M = 13{,}755$, or $F + 30M = 13{,}755$. However, the value of F cannot be determined. For example, $F = 13{,}005$ and $M = 25$ is possible, and $F = 13{,}035$ and $M = 24$ is possible; NOT sufficient.

(2) Given that Naoko used 400 minutes and was charged ¥13,125, it follows that

$F = 13{,}125$, since Naoko did not use more than 420 minutes; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS13949

301. Is the sum of the prices of the 3 books that Shana bought less than \$48 ?

(1) The price of the most expensive of the 3 books that Shana bought is less than \$17.

(2) The price of the least expensive of the 3 books that Shana bought is exactly \$3 less than the price of the second most expensive book.

Arithmetic Operations with integers; Order

Let B_1, B_2, and B_3 be the prices, in dollars and in numerical order, of the 3 books. Thus, $B_1 \le B_2 \le B_3$. Determine whether $B_1 + B_2 + B_3 < 48$.

(1) Given that $B_3 < 17$, it is possible that $B_1 + B_2 + B_3 < 48$ (choose $B_1 = B_2 = 15$ and $B_3 = 16$) and it is possible that $B_1 + B_2 + B_3 > 48$ (choose $B_1 = B_2 = 16$ and $B_3 = 16.5$); NOT sufficient.

(2) Given that $B_1 = B_2 - 3$, it is possible that $B_1 + B_2 + B_3 < 48$ (choose $B_1 = 3$, $B_2 = 6$, and $B_3 = 10$) and it is possible that $B_1 + B_2 + B_3 > 48$ (choose $B_1 = 3$, $B_2 = 6$, and $B_3 = 48$); NOT sufficient.

Taking (1) and (2) together, from (1) and the information given it follows that $B_3 < 17$ and $B_2 \le B_3 < 17$, or $B_2 < 17$. Also, from $B_2 < 17$ and (2), it follows that $B_1 < 17 - 3$, or $B_1 < 14$. Therefore, $B_1 + B_2 + B_3$ is the sum of 3 numbers—a number less than 14, a number less than 17, and a number less than 17—and thus $B_1 + B_2 + B_3 < 14 + 17 + 17 = 48$.

The correct answer is C;
both statements together are sufficient.

DS12943

302. If r and t are three-digit positive integers, is r greater than t ?

(1) The tens digit of r is greater than each of the three digits of t.

(2) The tens digit of r is less than either of the other two digits of r.

Arithmetic Properties of numbers

Let r_3, r_2, and r_1 be the hundreds, tens, and units digits, respectively, of the 3-digit integer r, and let t_3, t_2, and t_1 be the hundreds, tens, and units digits, respectively, of the 3-digit integer t.

(1) Given that r_2 is greater than each of t_3, t_2, and t_1, then r could be greater than t (for example, if $r = 242$ and $t = 222$) and r could be less than t (for example, if $r = 242$ and $t = 333$); NOT sufficient.

(2) Given that r_2 is less than either of r_1 or r_3, then r could be greater than t (for example, if $r = 212$ and $t = 111$) and r could be less than t (for example, if $r = 212$ and $t = 222$); NOT sufficient.

Taking (1) and (2) together, $r_3 > r_2$ follows from (2) and $r_2 > t_3$ follows from (1). Therefore, $r_3 > t_3$ and hence r is greater than t.

The correct answer is C;
both statements together are sufficient.

DS14788

303. Is the product of two positive integers x and y divisible by the sum of x and y ?

(1) $x = y$

(2) $x = 2$

Arithmetic Properties of numbers

(1) Given that $x = y$, then xy could be divisible by $x + y$ (for example, if $x = y = 2$, then $xy = 4$ is divisible by $x + y = 4$) and xy could fail to be divisible by $x + y$ (for example, if $x = y = 3$, then $xy = 9$ is not divisible by $x + y = 6$); NOT sufficient.

(2) Given that $x = 2$, then xy could be divisible by $x + y$ (for example, if $x = y = 2$, then $xy = 4$ is divisible by $x + y = 4$) and xy could fail to be divisible by $x + y$ (for example, if $x = 2$ and $y = 3$, then $xy = 6$ is not divisible by $x + y = 5$); NOT sufficient.

Taking (1) and (2) together, it follows that $x = y = 2$, and so $xy = 4$ is divisible by $x + y = 4$.

The correct answer is C;
both statements together are sufficient.

DS05330

304. A company makes and sells two products, P and Q. The costs per unit of making and selling P and Q are \$8.00 and \$9.50, respectively, and the selling prices per unit of P and Q are \$10.00 and \$13.00, respectively. In one month the company sold a total of 834 units of these products. Was the total profit on these items more than \$2,000.00 ?

(1) During the month, more units of P than units of Q were sold.

(2) During the month, at least 100 units of Q were sold.

Arithmetic Applied problems

It is given that the profit on each unit of P is \$10.00 – \$8.00 = \$2.00, and the profit on each unit of Q is \$13.00 – \$9.50 = \$3.50. Also if p and q represent the number of units of Products P and Q made and sold, then it is given that $p + q = 834$. Determining if the total profit on these items was more than \$2,000 requires determining how many units of at least one of the products were sold.

(1) This indicates that $p > q$, but does not give a specific value for p or q; NOT sufficient.

(2) This indicates that $q \geq 100$, but does not give a specific value for q; NOT sufficient.

Taking (1) and (2) together, $p > q$ and $q \geq 100$, so $p > 100$, but a specific value for p or q cannot be determined.

The correct answer is E;
both statements together are still not sufficient.

DS03045

305. Jill has applied for a job with each of two different companies. What is the probability that she will get job offers from both companies?

(1) The probability that she will get a job offer from neither company is 0.3.

(2) The probability that she will get a job offer from exactly one of the two companies is 0.5.

Arithmetic Probability

Let $P(2)$ be the probability that she will get a job offer from both companies, $P(1)$ be the probability that she will get a job offer from exactly one of the companies, and $P(0)$ be the probability that she will get a job offer from neither company. Then $P(2) + P(1) + P(0) = 1$. What is the value of $P(2)$?

(1) Given that $P(0) = 0.3$, it follows that $P(2) + P(1) + 0.3 = 1$, or $P(2) + P(1) = 0.7$. However, the value of $P(2)$ cannot be determined, since nothing is known about the value of $P(1)$ other than $0 \leq P(1) \leq 0.7$; NOT sufficient.

(2) (2) Given that $P(1) = 0.5$, it follows that $P(2) + 0.5 + P(0) = 1$, or $P(2) + P(0) = 0.5$. However, the value of $P(2)$ cannot be determined, since nothing is known about the value of $P(0)$ other than $0 \leq P(1) \leq 0.5$; NOT sufficient.

Taking (1) and (2) together, it follows that $P(2) + 0.5 + 0.3 = 1$, or $P(2) = 0.2$.

The correct answer is C;
both statements together are sufficient.

DS01257

306. A conveyor belt moves bottles at a constant speed of 120 centimeters per second. If the conveyor belt moves a bottle from a loading dock to an unloading dock, is the distance that the conveyor belt moves the bottle less than 90 meters? (1 meter = 100 centimeters)

(1) It takes the conveyor belt less than 1.2 minutes to move the bottle from the loading dock to the unloading dock.

(2) It takes the conveyor belt more than 1.1 minutes to move the bottle from the loading dock to the unloading dock.

Arithmetic Applied problems

Since the rate at which the conveyor belt moves is given as 120 centimeters per second, which is equivalent to $\frac{120}{100} = 1.2$ meters per second, the conveyor will move less than 90 meters if it moves for less than $\frac{90}{1.2} = 75$ seconds.

(1) This indicates that the length of time the conveyor belt moves is less than 1.2 minutes, which is equivalent to $(1.2)(60) = 72$ seconds and $72 < 75$; SUFFICIENT.

(2) This indicates that the length of time the conveyor belt moves is more than 1.1 minutes, which is equivalent to $(1.1)(60) = 66$ seconds, but does not indicate

how much more than 66 seconds the conveyor moves. If the conveyor moves for 70 seconds ($70 > 66$), for example, it moves less than 90 meters since $70 < 75$. However, if the conveyor belt moves for 80 seconds ($80 > 66$), then it moves more than 90 meters since $80 > 75$; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS02706

307. If x, y, and z are positive numbers, what is the value of the average (arithmetic mean) of x and z?

(1) $x - y = y - z$

(2) $x^2 - y^2 = z$

Algebra Statistics; Simplifying algebraic expressions

(1) Given that $x - y = y - z$, it follows that $x + z = 2y$, or $y = \frac{x+z}{2}$. Therefore, y is the average of x and z. However, it is not possible to determine the value of y. For example, $x = y = z = 3$ gives three positive numbers that satisfy $x - y = y - z$, and $x = y = z = 4$ gives three positive numbers that satisfy $x - y = y - z$; NOT sufficient.

(2) Given that $x^2 - y^2 = z$, it is not possible to determine the value of $\frac{x+z}{2}$, the average of the x and z. For example, if $x = 3$, $y = 2$, and $z = 5$, then $x^2 - y^2 = z$ is true (because $3^2 - 2^2 = 9 - 4 = 5$) and $\frac{x+z}{2} = \frac{3+5}{2} = 4$. However, if $x = 5$, $y = 4$, and $z = 9$, then $x^2 - y^2 = z$ is true (because $5^2 - 4^2 = 25 - 16 = 9$) and $\frac{x+z}{2} = \frac{5+9}{2} = 7$; NOT sufficient.

Taking (1) and (2) together, it follows that $x + z = 2y$ and $x^2 - y^2 = z$, so $x + (x^2 - y^2) = 2y$, or $x + x^2 = y^2 + 2y$. Adding 1 to both sides of this last equation gives $x + x^2 + 1 = y^2 + 2y + 1 = (y + 1)^2$, which simplifies the task of finding positive numbers x, y, and z that satisfy both (1) and (2) and for which the values of $\frac{x+z}{2} = y$ can be different.

For one example, letting $x = 1$ gives $x + x^2 + 1 = 3$, and thus $(y + 1)^2 = 3$. Solving for a positive value of y gives $y + 1 = \sqrt{3}$, or $y = \sqrt{3} - 1$, and by using $x + z = 2y$, the corresponding value of z is $2y - x = 2(\sqrt{3} - 1) - 1 = 2\sqrt{3} - 3$. Note that $2\sqrt{3} - 3$ is positive because $(2\sqrt{3})^2 = 4(3) = 12$ is greater than $3^2 = 9$. The values $x = 1$, $y = \sqrt{3} - 1$, and $z = 2\sqrt{3} - 3$ satisfy both (1) and (2), and for this choice of values of x, y, and z, the value of $\frac{x+z}{2} = y$ is $\sqrt{3} - 1$. For another example, letting $x = 2$ gives $x + x^2 + 1 = 7$, and thus $(y + 1)^2 = 7$. Solving for a positive value of y gives $y + 1 = \sqrt{7}$, or $y = \sqrt{7} - 1$, and by using $x + z = 2y$, the corresponding value of z is $2y - x = 2(\sqrt{7} - 1) - 2 = 2\sqrt{7} - 4$. Note that $2\sqrt{7} - 4$ is positive because $(2\sqrt{7})^2 = 4(7) = 28$ is greater than $4^2 = 16$. The values $x = 2$, $y = \sqrt{7} - 1$, and $z = 2\sqrt{7} - 4$ satisfy both (1) and (2), and for this choice of values of x, y, and z, the value of $\frac{x+z}{2} = y$ is $\sqrt{7} - 1$; NOT sufficient.

The correct answer is E;
both statements together are still not sufficient.



DS04428

308. The rectangular rug shown in the figure above has an accent border. What is the area of the portion of the rug that excludes the border?

(1) The perimeter of the rug is 44 feet.

(2) The width of the border on all sides is 1 foot.

Geometry Rectangles; Perimeter

(1) Given that the perimeter of the rug is 44 ft, then the rectangular rug could have area 121 ft^2 (length = 11 ft and width = 11 ft) and the rectangular rug could have area 96 ft^2 (length = 16 ft and width = 6 ft). Moreover, no information is given about the width of the accent border, so even if the area of the rectangular rug could be determined, it would still not be possible to determine the area of the portion of the rug that excludes the border; NOT sufficient.

(2) Given that the width of the border is 1 ft, the area of the portion of the rug that excludes the border is $(L-2)(W-2)$ ft^2, where L ft and W ft are the length and width, respectively, of the rug. If $L = W = 11$, then $(L-2)(W-2) = 81$. However, if $L = 16$ and $W = 6$, then $(L-2)(W-2) = 56$; NOT sufficient.

Taking (1) and (2) together, the area could be 81 ft^2 or 56 ft^2 as shown by the examples in (1) and (2).

The correct answer is E;
both statements together are still not sufficient.

DS07227

309. Terry holds 12 cards, each of which is red, white, green, or blue. If a person is to select a card randomly from the cards Terry is holding, is the probability less than $\frac{1}{2}$ that the card selected will be either red or white?

(1) The probability that the person will select a blue card is $\frac{1}{3}$.

(2) The probability that the person will select a red card is $\frac{1}{6}$.

Arithmetic Probability

Determine if the probability is less than $\frac{1}{2}$ that a card selected at random from 12 cards, each of which is red, white, green, or blue, will be either red or white. That is, determine whether the number of cards that are red or white is less than $\frac{1}{2}(12) = 6$.

(1) It is given that the probability is $\frac{1}{3}$ that the selected card will be blue. This means that $\frac{1}{3}(12) = 4$ of the cards are blue. However, there is no information about how many of the 8 remaining cards are red, how many are white, or how many are green. If, for example, there are 2 red cards, 2 white cards, and 4 green cards, then the number of cards that are red or white is $2 + 2 = 4$, which is less than 6. On the other hand, if there are 3 red cards, 4 white cards, and 1 green card, then the number of cards that are red or white is $3 + 4 = 7$, which is not less than 6; NOT sufficient.

(2) It is given that the probability is $\frac{1}{6}$ that the selected card will be red. This means that $\frac{1}{6}(12) = 2$ of the cards are red. However, there is no information about how many of the 10 remaining cards are white, how many are blue, or how many are green. If, for example, there are 3 white cards, 5 blue cards, and 2 green cards, then the number of cards that are red or white is $2 + 3 = 5$, which is less than 6. On the other hand, if there are 7 white cards, 1 blue card, and 2 green cards, then the number of cards that are red or white is $2 + 7 = 9$, which is not less than 6; NOT sufficient.

Taking (1) and (2) together, there are 4 blue cards and 2 red cards. However, there is no information about how many of the 6 remaining cards are white or how many are green. If, for example, there are 3 white cards and 3 green cards, then the number of cards that are red or white is $2 + 3 = 5$, which is less than 6. On the other hand, if there are 5 white cards and 1 green card, then the number of cards that are red or white is $2 + 5 = 7$, which is not less than 6.

The correct answer is E;
both statements together are still not sufficient.

DS06537

310. If $y \neq 2xz$, what is the value of $\frac{2xz + yz}{2xz - y}$?

(1) $2x + y = 3$

(2) $z = 2$

Algebra Simplifying algebraic expressions

The task is to determine the value of $\frac{2xz + yz}{2xz - y} - \frac{z(2x + y)}{2xz - y}$.

(1) This indicates $2x + y = 3$, and so $\frac{2xz + yz}{2xz - y} = \frac{3z}{2xz - y}$. However, this is not enough to determine the value of $\frac{2xz + yz}{2xz - y}$. For example, if $x = 1$, $y = 1$, and $z = 2$, then $2x + y = 3$ and $\frac{2xz + yz}{2xz - y} = 2$, but if $x = 0$,

$y = 3$, and $z = 2$, then $2x + y = 3$ and $\frac{2xz + yz}{2xz - y} = -2$; NOT sufficient.

(2) This indicates that $z = 2$, and so $\frac{2xz + yz}{2xz - y} = \frac{4x + 2y}{4x - y}$. However, this is not enough to determine the value of $\frac{2xz + yz}{2xz - y}$. For example, if $x = 1$, $y = 1$, and $z = 2$, then $\frac{2xz + yz}{2xz - y} = 2$, but if $x = 0$, $y = 3$, and $z = 2$, then $\frac{2xz + yz}{2xz - y} = -2$; NOT sufficient.

Taking (1) and (2) together is not enough to determine the value of $\frac{2xz + yz}{2xz - y}$ because the same examples used to show that (1) is not sufficient were also used to show that (2) is not sufficient.

The correct answer is E;
both statements together are still not sufficient.



DS04852

311. In the parallelogram shown, what is the value of x ?

(1) $y = 2x$

(2) $x + z = 120$

Geometry Angles

(1) Given that $y = 2x$ and the fact that adjacent angles of a parallelogram are supplementary, it follows that $180 = x + y = x + 2x = 3x$, or $180 = 3x$. Solving this equation gives $x = 60$; SUFFICIENT.

(2) Given that $x + z = 120$ and the fact that opposite angles of a parallelogram have the same measure, it follows that $120 = x + z = x + x$, or $120 = 2x$. Solving this equation gives $x = 60$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06096

312. In a product test of a common cold remedy, x percent of the patients tested experienced side effects from the use of the drug and y percent experienced relief of cold symptoms. What percent of the patients tested experienced both side effects and relief of cold symptoms?

(1) Of the 1,000 patients tested, 15 percent experienced neither side effects nor relief of cold symptoms.

(2) Of the patients tested, 30 percent experienced relief of cold symptoms without side effects.

Algebra Sets

One way to solve problems about sets is by using a contingency table. For the information given in this problem with T representing the total number of people tested, the table might be as follows:

	side effects	no side effects	totals
relief of symptoms			$\left(\frac{y}{100}\right)T$
no relief of symptoms			
totals	$\left(\frac{x}{100}\right)T$		T

(1) Using the information in (1) gives the following table, where 15% of 1,000 is 150.

	side effects	no side effects	totals
relief of symptoms			$\left(\frac{y}{100}\right)(1{,}000)$
no relief of symptoms		150	
totals	$\left(\frac{x}{100}\right)(1{,}000)$		1,000

If $x = 55$ and $y = 55$, then the table becomes

	side effects	no side effects	totals
relief of symptoms	250	300	550
no relief of symptoms	300	150	450
totals	550	450	1,000

and this gives $\frac{250}{1,000} = 25\%$ as the percent of patients tested who experienced both side effects and relief of cold symptoms.

However, if $x = 55$ and $y = 50$, then the table becomes

	side effects	no side effects	totals
relief of symptoms	200	300	500
no relief of symptoms	350	150	500
totals	550	450	1,000

and this gives $\frac{200}{1,000} = 20\%$ as the percent of patients tested who experienced both side effects and relief of cold symptoms; NOT sufficient.

(2) This indicates that 30% of the patients tested experienced relief of cold symptoms without side effects. The same examples that were used for (1) can be used here, also, to show that different values can be obtained for the percent of patients tested who experienced both side effects and relief of cold symptoms; NOT sufficient.

Since the examples satisfy both (1) and (2), these statements taken together are not sufficient to determine the percent of patients tested who experienced both side effects and relief of cold symptoms.

The correct answer is E;
both statements together are still not sufficient.

DS13588

313. Is $x < 5$?

(1) $x^2 > 5$

(2) $x^2 + x < 5$

Algebra Inequalities

(1) Given that $x^2 > 5$, then $x < 5$ can be true (for example, if $x = -3$) and $x < 5$ can be false (for example, if $x = 6$); NOT sufficient.

(2) Given that $x^2 + x < 5$, then $x < 5 - x^2$. Also, since $x^2 \geq 0$ is true for any real number x, it follows that $-x^2 \leq 0$, or after adding 5 to both sides, $5 - x^2 \leq 5$. From $x < 5 - x^2$ and $5 - x^2 \leq 5$, it follows that $x < 5$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS11257

314. Is zp negative?

(1) $pz^4 < 0$

(2) $p + z^4 = 14$

Arithmetic Properties of numbers

(1) Given that $pz^4 < 0$, zp can be negative (choose $p = -2$ and $z = 2$) and zp can be nonnegative (choose $p = -2$ and $z = -2$); NOT sufficient.

(2) Given that $p + z^4 = 14$, the same examples used to show that (1) is not sufficient can be used for (2); NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS04157

315. In each game of a certain tournament, a contestant either loses 3 points or gains 2 points. If Pat had 100 points at the beginning of the tournament, how many games did Pat play in the tournament?

(1) At the end of the tournament, Pat had 104 points.

(2) Pat played fewer than 10 games.

Arithmetic Computation with integers

Pat either lost 3 points or gained 2 points in each game she played. Therefore, since she had 100 points at the beginning of the tournament, her score at the end of the tournament was $100 - 3x + 2y$, where x and y represent, respectively, the number of games in which she lost 3 points and the number of games in which she gained 2 points. Determine the value of $x + y$.

(1) This indicates that $100 - 3x + 2y = 104$, but no information is given about the values of x or y. For example, the values of x and y could be 0 and 2, respectively, so that $x + y = 2$ or the values of x and y could be 2 and 5, respectively, so that $x + y = 7$; NOT sufficient.

(2) This indicates that Pat played fewer than 10 games, but in each of the examples above, Pat played fewer than 10 games; NOT sufficient.

Since the examples given satisfy both (1) and (2), taking (1) and (2) together does not give enough information to determine the value of $x + y$.

The correct answer is E;
both statements together are still not sufficient.

DS05631

316. At the beginning of the year, the Finance Committee and the Planning Committee of a certain company each had n members, and no one was a member of both committees. At the end of the year, 5 members left the Finance Committee and 3 members left the Planning Committee. How many members did the Finance Committee have at the beginning of the year?

(1) The ratio of the total number of members who left at the end of the year to the total number of members at the beginning of the year was 1:6.

(2) At the end of the year, 21 members remained on the Planning Committee.

Algebra Applied problems

(1) It is given that $\frac{8}{2n} = \frac{1}{6}$, since a total of $3 + 5 = 8$ members left at the end of the year and there were a total of $n + n = 2n$ members at the beginning of the year. It follows that $(8)(6) = (2n)(1)$, or $n = 24$; SUFFICIENT.

(2) It is given that $n - 3 = 21$, since 3 members from the original n members left the Planning Committee at the end of the year. It follows that $n = 21 + 3 = 24$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS03268

317. Can a certain rectangular sheet of glass be positioned on a rectangular tabletop so that it covers the entire tabletop and its edges are parallel to the edges of the tabletop?

(1) The tabletop is 36 inches wide by 60 inches long.

(2) The area of one side of the sheet of glass is 2,400 square inches.

Geometry Area

Determine whether the length and width of the sheet of glass are greater than or equal to the length and width of the tabletop.

(1) The length and width of the tabletop are given, but nothing can be determined about the dimensions of the sheet of glass; NOT sufficient.

(2) The area of the sheet of glass is given, but nothing can be determined about the dimensions of the tabletop; NOT sufficient.

(1) and (2) together are not sufficient because the information given does not specify the length and width of the sheet of glass. For example, the length and width of the sheet of glass could be 40 inches and 60 inches and, since $40 \geq 36$ and $60 \geq 60$, the answer to the question would be "Yes." On the other hand, the length and width of the sheet of glass could be 100 inches and 24 inches, and in this case the answer to the question would be "No" because 24 is not greater than or equal to 36 or 60.

The correct answer is E;
both statements together are still not sufficient.

DS15561

318. If $xy \neq 0$, is $x^3 + y^3 > 0$?

(1) $x + y > 0$

(2) $xy > 0$

Algebra Inequalities

(1) Given that $x + y > 0$, factor $x^3 + y^3$ as $(x + y)(x^2 - xy + y^2)$ and consider $x^2 - xy + y^2$. If $xy < 0$, then $x^2 - xy + y^2$ is the sum of $x^2 + y^2$ (which is greater than or equal to 0) and $-xy$ (which is greater than 0), and so $x^2 - xy + y^2 > 0$. (In fact, $x^2 + y^2 > 0$, since from $xy \neq 0$ it follows that $x \neq 0$ and $y \neq 0$.) If $xy > 0$, then $xy < 2xy$ and $2xy \leq x^2 + y^2$ (this follows from $0 \leq x^2 - 2xy + y^2$, which in turn follows from $0 \leq (x - y)^2$), and therefore $xy < x^2 + y^2$, or $x^2 - xy + y^2 > 0$. Finally, $xy = 0$ is not possible from the information given. Thus, from the information given, it is always the case that $x^2 - xy + y^2$ is positive, and hence $x^3 + y^3$ is the product of two positive quantities, $x + y$ and $x^2 - xy + y^2$, and therefore $x^3 + y^3 > 0$.

By using ideas involving functions and their graphs, a shorter argument can be given. The function $f(x) = x^3$ is a strictly increasing function, which is evident from its graph. Thus, if $a < b$, then $f(a) < f(b)$, or $a^3 < b^3$. From (1) it follows that $x > -y$. Therefore, $x^3 < (-y)^3$, or $x^3 < -y^3$, or $x^3 + y^3 < 0$; SUFFICIENT.

(2) Given that $xy > 0$, it is possible that $x^3 + y^3 > 0$ (choose $x = y = 1$) and it is possible that $x^3 + y^3 < 0$ (choose $x = y = -1$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS13541

319. Max purchased a guitar for a total of \$624, which consisted of the price of the guitar and the sales tax. Was the sales tax rate greater than 3 percent?

(1) The price of the guitar that Max purchased was less than \$602.

(2) The sales tax for the guitar that Max purchased was less than \$30.

Arithmetic Applied problems; Percents

Letting P be the price, in dollars, of the guitar and r% be the sales tax rate, it is given that $P\left(1 + \frac{r}{100}\right) = 624$. Determine if $r > 3$.

(1) Given that $P < 602$, then $624 = P\left(1 + \frac{r}{100}\right) < 602\left(1 + \frac{r}{100}\right)$, and so $1 + \frac{r}{100} > \frac{624}{602}$. Therefore, $\frac{r}{100} > \frac{624}{602} - 1 = \frac{22}{602} = \frac{11}{301} > \frac{3}{100}$, because $(11)(100) > (3)(301)$, and so $r > 100\left(\frac{3}{100}\right) = 3$; SUFFICIENT.

(2) Given that the sales tax was less than \$30, it is not possible to determine whether the sales tax rate was greater than 3%. Since $(0.03)(\$624) = \$18.72$, a 3% sales tax rate on the price corresponds to a sales tax that is less than \$18.72, since the price is less than \$624. Therefore, it is possible that the sales tax is less than \$30 and the sales tax rate is less than 3%, since any sales tax rate less than 3% would give a sales tax that is less than \$18.72 (and hence less than \$30). However, it is also possible that the sales tax is less than \$30 and the sales tax rate is 4% (which is greater than 3%), since a 4% sales tax rate implies $P\left(1 + \frac{4}{100}\right) = 624$, or $P = \frac{62,400}{104} = 600$, and thus the sales tax would be $0.04(\$600) = \$24 < \$30$; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS06027

320. What is the sum of a certain pair of consecutive odd integers?

(1) At least one of the integers is negative.

(2) At least one of the integers is positive.

Arithmetic Properties of numbers

(1) Given that at least one of the integers is negative, the sum could be -4 (if the integers were -3 and -1) and the sum could be 0 (if the integers were -1 and 1); NOT sufficient.

(2) Given that at least one of the integers is positive, the sum could be 0 (if the integers were -1 and 1) and the sum could be 4 (if the integers were 1 and 3); NOT sufficient.

Taking (1) and (2) together, the smaller of the two numbers cannot be less than –1 (otherwise (2) would not be true) and the larger of the two numbers cannot be greater than 1 (otherwise (1) would not be true). Therefore, the integers must be –1 and 1, and the sum must be 0.

The correct answer is C;
both statements together are sufficient.

DS08197

321. The sum of 4 different odd integers is 64. What is the value of the greatest of these integers?

(1) The integers are consecutive odd numbers.

(2) Of these integers, the greatest is 6 more than the least.

Arithmetic Properties of numbers

Determine the greatest of four odd integers whose sum is 64.

(1) This indicates that the integers are consecutive odd integers. Letting z represent the greatest of the four integers, it follows that $(z-6)+(z-4)+(z-2)+z=64$, from which a unique value of z can be determined; SUFFICIENT.

(2) Letting w, x, y, and z represent four different odd integers, where $w<x<y<z$, this indicates that $z-w=6$ or $w=z-6$. This means that $x=z-4$ and $y=z-2$ since w, x, y, and z must be different odd integers and it must be true that $w<x<y<z$. From $w+x+y+z=64$, it follows that $(z-6)+(z-4)+(z-2)+z=64$ from which a unique value of z can be determined; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS13130

322. Was the number of books sold at Bookstore X last week greater than the number of books sold at Bookstore Y last week?

(1) Last week, more than 1,000 books were sold at Bookstore X on Saturday and fewer than 1,000 books were sold at Bookstore Y on Saturday.

(2) Last week, less than 20 percent of the books sold at Bookstore X were sold on Saturday and more than 20 percent of the books sold at Bookstore Y were sold on Saturday.

Arithmetic Inequalities

Determine if Bookstore X sold more books last week than Bookstore Y.

(1) This indicates that Bookstore X sold more books on Saturday than Bookstore Y, but it gives no information about the numbers of books sold by the two bookstores on the other days of last week; NOT sufficient.

(2) This gives information about the percents of the books sold last week that were sold on Saturday, but it gives no information about the actual numbers of books sold at the two bookstores last week. Therefore, a comparison of the numbers of books sold last week by Bookstore X and Bookstore Y is not possible; NOT sufficient.

Taking (1) and (2) together, if x represents the number of books that Bookstore X sold on Saturday and T_X represents the total number of books that Bookstore X sold last week, then $x>1{,}000$ and $x<0.2T_X$. It follows that $1{,}000<0.2T_X$ and so $T_X>5{,}000$. Similarly, if y represents the number of books that Bookstore Y sold on Saturday and T_Y represents the total number of books that Bookstore Y sold last week, then $y<1{,}000$ and $y>0.2T_Y$. It follows that $0.2T_Y<1{,}000$, and so $T_Y<5{,}000$. Combining the inequalities gives $T_Y<5{,}000<T_X$, and so $T_Y<T_X$, which means that Bookstore X sold more books last week than Bookstore Y.

The correct answer is C;
both statements together are sufficient.

DS04540

323. From May 1 to May 30 in the same year, the balance in a checking account increased. What was the balance in the checking account on May 30 ?

(1) If, during this period of time, the increase in the balance in the checking account had been 12 percent, then the balance in the account on May 30 would have been $504.

(2) During this period of time, the increase in the balance in the checking account was 8 percent.

Arithmetic Applied problems; Percents

(1) Given that the amount on May 30 would have been \$504 if the increase had been 12%, it follows that the amount on May 1 was $\$\frac{504}{1.12} = \$450$. However, the balance on May 30 cannot be determined because the (actual) percent increase is not given. For example, if the increase had been 10%, then the balance on May 30 would have been (1.1)(\$450) = \$495, which is different from \$504; NOT sufficient.

(2) Given that the increase was 8%, the amount on May 30 could be \$108 (if the amount on May 1 was \$100) and the amount on May 30 could be \$216 (if the amount on May 1 was \$200); NOT sufficient.

Taking (1) and (2) together, it follows that the amount on May 30 was (\$450)(1.08) = \$486.

The correct answer is C;
both statements together are sufficient.

DS08365

324. A merchant discounted the sale price of a coat and the sale price of a sweater. Which of the two articles of clothing was discounted by the greater dollar amount?

(1) The percent discount on the coat was 2 percentage points greater than the percent discount on the sweater.

(2) Before the discounts, the sale price of the coat was \$10 less than the sale price of the sweater.

Arithmetic Applied problems; Percents

(1) Given that the discount on the coat was 2 percentage points greater than the discount on the sweater, the sweater could have been discounted by the greater dollar amount (\$30 sweater with 10% discount is a discount of \$3; \$20 coat with 12% discount is a discount of \$2.40) and the coat could have been discounted by the greater dollar amount (\$110 sweater with 10% discount is a discount of \$11; \$100 coat with 12% discount is a discount of \$12); NOT sufficient.

(2) Given that the coat's sale price was \$10 less than the sweater's sale price, the same examples used to show that (1) is not sufficient can be used for (2); NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS01168

325. If the positive integer n is added to each of the integers 69, 94, and 121, what is the value of n?

(1) $69 + n$ and $94 + n$ are the squares of two consecutive integers.

(2) $94 + n$ and $121 + n$ are the squares of two consecutive integers.

Algebra Computation with integers

Determine the value of the positive integer n.

(1) This indicates that $69 + n$ and $94 + n$ are the squares of two consecutive integers. Letting x and $(x + 1)$ represent the consecutive integers, it follows that $(x + 1)^2 - x^2 = x^2 + 2x + 1 - x^2 = 2x + 1$. Therefore, $(94 + n) - (69 + n) = 2x + 1$ or $25 = 2x + 1$, from which $x = 12$. Then $12^2 = 69 + n$ and $n = 144 - 69 = 75$ or $(12 + 1)^2 = 94 + n$ and $n = 169 - 94 = 75$; SUFFICIENT.

(2) This indicates that $94 + n$ and $121 + n$ are the squares of two consecutive integers. Letting x and $(x + 1)$ represent the consecutive integers, it follows that $(x + 1)^2 - x^2 = x^2 + 2x + 1 - x^2 = 2x + 1$. Therefore, $(121 + n) - (94 + n) = 2x + 1$ or $27 = 2x + 1$, from which $x = 13$. Then $13^2 = 94 + n$ and $n = 169 - 94 = 75$ or $(13 + 1)^2 = 121 + n$ and $n = 196 - 121 = 75$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS05269

326. Last year, in a certain housing development, the average (arithmetic mean) price of 20 new houses was $160,000. Did more than 9 of the 20 houses have prices that were less than the average price last year?

(1) Last year the greatest price of one of the 20 houses was $219,000.

(2) Last year the median of the prices of the 20 houses was $150,000.

Arithmetic Statistics

(1) Given that the greatest price was $219,000, it is possible that only one of the houses had a price less than $160,000. For example, the 20 houses that have an average price of $160,000 could have been 18 houses each at $160,000, one house at $219,000 = $160,000 + $59,000 and one house at $101,000 = $160,000 – $59,000. On the other hand, it is possible that 19 of the houses had a price less than $160,000. For example, the 20 houses that have an average price of $160,000 could have been 18 houses each at $159,000 = $160,000 – $1,000, one house at $219,000 = $160,000 + $59,000 and one house at $119,000 = $160,000 – $59,000 + $18,000. Therefore, it is not possible to determine whether more than 9 of the houses had prices less than $160,000; NOT sufficient.

(2) Given that the median of the 20 prices was $150,000, then when the prices are listed in numerical order from least to greatest, it follows that at least 10 of the prices are less than or equal to $150,000, and hence more than 9 of the prices are less than $160,000. For one of these 10 prices this is true because the median is the average of the 10th and 11th prices, and so the 10th price cannot be greater than $150,000, otherwise both the 10th and 11th prices would be greater than $150,000 and their average would be greater than $150,000. The other 9 of these 10 prices are the 1st through 9th prices, since each of these is less than or equal to the 10th price; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

DS14527

327. For a certain city's library, the average cost of purchasing each new book is $28. The library receives $15,000 from the city each year; the library also receives a bonus of $2,000 if the total number of items checked out over the course of the year exceeds 5,000. Did the library receive the bonus last year?

(1) The library purchased an average of 50 new books each month last year and received enough money from the city to cover this cost.

(2) The lowest number of items checked out in one month was 459.

Arithmetic Applied problems

(1) Given that the library purchased an average of 50 new books each month, for the entire year the library purchased a total of (50)(12) = 600 books for a total cost of (600)($28) = $16,800. Excluding any possible bonus, the library received $15,000 from the city. Since this amount received from the city is not enough to cover the cost of the books, and the information provided in (1) says that the total amount received from the city was enough to cover the cost of the books, it follows that the library received a bonus; SUFFICIENT.

(2) Given that the least number of books checked out in one month was 459, it follows that the total number of books checked out for the year was at least (12)(459) = 5,508. Since this is greater than 5,000, it follows that the total number of books checked out for the year was greater than 5,000 and the library received a bonus; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

DS00395

328. Each gift certificate sold yesterday by a certain bookstore cost either $10 or $50. If yesterday the bookstore sold more than 5 gift certificates that cost $50 each, what was the total number of gift certificates sold yesterday by the bookstore?

(1) Yesterday the bookstore sold fewer than 10 gift certificates that cost $10 each.

(2) The total cost of gift certificates sold yesterday by the bookstore was $460.

Arithmetic Computation with integers

Let x be the number of $10 gift certificates sold yesterday and let y be the number of $50 gift certificates sold yesterday. It is given that $y > 5$. Determine the value of $x + y$.

(1) It is given that $x < 10$, so x could be 0, 1, 2, 3, . . ., 9. Since $y > 5$, y could be 6, 7, 8, . . ., and then $x + y$ could be, for example, $0 + 6 = 6$ or $2 + 8 = 10$; NOT sufficient.

(2) It is given that $10x + 50y = 460$. Several values of x and y with $y > 5$ will satisfy this equation and give different values for $x + y$, as shown below; NOT sufficient.

x	y	$10x + 50y$	$x + y$
16	6	460	22
11	7	460	18
6	8	460	14
1	9	460	10

Taking (1) and (2) together, it can be seen from the table that all the requirements are satisfied by $x = 6$ and $y = 8$ and also by $x = 1$ and $y = 9$, but the value of $x + y$ is not uniquely determined; NOT sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS15045

329. Three dice, each of which has its 6 sides numbered 1 through 6, are tossed. The sum of the 3 numbers that are facing up is 12. Is at least 1 of these numbers 5 ?

(1) None of the 3 numbers that are facing up is divisible by 3.

(2) Of the numbers that are facing up, 2, but not all 3, are equal.

Arithmetic Properties of integers

When three dice, each with its 6 faces numbered 1 through 6, are tossed and the sum of the three integers facing up is 12, the possible outcomes are {1, 5, 6}, {2, 4, 6}, {2, 5, 5}, {3, 3, 6}, {3, 4, 5}, and {4, 4, 4}. Determine if, in the outcome described, at least one of the numbers is 5.

(1) This indicates that none of the numbers is divisible by three. The outcome could be {4, 4, 4}. In this case, none of the three numbers is 5. On the other hand, the outcome could be {2, 5, 5}. In this case, at least one of the numbers is 5; NOT sufficient.

(2) This indicates that two of the numbers, but not all three, are equal. The outcome could be {3, 3, 6}. In this case, none of the three numbers is 5. On the other hand, the outcome could be {2, 5, 5}. In this case, at least one of the numbers is 5; NOT sufficient.

Taking (1) and (2) together, of the possible outcomes {1, 5, 6}, {2, 4, 6}, {3, 3, 6}, and {3, 4, 5} are eliminated because they do not satisfy (1), and {4, 4, 4} is eliminated because it does not satisfy (2). This leaves only {2, 5, 5}, and at least one number is 5.

The correct answer is C;
both statements together are sufficient.

DS01324

330. On the number line, point R has coordinate r and point T has coordinate t. Is $t < 0$?

(1) $-1 < r < 0$

(2) The distance between R and T is equal to r^2.

Arithmetic Number line

Determine if t, the coordinate of point T on the number line, is negative.

(1) This indicates that r, the coordinate of point R on the number line, lies between -1 and 0, but gives no information about the location of T in relation to either point R or 0; NOT sufficient.

(2) This indicates that the distance between R and T, which can be expressed as $|r - t|$, is r^2. If $r = 2$, for example, then $|2 - t| = 4$ so $2 - t = 4$ or $2 - t = -4$. If $2 - t = 4$, then $t = -2$, which is negative. If $2 - t = -4$, then $t = 6$, which is not negative; NOT sufficient.

Taking (1) and (2) together, if $r - t \geq 0$, then $r - t = r^2$ and $t = r - r^2 = r(1 - r)$. Since $-1 < r < 0$, it follows that $1 < 1 - r < 2$, and so t is the product of a negative number (r) and a positive number ($1 - r$) and is therefore negative. Similarly, if $r - t < 0$, then $|r - t| = -(r - t) = t - r = r^2$ and $t = r^2 + r = r(r + 1)$. Since $-1 < r < 0$, it follows that $0 < r + 1 < 1$, and so t is the product of a negative number (r) and the positive number ($r + 1$) and is therefore negative.

The correct answer is C;
both statements together are sufficient.

DS06659

331. S is a set of points in the plane. How many distinct triangles can be drawn that have three of the points in S as vertices?

(1) The number of distinct points in S is 5.

(2) No three of the points in S are collinear.

Arithmetic Elementary combinatorics

(1) Given that the number of points in S is 5, the number of triangles can be 0 (if the points are collinear) and the number of triangles can be greater than 0 (if the points are not all collinear); NOT sufficient.

(2) Given that no three points of S are collinear, the number of triangles can be 1 (if S consists of 3 points) and the number of triangles can be 4 (if S consists of 4 points); NOT sufficient.

Taking (1) and (2) together, the number of distinct triangles must be $\binom{5}{3} = \frac{5!}{3!(5-3)!} = 10$, which is the number of combinations of 5 points taken 3 at a time.

The correct answer is C;
both statements together are sufficient.

DS16078

332. Stores L and M each sell a certain product at a different regular price. If both stores discount their regular price of the product, is the discount price at Store M less than the discount price at Store L ?

(1) At Store L the discount price is 10 percent less than the regular price; at Store M the discount price is 15 percent less than the regular price.

(2) At Store L the discount price is \$5 less than the regular store price; at Store M the discount price is \$6 less than the regular price.

Arithmetic Percents

Let L_r and L_d be the regular and discounted prices, respectively, at Store L, and let M_r and M_d be the regular and discounted prices, respectively, at Store M. Determine if $M_d < L_d$.

(1) Knowing that $L_d = (1 - 0.10)L_r = 0.90L_r$ and that $M_d = (1 - 0.15)M_r = 0.85M_r$ gives no information for comparing M_d and L_d; NOT sufficient.

(2) Knowing that $L_d = L_r - 5$ and that $M_d = M_r - 6$ gives no information for comparing M_d and L_d; NOT sufficient.

Taking (1) and (2) together gives $0.90L_r = L_r - 5$ and $0.85\ M_r = M_r - 6$, from which it follows that $0.10L_r = 5$ or $L_r = 50$ and $0.15M_r = 6$ or $M_r = 40$. Then $L_d = 50 - 5 = 45$ and $M_d = 40 - 6 = 34$. Therefore, $M_d < L_d$.

The correct answer is C;
both statements together are sufficient.

DS16529

333. If d denotes a decimal, is $d \geq 0.5$?

(1) When d is rounded to the nearest tenth, the result is 0.5.

(2) When d is rounded to the nearest integer, the result is 1.

Arithmetic Rounding; Estimating

(1) In this case, for example, the value of d could range from the decimal 0.45 to 0.54. Some of these, such as 0.51 or 0.52, are greater than or equal to 0.5, and others, such as 0.47 or 0.48, are less than 0.5; NOT sufficient.

(2) When the result of rounding d to the nearest integer is 1, d could range in value from the decimal 0.50 to 1.49, which are greater than or equal to 0.5; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS08231

334. In the two-digit integers 3■ and 2▲, the symbols ■ and ▲ represent different digits, and the product (3■)(2▲) is equal to 864. What digit does ■ represent?

(1) The sum of ■ and ▲ is 10.

(2) The product of ■ and ▲ is 24.

Arithmetic Properties of numbers

(1) Given that ■ + ▲ = 10, then the possible products ■▲ are (1)(9) = (9)(1) = 9, (2)(8) = (8)(2) = 16, (3)(7) = (7)(3) = 21, (4)(6) = (6)(4) = 24, and (5)(5) = 25. Since (3■)(2▲) = 864 has units digit 4, it follows that ■▲ must also have units digit 4. Since only two of the possible products of ■▲ has units digit 4, it follows that ■ = 4 and ▲ = 6, or ■ = 6 and ▲ = 4. If ■ = 4 and ▲ = 6, then (3■)(2▲) = (34) (26) = 884 ≠ 864, and if ■ = 6 and ▲ = 4, then (3■)(2▲) = (36)(24) = 864. Therefore, ■ = 6; SUFFICIENT.

(2) Given that ■▲ = 24, and that both ■ and ▲ are digits, there are four possibilities for the values of ■ and ▲.
(a) ■ = 3 and ▲ = 8, which gives (3■)(2▲) = (33)(28) = 924 ≠ 864;
(b) ■ = 4 and ▲ = 6, which gives (3■)(2▲) = (34)(26) = 884 ≠ 864;
(c) ■ = 6 and ▲ = 4, which gives (3■)(2▲) = (36)(24) = 864;
(d) ■ = 8 and ▲ = 3, which gives (3■)(2▲) = (38)(23) = 874 ≠ 864.
Therefore, ■ = 6; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.



DS07262

335. Two points, N and Q (not shown), lie to the right of point M on line ℓ. What is the ratio of the length of QN to the length of MQ?

(1) Twice the length of MN is 3 times the length of MQ.

(2) Point Q is between points M and N.

Algebra Order

(1) Given that twice the length of MN is 3 times the length of MQ, it follows that the points are ordered from left to right as M, Q, and N. Thus, letting $MQ = x$ and $QN = y$, it is given that $2(x + y) = 3x$ and the value of $\frac{y}{x}$ is to be determined. The given equation can be rewritten as $2x + 2y = 3x$, or $2y = x$, or $\frac{y}{x} = \frac{1}{2}$; SUFFICIENT.

(2) Given that Q is between M and N, the ratio of QN to MQ can be close to zero (if Q and N are close together and both far from M) and the ratio of QN to MQ can be large (if M and Q are close together and both far from N); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS05639

336. Did the sum of the prices of three shirts exceed $60?

(1) The price of the most expensive of the shirts exceeded $30.

(2) The price of the least expensive of the shirts exceeded $20.

Arithmetic Applied problems

(1) Given that the price of the most expensive shirt exceeded $30, the sum of the prices of the shirts can be under $60 (if the prices were $10, $10, and $35) and the sum of the prices of the shirts can be over $60 (if the prices were $10, $10, and $50); NOT sufficient.

(2) Given that the price of the least expensive shirt exceeded $20, it follows that the sum of the prices of the shirts exceeds 3($20) = $60; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS03057

337. What is the total number of coins that Bert and Claire have?

(1) Bert has 50 percent more coins than Claire.

(2) The total number of coins that Bert and Claire have is between 21 and 28.

Arithmetic Computation with integers

Determine the total number of coins Bert and Claire have. If B represents the number of coins that Bert has and C represents the number of coins that Claire has, determine $B + C$.

(1) Bert has 50% more coins than Claire, so $B = 1.5C$, and $B + C = 1.5C + C = 2.5C$, but the value of C can vary; NOT sufficient.

(2) The total number of coins Bert and Claire have is between 21 and 28, so $21 < B + C < 28$ and, therefore, $B + C$ could be 22, 23, 24, 25, 26, or 27; NOT sufficient.

Taking (1) and (2) together, $21 < 2.5C < 28$ and then $\frac{21}{2.5} < C < \frac{28}{2.5}$ or $8.4 < C < 11.2$.

If $C = 9$, then $B = (1.5)(9) = 13.5$; if $C = 10$, then $B = (1.5)(10) = 15$; and if $C = 11$, then $B = (1.5)(11) = 16.5$. Since B represents a number of coins, B is an integer. Therefore, $B = 15$, $C = 10$, and $B + C = 25$.

The correct answer is C;
both statements together are sufficient.

DS05668

338. A telephone station has x processors, each of which can process a maximum of y calls at any particular time, where x and y are positive integers. If 500 calls are sent to the station at a particular time, can the station process all of the calls?

(1) $x = 600$

(2) $100 < y < 200$

Algebra Applied problems

At a particular time, the telephone station can process a maximum of xy calls, where x and y are positive integers. Determine whether $xy \geq 500$.

(1) Given that $x = 600$, it follows that $xy \geq 600$ since $y \geq 1$ (y is a positive integer); SUFFICIENT.

(2) Given that $100 < y < 200$, $xy < 500$ is possible (if $x = 3$ and $y = 150$) and $xy \geq 500$ is possible (if $x = 10$ and $y = 150$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

	Price per Flower
Roses	\$1.00
Daisies	\$0.50

DS07953

339. Kim and Sue each bought some roses and some daisies at the prices shown above. If Kim bought the same total number of roses and daisies as Sue, was the price of Kim's purchase of roses and daisies higher than the price of Sue's purchase of roses and daisies?

(1) Kim bought twice as many daisies as roses.

(2) Kim bought 4 more roses than Sue bought.

Algebra Applied problems

Let R_K be the number of roses that Kim bought, let R_S be the number of roses that Sue bought, and let T be the total number of roses and daisies each bought. Then Kim bought $(T - R_K)$ daisies and Sue bought $(T - R_S)$ daisies.

For the roses and daisies, Kim paid a total of $\$[R_K + \frac{1}{2}(T - R_K)] = \$\frac{1}{2}(R_K + T)$ and Sue paid a total of $\$[R_S + \frac{1}{2}(T - R_S)] = \$\frac{1}{2}(R_S + T)$.

Determine whether $\frac{1}{2}(R_K + T) > \frac{1}{2}(R_S + T)$, or equivalently, determine whether $R_K > R_S$.

(1) Given that $T - R_K = 2R_K$, or $T = 3R_K$, it is not possible to determine whether $R_K > R_S$ because no information is provided about the value of R_S; NOT sufficient.

(2) Given that $R_K = R_S + 4$, it follows that $R_K > R_S$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS14406

340. Jazz and blues recordings accounted for 6 percent of the $840 million revenue from the sales of recordings in Country Y in 2000. What was the revenue from the sales of jazz and blues recordings in Country Y in 1998 ?

(1) Jazz and blues recordings accounted for 5 percent of the revenue from the sales of recordings in Country Y in 1998.

(2) The revenue from the sales of jazz and blues recordings in Country Y increased by 40 percent from 1998 to 2000.

Arithmetic Percents

It is given that jazz and blues recordings accounted for 6% of the $840 million revenue of recordings in Country Y in 2000. Determine the revenue from the sales of jazz and blues recordings in 1998.

(1) This indicates that jazz and blues recordings accounted for 5% of the revenue from the sales of recordings in Country Y in 1998. However, no information is given to indicate what the revenue from the sales of recordings was in 1998. Therefore, the revenue from sales of jazz and blues recordings in 1998 cannot be determined; NOT sufficient.

(2) This indicates that the revenue from the sales of jazz and blues recordings in Country Y increased by 40% from 1998 to 2000. Letting R_{1998} and R_{2000} represent the revenue from the sales of jazz and blues recordings in 1998 and 2000, respectively, then $R_{2000} = 1.4R_{1998}$, and so $(0.06)(\$840 \text{ million}) = 1.4R_{1998}$, from which R_{1998} can be determined; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS06315

341. On a certain nonstop trip, Marta averaged x miles per hour for 2 hours and y miles per hour for the remaining 3 hours. What was her average speed, in miles per hour, for the entire trip?

(1) $2x + 3y = 280$

(2) $y = x + 10$

Algebra Rate problems

Marta traveled a total of $(2x + 3y)$ miles in $2 + 3 = 5$ hours for an average speed of $\left(\frac{2x+3y}{5}\right)$ miles per hour. Determine the value of $\frac{2x+3y}{5}$.

(1) Given that $2x + 3y = 280$, it follows that $\frac{2x+3y}{5} = \frac{280}{5}$; SUFFICIENT.

(2) Given that $y = x + 10$, it follows that $\frac{2x+3y}{5} - \frac{2x+3(x+10)}{5} = x + 6$.
Therefore, the value of $\frac{2x+3y}{5}$ can be 56 (if $x = 50$ and $y = 60$) and the value of $\frac{2x+3y}{5}$ can be 61 (if $x = 55$ and $y = 65$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS12730

342. If x is a positive integer, what is the value of $\sqrt{x+24} - \sqrt{x}$?

(1) $\sqrt{x}$ is an integer.

(2) $\sqrt{x+24}$ is an integer.

Arithmetic Operations with radicals

Determine the value of $\sqrt{x+24} - \sqrt{x}$.

(1) This indicates that x is a perfect square. If $x = 1$, then $\sqrt{x+24} - \sqrt{x} = \sqrt{1+24} - \sqrt{1} = 5 - 1 = 4$. However, if $x = 25$, then $\sqrt{x+24} - \sqrt{x} = \sqrt{25+24} - \sqrt{25}$
$= 7 - 5 = 2$;
NOT sufficient.

(2) This indicates that $\sqrt{x+24}$ is an integer. In each of the examples above, $\sqrt{x+24}$ is an integer; NOT sufficient.

Because the same examples were used to establish that neither (1) nor (2) is sufficient, it is not possible to determine the value of $\sqrt{x+24} - \sqrt{x}$ from the given information.

The correct answer is E;
both statements together are still not sufficient.

DS13982

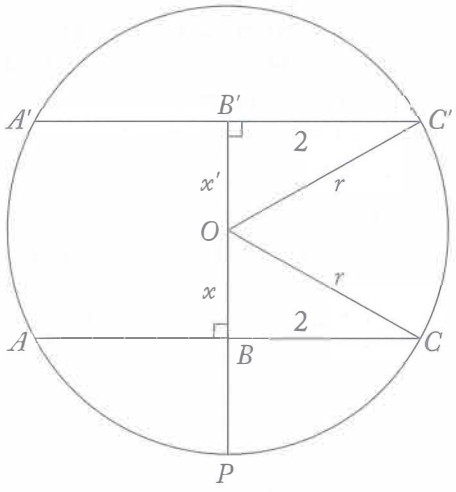
343. A tank is filled with gasoline to a depth of exactly 2 feet. The tank is a cylinder resting horizontally on its side, with its circular ends oriented vertically. The inside of the tank is exactly 6 feet long. What is the volume of the gasoline in the tank?

(1) The inside of the tank is exactly 4 feet in diameter.

(2) The top surface of the gasoline forms a rectangle that has an area of 24 square feet.

Geometry Cylinders; Volume

(1) Given that the diameter of the cylindrical tank is 4 ft, it follows that its radius is 2 ft. Therefore, the radius and height of the cylindrical tank are known, and hence the volume of the gasoline is half the volume of the cylinder, or $\frac{1}{2}\pi r^2 h = \frac{1}{2}\pi\,(2\text{ ft})^2(6\text{ ft}) = 12\pi\text{ ft}^3$; SUFFICIENT.



(2) Given that the rectangular surface of the gasoline forms a rectangle with area 24 ft^2 and length 6 ft, the width of the rectangular surface is 4 ft. Let r ft be the radius of the cylinder. The figure shows a vertical cross-section of the cylindrical tank—a circle with center O—with the two possibilities for the location of the top surface of the gasoline: the depth of the gasoline is less than or equal to r ft ($\overline{AC}$ marks the location of the top surface of the gasoline and $BP = 2$ ft is the depth of the gasoline) or the depth of the gasoline is greater than or equal to r ft ($\overline{A'C'}$ marks the location of the top surface of the gasoline and $B'P = 2$ ft is the depth of the gasoline). For the first possibility, ΔOBC is a right triangle with hypotenuse $\overline{OC}$ of length r ft, so letting $OB = x$ ft, it follows from the Pythagorean theorem that $r^2 = x^2 + 2^2 = x^2 + 4$. However, because $r = OB + BP = x + 2$, it follows that $r^2 = (x+2)^2 = x^2 + 4x + 4$. Therefore, $x^2 + 4 = x^2 + 4x + 4$, so $4x = 0$, $x = 0$, and $\overline{AC}$ is a diameter of the circle. Thus, the diameter of the cylindrical tank is 4 ft, and by the same reasoning used in (1), the volume of the gasoline can be determined. For the second possibility, $\Delta OB'C'$ is a right triangle with hypotenuse $\overline{OC'}$ of length r ft, so letting $B'O = x'$ ft, it follows from the Pythagorean theorem that $r^2 = (x')^2 + 2^2 = (x')^2 + 4$. However, because $r = B'P - B'O = 2 - x'$, it follows that $r^2 = (2 - x')^2 = 4 - 4x' + (x')^2$. Therefore, $(x')^2 + 4 = 4 - 4x' + (x')^2$, so $0 = -4x'$, $x' = 0$, and $\overline{A'C'}$ is a diameter of the circle. Thus, the diameter of the cylindrical tank is 4 ft, and by the same reasoning used in (1), the volume of the gasoline can be determined; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.



DS16197

344. Of the four numbers represented on the number line above, is r closest to zero?

(1) $q = -s$

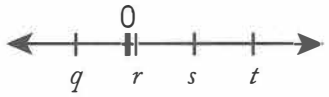
(2) $-t < q$

Algebra Order

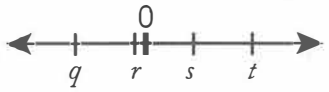


Referring to the figure above, in which it may be assumed that q, r, s, and t are different numbers, determine if r is closest to 0.

(1) Since $q = -s$, one of q and s is positive and the other is negative. Since s is to the right of q, then s is positive and q is negative. Also, 0 is halfway between q and s, so q and s are the same distance from 0. If $r = 0$, then, of q, r, s, and t, r is closest to 0 because it IS 0. If $r \neq 0$ then either (i) $q < 0 < r < s < t$ or (ii) $q < r < 0 < s < t$.



(i) If $q < 0 < r < s < t$, as shown above, r is closer to 0 than s is because r is between 0 and s, and r is clearly closer to 0 than t is because t is farther away from 0 than s is. Also, since q and s are the same distance from 0 and r is closer to 0 than s is, then r is closer to 0 than q is. Therefore, r is closest to 0.



(ii) If $q < r < 0 < s < t$, as shown above, r is closer to 0 than q is because r is between 0 and q. Also, r is closer to 0 than s is because r is closer to 0 than q is and q and s are the same distance from 0. Moreover, r is closer to 0 than t is because t is farther away from 0 than s is. Therefore, r is closest to 0.

In each case, r is closest to 0; SUFFICIENT.

(2) If $-t < q$, then $-t$ is to the left of q. If $t = 5$, $s = 4$, $r = 3$, and $q = -2$, then $-5 < -2$, so (2) is satisfied. In this case, q is closest to 0. On the other hand, if $t = 5$, $s = 4$, $r = -1$, and $q = -2$, then $-5 < -2$, so (2) is satisfied, but r is closest to 0; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS18414

345. A group consisting of several families visited an amusement park where the regular admission fees were ¥5,500 for each adult and ¥4,800 for each child. Because there were at least 10 people in the group, each paid an admission fee that was 10% less than the regular admission fee. How many children were in the group?

(1) The total of the admission fees paid for the adults in the group was ¥29,700.

(2) The total of the admission fees paid for the children in the group was ¥4,860 more than the total of the admission fees paid for the adults in the group.

Arithmetic Simultaneous equations

Determine the number of children in a group of at least 10 people who visited an amusement park.

(1) This indicates that $(0.9)(5{,}500)A = 29{,}700$, where A represents the number of adults in the group. From this, the number of adults can be determined. However, the number of children in the group cannot be determined without additional information about the exact number of people in the group; NOT sufficient.

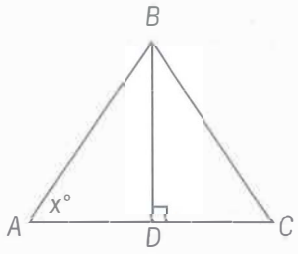
(2) If C and A represent the numbers of children and adults, respectively, in the group, this indicates that $(0.9)(4{,}800)C = (0.9)(5{,}500)A + 4{,}860$, or $48C = 55A + 54$, or $48C - 55A = 54$, which is a single equation with two variables from which unique values of C and A cannot be determined, even under the assumptions that C and A are integers such that $C + A \geq 10$. For example, the values of C and A could be 8 and 6, respectively, since $(48)(8) - (55)(6) = 54$, or the values of C and A could be 63 and 54, respectively, since $(48)(63) - (55)(54) = 54$; NOT sufficient.

Taking (1) and (2) together, it follows that

$$A = \frac{29{,}700}{(0.9)(5{,}500)} = 6 \text{ and}$$

$$C = \frac{(0.9)(5{,}500)(6) + 4{,}860}{(0.9)(4{,}800)} = 8.$$

The correct answer is C;
both statements together are sufficient.



DS16536

346. What is the area of triangular region *ABC* above?

(1) The product of *BD* and *AC* is 20.

(2) $x = 45$

Geometry Triangles; Area

The area of $\Delta ABC = \frac{BD \times AC}{2}$.

(1) The product of *BD* and *AC* is given as 20, so the area of ΔABC is $\frac{20}{2}$ or 10; SUFFICIENT.

(2) With the measurement of x being 45, it is concluded that ΔABD is a 45–45–90 right triangle, where the length of side *BD* is equal to the length of side *AD*. However, with no lengths of any side known, there is not enough information to calculate the area; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS05265

347. In the *xy*-coordinate plane, is point *R* equidistant from points (–3,–3) and (1,–3) ?

(1) The *x*-coordinate of point *R* is –1.

(2) Point *R* lies on the line $y = -3$.

Algebra Coordinate geometry

Determine if point *R* is equidistant from (–3,–3) and (1,–3). Letting $R = (x,y)$, then *R* will be equidistant from (–3,–3) and (1,–3) if and only if *R* lies on the perpendicular bisector of the line segment with endpoints (–3,–3) and (1,–3), or equivalently, if and only if *R* lies on the vertical line that consists of all points with *x*-coordinate equal to $\frac{-3+1}{2} = -1$. Therefore, determine if $x = -1$.

(1) Given that $x = -1$, then *R* will be equidistant from (–3,–3) and (1,–3); SUFFICIENT.

(2) Given that $y = -3$, then both $x = -1$ and $x \neq -1$ are possible; NOT sufficient.

Alternatively, letting $R = (x,y)$, then *R* will be equidistant from (–3,–3) and (1,–3) if and only if the distance between (x,y) and (–3,–3) is the same as the distance between (x,y) and (1,–3), or if and only if $\sqrt{(x+3)^2 + (y+3)^2} = \sqrt{(x-1)^2 + (y+3)^2}$ or if and only if $(x+3)^2 + (y+3)^2 = (x-1)^2 + (y+3)^2$, or if and only if $(x+3)^2 = (x-1)^2$.

(1) Given that $x = -1$, then $(-1+3)^2 = 4$ and $(-1-1)^2 = 4$; SUFFICIENT.

(2) Given that $y = -3$, then it is impossible to determine if $(x+3)^2 = (x-1)^2$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS09603

348. What is the ratio of the average (arithmetic mean) height of students in class X to the average height of students in class Y ?

(1) The average height of the students in class X is 120 centimeters.

(2) The average height of the students in class X and class Y combined is 126 centimeters.

Arithmetic Statistics

(1) Given that the average height of the students in class X is 120 cm, but given no information about the average height of the students in class Y, the desired ratio cannot be determined; NOT sufficient.

(2) Given that the average height of the students in class X and class Y combined is 126 cm, the ratio of the averages of the individual classes cannot be determined. For example, if class X consists of 10 students, each of whom has height 120 cm, and class Y consists of 10 students, each of whom has height 132 cm, then the average height of the students in class X and class Y combined is $\frac{10(120)+10(132)}{20} = 126$ cm

and the ratio of the averages of the individual classes is $\frac{120}{132}$. However, if class X consists of 10 students, each of whom has height 120 cm and class Y consists of 20 students, each of whom has height 129 cm, then the average height of the students in class X and class Y combined is $\frac{10(120)+20(129)}{30} = 126$ cm and the ratio of the averages of the individual classes is $\frac{120}{129}$; NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (2) is not sufficient include the information from (1).

The correct answer is E;
both statements together are still not sufficient.

DS04631

349. Is the positive two-digit integer N less than 40 ?

(1) The units digit of N is 6 more than the tens digit.

(2) N is 4 less than 4 times the units digit.

Arithmetic Place value

Determine if the two-digit integer N is less than 40. Letting the tens digit be t and the units digit be u, then $N = 10t + u$. Determine if $10t + u < 40$.

(1) Given that $u = t + 6$, then $N = 10t + (t + 6) = 11t + 6$. Since u is a digit, $u = t + 6 \leq 9$, so $t \leq 3$. Therefore, $N = 11t + 6 \leq 11(3) + 6 = 39$; SUFFICIENT.

(2) Given that $N = 4u - 4$, then since u is a digit and $u \leq 9$, it follows that $N = 4u - 4 \leq 4(9) - 4 = 32$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06318

350. If $2^{x+y} = 4^8$, what is the value of y ?

(1) $x^2 = 81$

(2) $x - y = 2$

Algebra Exponents

Since $4^8 = (2^2)^8 = 2^{16}$, the equation $2^{x+y} = 4^8$ becomes $2^{x+y} = 2^{16}$, which is equivalent to $x + y = 16$.

(1) Given that $x^2 = 81$, then both $x = 9$ and $x = -9$ are possible. Therefore, $y = 7$ is possible (choose $x = 9$) and $y = 25$ is possible (choose $x = -9$); NOT sufficient.

(2) Given that $x - y = 2$, it follows that $-x + y = -2$. Adding the last equation to $x + y = 16$ gives $2y = 14$, or $y = 7$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS03680

351. Each week a certain salesman is paid a fixed amount equal to \$300, plus a commission equal to 5 percent of the amount of his sales that week over \$1,000. What is the total amount the salesman was paid last week?

(1) The total amount the salesman was paid last week is equal to 10 percent of the amount of his sales last week.

(2) The salesman's sales last week totaled \$5,000.

Algebra Applied problems

Let P be the salesman's pay for last week and let S be the amount of his sales last week. Then $P = 300 + 0.05(S - 1{,}000)$. Determine the value of P.

(1) Given $P = 0.10S$, then $0.10S = 300 + 0.05(S - 1{,}000)$. This equation can be solved for a unique value of S, from which the value of P can be determined; SUFFICIENT.

(2) Given $S = 5{,}000$, then $P = 300 + 0.05(5{,}000 - 1{,}000)$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS01383

352. At a bakery, all donuts are priced equally and all bagels are priced equally. What is the total price of 5 donuts and 3 bagels at the bakery?

(1) At the bakery, the total price of 10 donuts and 6 bagels is $12.90.

(2) At the bakery, the price of a donut is $0.15 less than the price of a bagel.

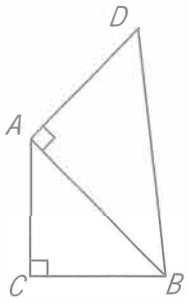
Algebra Simultaneous equations

Let x be the price, in dollars, of each donut and let y be the price, in dollars, of each bagel. Find the value of $5x + 3y$.

(1) Given that $10x + 6y = 12.90$, since $5x + 3y = \frac{1}{2}(10x + 6y)$, it follows that $5x + 3y = \frac{1}{2}(12.90)$; SUFFICIENT.

(2) Given that $x = y - 0.15$, then $5x + 3y = 5(y - 0.15) + 3y = 8y - 0.75$, which varies as y varies; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.



DS06869

353. In the figure above, is the area of triangular region *ABC* equal to the area of triangular region *DBA*?

(1) $(AC)^2 = 2(AD)^2$

(2) ΔABC is isosceles.

Geometry Triangles; Area

Determine whether $\frac{1}{2}(AC)(CB) = \frac{1}{2}(AD)(AB)$ or, equivalently, whether $(AC)(CB) = (AD)(AB)$.

(1) Given that $(AC)^2 = 2(AD)^2$, then $\sqrt{2}$ and 1 could be the values of AC and AD, respectively. If, for example, $CB = \sqrt{2}$, then $AB = \sqrt{\left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2} = 2$ by the Pythagorean theorem applied to ΔABC. Hence, $(AC)(CB) = (\sqrt{2})(\sqrt{2})$ and $(AD)(AB) = (1)(2)$, from which it follows that $(AC)(CB) = (AD)(AB)$. On the other hand, if $CB = 1$, then $AB = \sqrt{\left(\sqrt{2}\right)^2 + 1^2} = \sqrt{3}$ by the Pythagorean theorem applied to ΔABC. Hence, $(AC)(CB) = (\sqrt{2})(1)$ and $(AD)(AB) = (1)(\sqrt{3})$, from which it follows that $(AC)(CB) \neq (AD)(AB)$; NOT sufficient.

(2) Given that ΔABC is isosceles then, by varying the value of AD, the area of ΔABC may or may not be equal to the area of ΔDBA. For example, if $AC = CB = 1$, then $(AC)(CB) = (1)(1) = 1$ and $AB = \sqrt{2}$ by the Pythagorean theorem applied to ΔABC. If $AD = \frac{1}{\sqrt{2}}$, then $(AD)(AB) = \left(\frac{1}{\sqrt{2}}\right)(\sqrt{2}) = 1$, and $(AC)(CB) = (AD)(AB)$. On the other hand, if $AD = 1$, $(AD)(AB) = (1)(\sqrt{2}) = \sqrt{2}$ and $(AC)(CB) \neq (AD)(AB)$; NOT sufficient.

Given (1) and (2) together, let $AC = CB = x$ be the length of the legs of the isosceles triangle ABC. Then, from $(AC)^2 = 2(AD)^2$, it follows that $x^2 = 2(AD)^2$, and hence $AD = \frac{x}{\sqrt{2}}$. Also, by the Pythagorean theorem applied to ΔABC, it follows that $AB = \sqrt{x^2 + x^2} = x\sqrt{2}$. Therefore, it follows that $(AC)(CB) = (x)(x) = x^2$ and $(AD)(AB) = \left(\frac{x}{\sqrt{2}}\right)\left(x\sqrt{2}\right) = x^2$, and so $(AC)(CB) = (AD)(AB)$.

The correct answer is C;
both statements together are sufficient.

DS08105

354. If r and s are positive integers, can the fraction $\frac{r}{s}$ be expressed as a decimal with only a finite number of nonzero digits?

(1) s is a factor of 100.

(2) r is a factor of 100.

Arithmetic Properties of numbers

Determine if $\frac{r}{s}$, where r and s are positive integers, can be expressed as a decimal with a finite number of nonzero decimal digits.

(1) It is given that s is a factor of 100 and so $s = 1, 2, 4, 5, 10, 20, 25, 50$, or 100. This means that $\frac{r}{s}$ must be one of the quotients $\frac{r}{1}, \frac{r}{2}, \frac{r}{4}, \frac{r}{5}, \frac{r}{10}, \frac{r}{20}, \frac{r}{25}, \frac{r}{50}$ or $\frac{r}{100}$. Thus, $\frac{r}{s}$ must be one of the products $r(1)$, $r(0.5)$, $r(0.25)$, $r(0.20)$, $r(0.1)$, $r(0.05)$ $r(0.04)$, $r(0.02)$, or $r(0.01)$. In each case, $\frac{r}{s}$ is the product of an integer and a decimal with a finite number of nonzero digits, and hence, $\frac{r}{s}$ can be expressed as a decimal with a finite number of nonzero digits. In fact, it suffices to note this is true for $\frac{r}{100}$, since each of the other possibilities is a positive integer times $\frac{r}{100}$ (for example, $\frac{r}{20}$ is 5 times $\frac{r}{100}$); SUFFICIENT.

(2) It is given that r is a factor of 100. If $r = 4$ and $s = 5$, then $\frac{r}{s} = 0.8$, which is a decimal with a finite number of nonzero digits. On the other hand, if $r = 4$ and $s = 7$, then $\frac{r}{s} = 0.\overline{571428}$, which is not a decimal with a finite number of nonzero digits; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS16384

355. If $r > 0$ and $s > 0$, is $\frac{r}{s} < \frac{s}{r}$?

(1) $\frac{r}{3s} = \frac{1}{4}$

(2) $s = r + 4$

Algebra Ratios

Given positive numbers r and s, determine if $\frac{r}{s} < \frac{s}{r}$.

(1) If $\frac{r}{3s} = \frac{1}{4}$, then $\frac{r}{s} = \frac{3}{4}$, $\frac{s}{r} = \frac{4}{3}$, and so $\frac{r}{s} < \frac{s}{r}$, since $\frac{3}{4} < \frac{4}{3}$; SUFFICIENT.

(2) If $s = r + 4$, then $\frac{r}{s} = \frac{r}{r+4}$ and $\frac{s}{r} = \frac{r+4}{r}$. Since $r + 4 > r$, $\frac{r}{r+4} < 1$ and $\frac{r+4}{r} > 1$, so $\frac{r}{s} < \frac{s}{r}$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06789

356. If k is an integer such that $56 < k < 66$, what is the value of k?

(1) If k were divided by 2, the remainder would be 1.

(2) If $k + 1$ were divided by 3, the remainder would be 0.

Arithmetic Properties of integers

Determine the value of the integer k, where $56 < k < 66$.

(1) It is given that the remainder is 1 when k is divided by 2, which implies that k is odd. Therefore, the value of k can be 57, 59, 61, 63, or 65; NOT sufficient.

(2) It is given that the remainder is 0 when $k + 1$ is divided by 3, which implies that $k + 1$ is divisible by 3. Since $56 < k < 66$ (equivalently, $57 < k + 1 < 67$), the value of $k + 1$ can be 60, 63, or 66 so the value of k can be 59, 62, or 65; NOT sufficient.

Taking (1) and (2) together, 59 and 65 appear in both lists of possible values for k; NOT sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS13965

357. If x is a positive integer, then is x prime?

(1) $3x + 1$ is prime.

(2) $5x + 1$ is prime.

Arithmetic Properties of numbers

Determine if the positive integer x is prime.

(1) This indicates that $3x + 1$ is prime. If $x = 2$, then $3x + 1 = 7$ is prime and so is x. However, if $x = 6$, then $3x + 1 = 19$ is prime, but 6 is not prime; NOT sufficient.

(2) This indicates that $5x + 1$ is prime. If $x = 2$, then $5x + 1 = 11$ is prime and so is x. However, if $x = 6$, then $5x + 1 = 31$ is prime, but 6 is not prime; NOT sufficient.

Because the same examples were used to establish that neither (1) nor (2) is sufficient, it is not possible to determine whether x is prime from the given information.

The correct answer is E;
both statements together are still not sufficient.

$k, n, 12, 6, 17$

DS00172

358. What is the value of n in the list above?

(1) $k < n$

(2) The median of the numbers in the list is 10.

Arithmetic Statistics

Given the list $k, n, 12, 6, 17$, determine the value of n.

(1) Although $k < n$, no information is given about the value of k or n; NOT sufficient.

(2) Since the median of the numbers in the list is 10 and there are 5 numbers in the list, 10 is one of those 5 numbers. Therefore, $n = 10$ or $k = 10$. If $n = 10$, then the value of n has been determined. However, if $k = 10$, then n can be any number that is 10 or less, so the value of n cannot be determined; NOT sufficient.

Taking (1) and (2) together, if $k < n$ and the median of the list is 10, then 12 and 17 are to the right of the median and the list in ascending order is either $6, k, n, 12, 17$ or $k, 6, n, 12, 17$. In either case, n is the middle number, and since the median is 10, $n = 10$.

The correct answer is C;
both statements together are sufficient.

DS07508

359. If x and y are integers, what is the value of $x + y$?

(1) $3 < \frac{x+y}{2} < 4$

(2) $2 < x < y < 5$

Arithmetic Computation with integers

Determine the value of $x + y$ for integers x and y.

(1) Given that $3 < \frac{x+y}{2} < 4$, then $6 < x + y < 8$. Since $x + y$ is an integer and 7 is the only integer greater than 6 and less than 8, it follows that the value of $x + y$ is 7; SUFFICIENT.

(2) Given that $2 < x < y < 5$, then $2 < x < 5$, and hence $x = 3$ or $x = 4$. Likewise, $2 < y < 5$, and hence $y = 3$ or $y = 4$. With the additional restriction that $x < y$, it follows that $x = 3$ and $y = 4$, and thus $x + y = 7$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS00764

360. Last year, if Arturo spent a total of \$12,000 on his mortgage payments, real estate taxes, and home insurance, how much did he spend on his real estate taxes?

(1) Last year, the total amount that Arturo spent on his real estate taxes and home insurance was $33\frac{1}{3}$ percent of the amount that he spent on his mortgage payments.

(2) Last year, the amount that Arturo spent on his real estate taxes was 20 percent of the total amount he spent on his mortgage payments and home insurance.

Arithmetic Applied problems

Let M, R, and H be the amounts that Arturo spent last year on mortgage payments, real estate taxes, and home insurance, respectively. Given that $M + R + H = 12{,}000$, determine the value of R.

(1) Given that $R + H = \frac{1}{3}M$ and $M + R + H = 12{,}000$, then $M + \frac{1}{3}M = 12{,}000$, or $M = 9{,}000$. However, the value of R cannot be determined, since it is possible that $R = 2{,}000$ (use $M = 9{,}000$ and $H = 1{,}000$) and it is possible that $R = 1{,}000$ (use $M = 9{,}000$ and $H = 2{,}000$); NOT sufficient.

(2) Given that $R = \frac{1}{5}(M + H)$, or $5R = M + H$ and $M + R + H = 12{,}000$, which can be rewritten as $(M + H) + R = 12{,}000$, then $5R + R = 12{,}000$, or $R = 2{,}000$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS06038

361. If a, b, c, and d are positive numbers, is $\frac{a}{b} < \frac{c}{d}$?

(1) $0 < \frac{c-a}{d-b}$

(2) $\left(\frac{ad}{bc}\right)^2 < \frac{ad}{bc}$

Algebra Inequalities

Determine whether $\frac{a}{b} < \frac{c}{d}$, where a, b, c, and d are positive numbers.

(1) Given that $0 < \frac{c-a}{d-b}$, then $a = 2$, $b = 3$, $c = 6$, and $d = 8$ are possible values for a, b, c, and d because $\frac{c-a}{d-b} = \frac{6-2}{8-3} = \frac{4}{5}$ and $\frac{4}{5} > 0$. For these values, $\frac{a}{b} < \frac{c}{d}$ is true because $\frac{2}{3} < \frac{6}{8}$. On the other hand, $a = 4$, $b = 6$, $c = 2$, and $d = 3$ are also possible values of a, b, c, and d because $\frac{c-a}{d-b} = \frac{2-4}{3-6} = \frac{2}{3}$ and $\frac{2}{3} > 0$. For these values, $\frac{a}{b} < \frac{c}{d}$ is false because $\frac{4}{6} = \frac{2}{3}$; NOT sufficient.

(2) Given that $\left(\frac{ad}{bc}\right)^2 < \frac{ad}{bc}$, then

$\frac{ad}{bc} < 1$ dividing both sides by the positive number $\frac{ad}{bc}$

$\frac{ad}{b} < c$ multiplying both sides by the positive number c

$\frac{a}{b} < \frac{c}{d}$ dividing both sides by the positive number d; SUFFICIENT

The correct answer is B;
statement 2 alone is sufficient.

DS12008

362. Is the number of members of Club X greater than the number of members of Club Y ?

(1) Of the members of Club X, 20 percent are also members of Club Y.

(2) Of the members of Club Y, 30 percent are also members of Club X.

Arithmetic Sets

Let a be the number of members in Club X that do not belong to Club Y, let b be the number of members in Club Y that do not belong to Club X, and let c be the number of members that belong to both Club X and to Club Y. Determine whether $a + c > b + c$, or equivalently, whether $a > b$.

(1) If $a = 80$, $b = 79$, and $c = 20$, then 20 percent of the members of Club X are also members of Club Y (because $c = 20$ is 20 percent of $a + c = 100$) and $a > b$ is true. However, if $a = 80$, $b = 80$, and $c = 20$, then 20 percent of the members of Club X are also members of Club Y (because $c = 20$ is 20 percent of $a + c = 100$) and $a > b$ is false. Therefore, it cannot be determined whether $a > b$; NOT sufficient.

(2) If $a = 71$, $b = 70$, and $c = 30$, then 30 percent of the members of Club Y are also members of Club X (because $c = 30$ is 30 percent of $b + c = 100$) and $a > b$ is true. However, if $a = 70$, $b = 70$, and $c = 30$, then 30 percent of the members of Club Y are also members of Club X (because $c = 30$ is 30 percent of $b + c = 100$) and $a > b$ is false. Therefore, it cannot be determined whether $a > b$; NOT sufficient.

Now assume both (1) and (2). From (1) it follows that $\frac{c}{a+c} = 0.20 = \frac{1}{5}$, or $5c = a + c$, and so $a = 4c$. From (2) it follows that $\frac{c}{b+c} = 0.30 = \frac{3}{10}$, or $10c = 3b + 3c$, and so $7c = 3b$ and $b = \frac{7}{3}c$. Since $4c > \frac{7}{3}c$ (from the statements it can be deduced that $c > 0$), it follows that $a > b$. Therefore, (1) and (2) together are sufficient.

The correct answer is C;
both statements together are sufficient.



DS16361

363. On the number line above, p, q, r, s, and t are five consecutive even integers in increasing order. What is the average (arithmetic mean) of these five integers?

(1) $q + s = 24$

(2) The average (arithmetic mean) of q and r is 11.

Arithmetic Properties of numbers

Since p, q, r, s, and t are consecutive even integers listed in numerical order, the 5 integers can also be given as p, $p+2$, $p+4$, $p+6$, and $p+8$. Determine the average of these 5 integers, which is the value of $\frac{p+(p+2)+(p+4)+(p+6)+(p+8)}{5} = \frac{5p+20}{5} = p+4$

(1) Given that $q + s = 24$, then $(p + 2) + (p + 6) = 24$. Therefore, $2p + 8 = 24$, or $p = 8$, and hence $p + 4 = 12$; SUFFICIENT.

(2) Given that $\frac{q+r}{2} = 11$, then $q + r = (2)(11) = 22$, or $(p + 2) + (p + 4) = 22$. Therefore, $2p + 6 = 22$, or $p = 8$, and hence $p + 4 = 12$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS06657

364. If $\lceil x \rceil$ denotes the least integer greater than or equal to x, is $\lceil x \rceil = 0$?

(1) $-1 < x < 1$

(2) $x < 0$

Algebra Functions

Determine if $\lceil x \rceil$, the least integer greater than or equal to x, is equal to 0, which is the same as determining if x satisfies $-1 < x \leq 0$.

(1) Given that $-1 < x < 1$, then it is possible that $\lceil x \rceil = 0$ (for example, if $x = 0$, then $\lceil x \rceil = 0$) and it is possible that $\lceil x \rceil \neq 0$ (for example, if $x = \frac{1}{2}$, then $\lceil x \rceil = 1 \neq 0$); NOT sufficient.

(2) Given that $x < 0$, then it is possible that $\lceil x \rceil = 0$ (for example, if $x = -\frac{1}{2}$, then $\lceil x \rceil = 0$) and it is possible that $\lceil x \rceil \neq 0$ (for example, if $x = -5$, then $\lceil x \rceil = -5 \neq 0$); NOT sufficient.

Taking (1) and (2) together gives $-1 < x < 0$, which implies that $\lceil x \rceil = 0$.

The correct answer is C;
both statements together are sufficient.

DS12718

365. If x and y are integers, is $x > y$?

(1) $x + y > 0$

(2) $y^x < 0$

Arithmetic Properties of integers

Determine if the integer x is greater than the integer y.

(1) It is given that $x + y > 0$, and so $-x < y$. If, for example, $x = -3$ and $y = 4$, then $x + y = -3 + 4 = 1 > 0$ and $x < y$. On the other hand, if $x = 4$ and $y = -3$, then $x + y = 4 - 3 = 1 > 0$ and $x > y$; NOT sufficient.

(2) It is given that $y^x < 0$, so $y < 0$. If, for example, $x = 3$ and $y = -2$, then $(-2)^3 = -8 < 0$ and $x > y$. On the other hand, if $x = -3$ and $y = -2$, then $(-2)^{-3} = -\frac{1}{8} < 0$ and $x < y$; NOT sufficient.

Taking (1) and (2) together, from (2) y is negative and from (1) $-x$ is less than y. Therefore, $-x$ is negative, and hence x is positive. Since x is positive and y is negative, it follows that $x > y$.

The correct answer is C;
both statements together are sufficient.

DS03046

366. If r and s are the roots of the equation $x^2 + bx + c = 0$, where b and c are constants, is $rs < 0$?

(1) $b < 0$

(2) $c < 0$

Algebra Second-degree equations

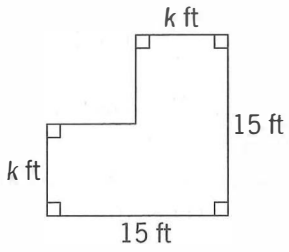
Determine whether the product of the roots to $x^2 + bx + c = 0$, where b and c are constants, is negative.

If r and s are the roots of the given equation, then $(x - r)(x - s) = x^2 + bx + c$. This implies that $x^2 - (r + s)x + rs = x^2 + bx + c$, and so $rs = c$. Therefore, rs is negative if and only if c is negative.

(1) Given that $b < 0$, then c could be negative or positive. For example, if $b = -1$ and $c = -6$, then the given equation would be $x^2 - x - 6 = (x - 3)(x + 2) = 0$, and the product of its roots would be $(3)(-2)$, which is negative. On the other hand, if $b = -6$ and $c = 5$, then the given equation would be $x^2 - 6x + 5 = (x - 5)(x - 1) = 0$, and the product of its roots would be $(5)(1)$, which is positive; NOT sufficient.

(2) Given that $c < 0$, it follows from the explanation above that $rs < 0$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.



DS02888

367. The figure above represents an L-shaped garden. What is the value of k?

(1) The area of the garden is 189 square feet.

(2) The perimeter of the garden is 60 feet.

Geometry Polygons

(1) Given that the area of the garden is 189 ft^2, and observing that area of the garden can be calculated by imagining the garden as a square with dimensions 15 ft by 15 ft from which its upper-right square corner with dimensions $(15 - k)$ ft by $(15 - k)$ ft is removed, it follows that $189 = (15)^2 - (15 - k)^2$. Therefore, $(15 - k)^2 = 225 - 189 = 36$, so $15 - k = 6$ or $15 - k = -6$. The figure implies $15 - k > 0$, so it follows that $15 - k = 6$, or $k = 9$; SUFFICIENT.

(2) Given that the perimeter of the garden is 60 ft, and because $15 + 15 + k + (15 - k) + (15 - k) + k = 60$, any value of k between 0 and 15 is possible, since for any such value of k the sum of the lengths of all the sides would be 60 ft; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS01049

368. The only articles of clothing in a certain closet are shirts, dresses, and jackets. The ratio of the number of shirts to the number of dresses to the number of jackets in the closet is 9:4:5, respectively. If there are more than 7 dresses in the closet, what is the total number of articles of clothing in the closet?

(1) The total number of shirts and jackets in the closet is less than 30.

(2) The total number of shirts and dresses in the closet is 26.

Arithmetic Ratio and proportion

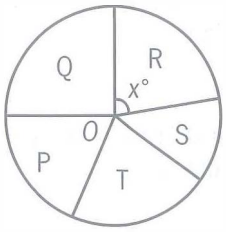
Letting s, d, and j represent, respectively, the numbers of shirts, dresses, and jackets in the closet, then $s = 9x$, $d = 4x$, and $j = 5x$, where x is a positive integer. It is given that $4x > 7$, and so $4x \geq 8$ or $x \geq 2$ since x is an integer. Determine the value of $9x + 4x + 5x = 18x$.

(1) This indicates that $9x + 5x < 30$, and so $14x \leq 28$ or $x \leq 2$ since x is an integer. It follows from $x \geq 2$ and $x \leq 2$ that $x = 2$ and $18x = 36$; SUFFICIENT.

(2) This indicates that $9x + 4x = 26$ or $13x = 26$, or $x = 2$. It follows that $18x = 36$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

TOTAL EXPENSES FOR THE FIVE DIVISIONS OF COMPANY H



DS16542

369. The figure above represents a circle graph of Company H's total expenses broken down by the expenses for each of its five divisions. If *O* is the center of the circle and if Company H's total expenses are \$5,400,000, what are the expenses for Division R ?

(1) $x = 94$

(2) The total expenses for Divisions S and T are twice as much as the expenses for Division R.

Geometry Circles

In this circle graph, the expenses of Division R are equal to the value of $\frac{x}{360}$ multiplied by \$5,400,000, or \$15,000x. Therefore, it is necessary to know the value of x in order to determine the expenses for Division R.

(1) The value of x is given as 94, so the expenses of Division R can be determined; SUFFICIENT.

(2) This gives a comparison among the expenses of some of the divisions of Company H, but no information is given about the value of x; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS13641

370. If x is negative, is $x < -3$?

(1) $x^2 > 9$

(2) $x^3 < -9$

Algebra Properties of numbers

(1) Given that $x^2 > 9$, it follows that $x < -3$ or $x > 3$, a result that can be obtained in a variety of ways. For example, consider the equivalent inequality $(|x|)^2 > 9$ that reduces to $|x| > 3$, or consider when the two factors of $x^2 - 9$ are both positive and when the two factors of $x^2 - 9$ are both negative, or consider where the graph of the parabola $y = x^2 - 9$ is above the x-axis, etc. Since it is also given that x is negative, it follows that $x < -3$; SUFFICIENT.

(2) Given that $x^3 < -9$, if $x = -4$, then $x^3 = -64$, and so $x^3 < -9$ and it is true that $x < -3$. However, if $x = -3$, then $x^3 = -27$, and so $x^3 < -9$, but it is not true that $x < -3$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS04897

371. What is the number of cans that can be packed in a certain carton?

(1) The interior volume of this carton is 2,304 cubic inches.

(2) The exterior of each can is 6 inches high and has a diameter of 4 inches.

Geometry Rectangular solids and cylinders

(1) No information about the size of the cans is given; NOT sufficient.

(2) No information about the size of the carton is given; NOT sufficient.

Taking (1) and (2) together, there is still not enough information to answer the question. If the carton is a rectangular solid that is 1 inch by 1 inch by 2,304 inches and the cans are cylindrical with the given dimensions, then 0 cans can be packed into the carton. However, if the carton is a rectangular solid that is 16 inches by 12 inches by 12 inches and the cans are cylindrical with the given dimensions, then 1 or more cans can be packed into the carton.

The correct answer is E; both statements together are still not sufficient.

r	s	t
u	v	w
x	y	z

DS08301

372. Each of the letters in the table above represents one of the numbers 1, 2, or 3, and each of these numbers occurs exactly once in each row and exactly once in each column. What is the value of r?

(1) $v + z = 6$

(2) $s + t + u + x = 6$

Arithmetic Properties of numbers

In the following discussion, "row/column convention" means that each of the numbers 1, 2, and 3 appears exactly once in any given row and exactly once in any given column.

(1) Given that $v + z = 6$, then both v and z are equal to 3, since no other sum of the possible values is equal to 6. Applying the row/column convention to row 2, and then to row 3, it follows that neither u nor x can be 3. Since neither u nor x can be 3, the row/column convention applied to column 1 forces r to be 3; SUFFICIENT.

(2) If $u = 3$, then $s + t + x = 3$. Hence, $s = t = x = 1$, since the values these variables can have does not permit another possibility. However, this assignment of values would violate the row/column convention for row 1, and thus u cannot be 3. If $x = 3$, then $s + t + u = 3$. Hence, $s = t = u = 1$, since the values these variables can have does not permit another possibility. However, this assignment of values would violate the row/column convention for row 1, and thus x cannot be 3. Since neither u nor x can be 3, the row/column convention applied to column 1 forces r to be 3; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS00328

373. Material A costs \$3 per kilogram, and Material B costs \$5 per kilogram. If 10 kilograms of Material K consists of x kilograms of Material A and y kilograms of Material B, is $x > y$?

(1) $y > 4$

(2) The cost of the 10 kilograms of Material K is less than \$40.

Algebra Inequalities

Since $x + y = 10$, the relation $x > y$ is equivalent to $x > 10 - x$, or $x > 5$.

(1) The given information is consistent with $x = 5.5$ and $y = 4.5$, and the given information is also consistent with $x = y = 5$. Therefore, it is possible for $x > y$ to be true and it is possible for $x > y$ to be false; NOT sufficient.

(2) Given that $3x + 5y < 40$, or $3x + 5(10 - x) < 40$, then $3x - 5x < 40 - 50$. It follows that $-2x < -10$, or $x > 5$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS16164

374. At what speed was a train traveling on a trip when it had completed half of the total distance of the trip?

(1) The trip was 460 miles long and took 4 hours to complete.

(2) The train traveled at an average rate of 115 miles per hour on the trip.

Arithmetic Applied problems

Determine the speed of the train when it had completed half the total distance of the trip.

(1) Given that the train traveled 460 miles in 4 hours, the train could have traveled at the constant rate of 115 miles per hour for 4 hours, and thus it could have been traveling 115 miles per hour when it had completed half the total distance of the trip. However, the train could have traveled 150 miles per hour for the first 2 hours (a distance of 300 miles) and 80 miles per hour for the last 2 hours (a distance of 160 miles), and thus it could have been traveling 150 miles per hour when it had completed half the total distance of the trip; NOT sufficient.

(2) Given that the train traveled at an average rate of 115 miles per hour, each of the possibilities given in the explanation for (1) could occur, since 460 miles in 4 hours gives an average speed of $\frac{460}{4} = 115$ miles per hour; NOT sufficient.

Assuming (1) and (2), each of the possibilities given in the explanation for (1) could occur. Therefore, (1) and (2) together are not sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS12047

375. Tom, Jane, and Sue each purchased a new house. The average (arithmetic mean) price of the three houses was $120,000. What was the median price of the three houses?

(1) The price of Tom's house was $110,000.

(2) The price of Jane's house was $120,000.

Arithmetic Statistics

Let T, J, and S be the purchase prices for Tom's, Jane's, and Sue's new houses. Given that the average purchase price is 120,000, or $T + J + S = 3(120{,}000)$, determine the median purchase price.

(1) Given $T = 110{,}000$, the median could be 120,000 (if $J = 120{,}000$ and $S = 130{,}000$) or 125,000 (if $J = 125{,}000$ and $S = 125{,}000$); NOT sufficient.

(2) Given $J = 120{,}000$, the following two cases include every possibility consistent with $T + J + S = (3)(120{,}000)$, or $T + S = (2)(120{,}000)$.

(i) $T = S = 120{,}000$

(ii) One of T or S is less than 120,000 and the other is greater than 120,000.

In each case, the median is clearly 120,000; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS13958

376. What is the value of x if $x^3 < x^2$?

(1) $-2 < x < 2$

(2) x is an integer greater than –2.

Algebra Inequalities

The inequality $x^3 < x^2$ is equivalent to $x^3 - x^2 < 0$, or $x^2(x - 1) < 0$. Since this inequality is false for $x = 0$, it follows that $x \neq 0$, and hence $x^2 > 0$. Therefore, $x^2(x - 1) < 0$ can only hold if $x - 1 < 0$, or if $x < 1$. Thus, the problem is equivalent to determining the value of x given that $x \neq 0$ and $x < 1$.

(1) Given that $-2 < x < 2$, it is not possible to determine the value of x. For example, the value of x could be –1 (note that $-1 < 1$) and the value of x could be 0.5 (note that $0.125 < 0.25$); NOT sufficient.

(2) Given that the value of x is an integer greater than –2, then the value of x must be among the integers $-1, 0, 1, 2, 3, \ldots$. However, from the discussion above, $x \neq 0$ and $x < 1$, so the value of x can only be –1; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS08451

377. For any integers x and y, min(x, y) and max(x, y) denote the minimum and the maximum of x and y, respectively. For example, min(5, 2) = 2 and max(5, 2) = 5. For the integer w, what is the value of min(10, w) ?

(1) $w = \max(20, z)$ for some integer z.

(2) $w = \max(10, w)$

Arithmetic Properties of numbers

If $w \geq 10$, then $(10, w) = 10$, and if $w < 10$, then $(10, w) = w$. Therefore, the value of $\min(10, w)$ can be determined if the value of w can be determined.

(1) Given that $w = \max(20, z)$ then $w \geq 20$. Hence, $w \geq 10$, and so $\min(10, w) = 10$; SUFFICIENT.

(2) Given that $w = \max(10, w)$, then $w \geq 10$, and so $\min(10, w) = 10$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS01473

378. A certain bookcase has 2 shelves of books. On the upper shelf, the book with the greatest number of pages has 400 pages. On the lower shelf, the book with the least number of pages has 475 pages. What is the median number of pages for all of the books on the 2 shelves?

(1) There are 25 books on the upper shelf.

(2) There are 24 books on the lower shelf.

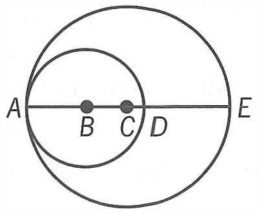
Arithmetic Statistics

(1) The information given says nothing about the number of books on the lower shelf. If there are fewer than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the upper shelf or the average number of pages in two books on the upper shelf. Hence, the median will be at most 400. If there are more than 25 books on the lower shelf, then the median number of pages will be the number of pages in one of the books on the lower shelf or the average number of pages in two books on the lower shelf. Hence, the median will be at least 475; NOT sufficient.

(2) An analysis very similar to that used in (1) shows the information given is not sufficient to determine the median; NOT sufficient.

Given both (1) and (2), it follows that there is a total of 49 books. Therefore, the median will be the 25th book when the books are ordered by number of pages. Since the 25th book in this ordering is the book on the upper shelf with the greatest number of pages, the median is 400. Therefore, (1) and (2) together are sufficient.

The correct answer is C;
both statements together are sufficient.



DS12070

379. In the figure above, points A, B, C, D, and E lie on a line. A is on both circles, B is the center of the smaller circle, C is the center of the larger circle, D is on the smaller circle, and E is on the larger circle. What is the area of the region inside the larger circle and outside the smaller circle?

(1) $AB = 3$ and $BC = 2$

(2) $CD = 1$ and $DE = 4$

Geometry Circles

If R is the radius of the larger circle and r is the radius of the smaller circle, then the desired area is $\pi R^2 - \pi r^2$. Thus, if both the values of R and r can be determined, then the desired area can be determined.

(1) Given that $AB = r = 3$ and $BC = 2$, then $AB + BC = R = 3 + 2 = 5$; SUFFICIENT.

(2) Given that $CD = 1$ and $DE = 4$, then $CD + DE = R = 1 + 4 = 5$. Since $\overline{AE}$ is a diameter of the larger circle, then $AD + DE = 2R$. Also, since $\overline{AD}$ is a diameter of the smaller circle, then $AD = 2r$. Thus, $2r + DE = 2R$ or $2r + 4 = 10$, and so $r = 3$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS08995

380. In planning for a trip, Joan estimated both the distance of the trip, in miles, and her average speed, in miles per hour. She accurately divided her estimated distance by her estimated average speed to obtain an estimate for the time, in hours, that the trip would take. Was her estimate within 0.5 hour of the actual time that the trip took?

(1) Joan's estimate for the distance was within 5 miles of the actual distance.

(2) Joan's estimate for her average speed was within 10 miles per hour of her actual average speed.

Arithmetic Applied problems; Estimating

(1) Given that Joan's estimate for the distance was within 5 miles of the actual distance, it is not possible to determine whether her estimate for the time was within 0.5 hour without information about her estimated average speed. For example, if her estimated distance was 20 miles and was within 5 miles of the actual distance, then the actual distance would be between 15 and 25 miles. If her estimated speed was 20 miles per hour (mph) and was within, say, 10 mph of her actual speed, then her actual speed would be between 10 and 30 mph. Her estimated time would then be $\frac{20}{20} = 1.0$ hour and the actual time would be between $\frac{15}{30} = 0.5$ hour (least distance over greatest speed) and $\frac{25}{10} = 2.5$ hours (greatest distance over least speed). Since 1.0 hour is between 0.5 hour and 2.5 hours, her estimate for the time could equal the actual time, and thus it is possible that her estimate of the time is within 0.5 hour of the actual time. However, her estimate for the time could be as much as $2.5 - 1.0 = 1.5$ hours over the actual time, and thus it is possible that her estimate of the time is not within 0.5 hour; NOT sufficient.

(2) Given that Joan's estimate for her average speed was within 10 miles per hour of her actual average speed, the same examples used in (1) can be used to show that it cannot be determined whether her estimate for the time would be within 0.5 hour of the actual time; NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E;
both statements together are still not sufficient.

DS12239

381. A certain list consists of 3 different numbers. Does the median of the 3 numbers equal the average (arithmetic mean) of the 3 numbers?

(1) The range of the 3 numbers is equal to twice the difference between the greatest number and the median.

(2) The sum of the 3 numbers is equal to 3 times one of the numbers.

Arithmetic Statistics

Let the numbers be x, y, and z so that $x < y < z$. Determine whether $y = \frac{x+y+z}{3}$, or equivalently, whether $3y = x + y + z$, or equivalently, whether $2y = x + z$.

(1) Given that the range is equal to twice the difference between the greatest number and the median, it follows that $z - x = 2(z - y)$, or $z - x = 2z - 2y$, or $2y = x + z$; SUFFICIENT.

(2) Given that the sum of the 3 numbers equals 3 times one of the numbers, it follows that $x + y + z = 3x$ or $x + y + z = 3y$ or $x + y + z = 3z$. If $x + y + z = 3x$, then $y + z = 2x$, or $(y - x) + (z - x) = 0$. Also, if $x + y + z = 3z$, then $x + y = 2z$, or $0 = (z - x) + (z - y)$. In each of these cases, the sum of two positive numbers is zero, which is impossible. Therefore, it must be true that $x + y + z = 3y$, from which it follows that $x + z = 2y$, and hence by the initial comments, the median of the 3 numbers equals the average of the 3 numbers; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS12806

382. Line ℓ lies in the xy-plane and does not pass through the origin. What is the slope of line ℓ?

(1) The x-intercept of line ℓ is twice the y-intercept of line ℓ.

(2) The x- and y-intercepts of line ℓ are both positive.

Geometry Coordinate geometry

Since the line does not pass through the origin, the line is either vertical and given by the equation $x = c$ for some constant c such that $c \neq 0$, or the line is not vertical and given by the equation $y = mx + b$ for some constants m and b such that $b \neq 0$. Determine whether the line is not vertical, and if so, determine the slope of the line, which is the value of m.

(1) Given that the x-intercept of the line is twice the y-intercept of the line, it follows that the line is not vertical, since a vertical line that does not pass through the origin will not have a y-intercept. Thus, the line is given by the equation $y = mx + b$. The x-intercept of the line is the solution to $0 = mx + b$, or $mx = -b$, which has solution $x = -\frac{b}{m}$, and the y-intercept of the line is b. Therefore, $-\frac{b}{m} = 2b$. Since $b \neq 0$, both sides of the last equation can be divided by b to get $-\frac{1}{m} = 2$, or $m = -1/2$; SUFFICIENT.

(2) Given that the x- and y-intercepts of the line are both positive, it is not possible to determine the slope of the line. For example, if the line is given by $y = -x + 1$, then the x-intercept is 1 (solve $-x + 1 = 0$), the y-intercept is 1 ($b = 1$), and the slope is -1. However, if the line is given by $y = -2x + 2$, then the x-intercept is 1 (solve $-2x + 2 = 0$), the y-intercept is 2 ($b = 2$), and the slope is -2; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

$$y = ax - 5$$
$$y = x + 6$$
$$y = 3x + b$$

DS07713

383. In the xy-plane, the straight-line graphs of the three equations above each contain the point (p,r). If a and b are constants, what is the value of b?

(1) $a = 2$

(2) $r = 17$

Algebra Coordinate geometry

Since (p,r) is on each of the lines, each of the following three equations is true:

(i) $r = ap - 5$

(ii) $r = p + 6$

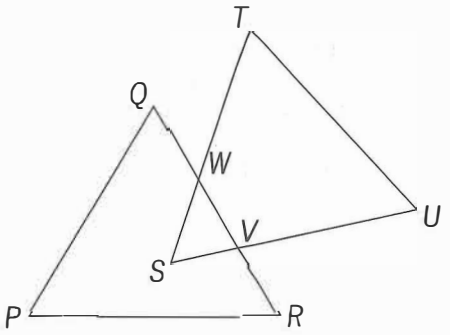
(iii) $r = 3p + b$

Determine the value of b.

(1) Given that $a = 2$, Equations (i) and (ii) become $r = 2p - 5$ and $r = p + 6$. Subtracting equations gives $0 = p - 11$, or $p = 11$. Now using $r = p + 6$, it follows that $r = 11 + 6 = 17$. Finally, using $p = 11$ and $r = 17$ in Equation (iii) gives $17 = 3(11) + b$, or $b = 17 - 33 = -16$; SUFFICIENT.

(2) Given that $r = 17$, Equation (ii) becomes $17 = p + 6$, and so $p = 17 - 6 = 11$. Using $r = 17$ and $p = 11$, Equation (iii) becomes $17 = 3(11) + b$, or $b = 17 - 33 = -16$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.



DS06861

384. In the figure above, PQR and STU are identical equilateral triangles, and $PQ = 6$. What is the perimeter of polygon $PQWTUVR$?

(1) Triangle SWV has perimeter 9.

(2) VW has length 3.5.

Geometry Triangles; Perimeter

(1) Given that triangle SWV has perimeter 9, the perimeter of polygon $PQWTUVR$ can be determined, since the perimeter of polygon $PQWTUVR$ equals the sum of the perimeters of the two equilateral triangles,

which is 3(6) + 3(6) = 36, minus the perimeter of triangle *SWV*; SUFFICIENT.

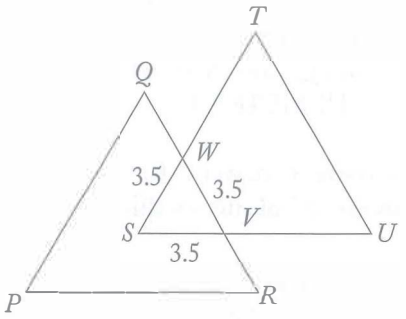


Figure 1

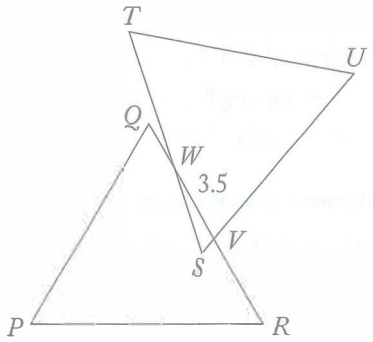


Figure 2

(2) Given that $VW = 3.5$, and using the fact that the perimeter of polygon *PQWTUVR* equals the sum of the perimeters of the two equilateral triangles, which is 3(6) + 3(6) = 36, minus the perimeter of triangle *SWV*, it is not possible to determine the perimeter of polygon *PQWTUVR*. For example, the perimeter of polygon *PQWTUVR* could be 36 – 3(3.5) = 25.5 (arrange triangles *PQR* and *STU* so that $VW = WS = SV = 3.5$, as shown in Figure 1), and the perimeter of polygon *PQWTUVR* could be greater than 25.5 (arrange triangles *PQR* and *STU* so that $VW = 3.5$, *WS* is slightly greater than 3.5, and *SV* is close to 0, as shown in Figure 2); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS09973

385. The range of the numbers in set *S* is x, and the range of the numbers in set *T* is y. If all of the numbers in set *T* are also in set *S*, is x greater than y?

(1) Set *S* consists of 7 numbers.

(2) Set *T* consists of 6 numbers.

Arithmetic Statistics

Set *S* has a range of x, set *T* has a range of y, and *T* is a subset of *S*. Determine if x is greater than y.

(1) It is given that *S* contains exactly 7 numbers, but nothing additional is known about *T*. Thus, if $S = \{1, 2, 3, 4, 5, 6, 7\}$ and $T = \{1, 2, 3, 4, 5, 6\}$, then $x = 7 - 1 = 6$, $y = 6 - 1 = 5$, and x is greater than y. On the other hand, if $S = \{1, 2, 3, 4, 5, 6, 7\}$ and $T = \{1, 3, 4, 5, 6, 7\}$, then $x = 7 - 1 = 6$, $y = 7 - 1 = 6$, and x is not greater than y; NOT sufficient.

(2) It is given that *T* contains exactly 6 numbers, but nothing additional is known about *T*. Since the same examples given in (1) can also be used in (2), it cannot be determined if x is greater than y; NOT sufficient.

Taking (1) and (2) together, the examples used in (1) can be used to show that it cannot be determined if x is greater than y.

The correct answer is E;
both statements together are still not sufficient.

DS13857

386. The hypotenuse of a right triangle is 10 cm. What is the perimeter, in centimeters, of the triangle?

(1) The area of the triangle is 25 square centimeters.

(2) The 2 legs of the triangle are of equal length.

Geometry Triangles

If x and y are the lengths of the legs of the triangle, then it is given that $x^2 + y^2 = 100$. Determining the value of $x + y + 10$, the perimeter of the triangle, is equivalent to determining the value of $x + y$.

(1) Given that the area is 25, then $\frac{1}{2}xy = 25$, or $xy = 50$. Since $(x + y)^2 = x^2 + y^2 + 2xy$, it follows that $(x + y)^2 = 100 + 2(50)$, or $x + y = \sqrt{200}$; SUFFICIENT.

(2) Given that $x = y$, since $x^2 + y^2 = 100$, it follows that $2x^2 = 100$, or $x = \sqrt{50}$. Hence, $x + y = x + x = 2x = 2\sqrt{50}$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

Shipment	S1	S2	S3	S4	S5	S6
Fraction of the Total Value of the Six Shipments	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{3}{20}$	$\frac{2}{15}$	$\frac{1}{10}$

DS01427

387. Six shipments of machine parts were shipped from a factory on two trucks, with each shipment entirely on one of the trucks. Each shipment was labeled either S1, S2, S3, S4, S5, or S6. The table shows the value of each shipment as a fraction of the total value of the six shipments. If the shipments on the first truck had a value greater than $\frac{1}{2}$ of the total value of the six shipments, was S3 shipped on the first truck?

(1) S2 and S4 were shipped on the first truck.

(2) S1 and S6 were shipped on the second truck.

Arithmetic Operations on rational numbers

Given that the shipments on the first truck had a value greater than $\frac{1}{2}$ of the total value of the 6 shipments, determine if S3 was shipped on the first truck.

To avoid dealing with fractions, it will be convenient to create scaled values of the shipments by multiplying each fractional value by 60, which is the least common denominator of the fractions. Thus, the scaled values associated with S1, S2, S3, S4, S5, and S6 are 15, 12, 10, 9, 8, and 6, respectively. The given information is that the scaled value of the shipments on the first truck is greater than $\left(\frac{1}{2}\right)(60) = 30$.

(1) Given that the first truck includes shipments with scaled values 12 and 9, it may or may not be the case that S3 (the shipment with scaled value 10) is on the first truck. For example, the first truck could contain only S2, S3, and S4, for a total scaled value $12 + 10 + 9 = 31 > 30$. Or, the first truck could contain only S1, S2, and S4, for a total scaled value $15 + 12 + 9 = 36 > 30$; NOT sufficient.

(2) Given that the second truck includes shipments with scaled values 15 and 6, the second truck cannot contain S3. Otherwise, the second truck would contain shipments with scaled values 15, 6, and 10, for a total scaled value $15 + 6 + 10 = 31$, leaving at most a total scaled value 29 (which is not greater than 30) for the first truck; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS11723

388. If x, y, and z are three-digit positive integers and if $x = y + z$, is the hundreds digit of x equal to the sum of the hundreds digits of y and z?

(1) The tens digit of x is equal to the sum of the tens digits of y and z.

(2) The units digit of x is equal to the sum of the units digits of y and z.

Arithmetic Place value

Letting $x = 100a + 10b + c$, $y = 100p + 10q + r$, and $z = 100t + 10u + v$, where a, b, c, p, q, r, t, u, and v are digits, determine if $a = p + t$.

(1) It is given that $b = q + u$ (which implies that $c + v \leq 9$ because if $c + v > 9$, then in the addition process a ten would need to be carried over to the tens column and b would be $q + u + 1$). Since b is a digit, $0 \leq b \leq 9$. Hence, $0 \leq q + u \leq 9$, and so $0 \leq 10(q + u) \leq 90$. Therefore, in the addition process, there are no hundreds to carry over from the tens column to the hundreds column, so $a = p + t$; SUFFICIENT.

(2) It is given that $c = r + v$. If $x = 687$, $y = 231$, and $z = 456$, then, $y + z = 231 + 456 = 687 = x$, $r + v = 1 + 6 = 7 = c$, and $p + t = 2 + 4 = 6 = a$. On the other hand, if $x = 637$, $y = 392$, and $z = 245$, then $y + z = 392 + 245 = 637 = x$, $r + v = 2 + 5 = 7 = c$, and $p + t = 3 + 2 = 5 \neq 6 = a$; NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

	Favorable	Unfavorable	Not Sure
Candidate M	40	20	40
Candidate N	30	35	35

DS05162

389. The table above shows the results of a survey of 100 voters who each responded "Favorable" or "Unfavorable" or "Not Sure" when asked about their impressions of Candidate M and of Candidate N. What was the number of voters who responded "Favorable" for both candidates?

(1) The number of voters who did not respond "Favorable" for either candidate was 40.

(2) The number of voters who responded "Unfavorable" for both candidates was 10.

Arithmetic Sets

If x is the number of voters who responded "Favorable" for both candidates, then it follows from the table that the number of voters who responded "Favorable" to at least one candidate is $40 + 30 - x = 70 - x$. This is because $40 + 30$ represents the number of voters who responded "Favorable" for Candidate M added to the number of voters who responded "Favorable" for Candidate N, a calculation that counts twice each of the x voters who responded "Favorable" for both candidates.

(1) Given that there were 40 voters who did not respond "Favorable" for either candidate and there were 100 voters surveyed, the number of voters who responded "Favorable" to at least one candidate is $100 - 40 = 60$. Therefore, from the comments above, it follows that $70 - x = 60$, and hence $x = 10$; SUFFICIENT.

(2) The information given affects only the numbers of voters in the categories "Unfavorable" for Candidate M only, "Unfavorable" for Candidate N only, and "Unfavorable" for both candidates. Thus, the numbers of voters in the categories "Favorable" for Candidate M only, "Favorable" for Candidate N only, and "Favorable" for both candidates are not affected. Since these latter categories are only constrained to have certain integer values that have a total sum of $70 - x$, more than one possibility exists for the value of x. For example, the numbers of voters in the categories "Favorable" for Candidate M only, "Favorable" for Candidate N only, and "Favorable" for both candidates could be 25, 15, and 15, respectively, which gives $70 - x = 25 + 15 + 15$, or $x = 15$. However, the numbers of voters in the categories "Favorable" for Candidate M only, "Favorable" for Candidate N only, and "Favorable" for both candidates could be 30, 20, and 10, respectively, which gives $70 - x = 30 + 20 + 10$, or $x = 10$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

DS00340

390. A school administrator will assign each student in a group of n students to one of m classrooms. If $3 < m < 13 < n$, is it possible to assign each of the n students to one of the m classrooms so that each classroom has the same number of students assigned to it?

(1) It is possible to assign each of $3n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

(2) It is possible to assign each of $13n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

Arithmetic Properties of numbers

Determine if n is divisible by m.

(1) Given that $3n$ is divisible by m, then n is divisible by m if $m = 9$ and $n = 27$ (note that $3 < m < 13 < n$, $3n = 81$, and $m = 9$, so $3n$ is divisible by m) and n is not divisible by m if $m = 9$ and $n = 30$ (note that $3 < m < 13 < n$, $3n = 90$, and $m = 9$, so $3n$ is divisible by m); NOT sufficient.

(2) Given that $13n$ is divisible by m, then $13n = qm$, or $\frac{n}{m} = \frac{q}{13}$, for some integer q.

Since 13 is a prime number that divides qm (because $13n = qm$) and 13 does not divide m (because $m < 13$), it follows that 13 divides q.

Therefore, $\frac{q}{13}$ is an integer, and since $\frac{n}{m} = \frac{q}{13}$, then $\frac{n}{m}$ is an integer. Thus, n is divisible by m; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS07441

391. If q, s, and t are all different numbers, is $q < s < t$?

(1) $t - q = |t - s| + |s - q|$

(2) $t > q$

Algebra Absolute value

(1) It is given that $t - q = |t - s| + |s - q|$, which can be rewritten without absolute values in four mutually exclusive and collectively exhaustive cases by making use of the algebraic definition of absolute value. Recall that $|x| = x$ if $x > 0$, and $|x| = -x$ if $x < 0$. Thus, for example, if $t - s < 0$, then $|t - s| = -(t - s)$.
Case 1: $t > s$ and $s > q$. In this case, $t - s > 0$ and $s - q > 0$, and so $t - q = |t - s| + |s - q|$ is equivalent to $t - q = (t - s) + (s - q)$, which is an identity. Therefore, the case for which $t > s$ and $s > q$ is consistent with the given information and the assumption $t - q = |t - s| + |s - q|$.
Case 2: $t > s$ and $s < q$. In this case, $t - s > 0$ and $s - q < 0$, and so $t - q = |t - s| + |s - q|$ is equivalent to $t - q = (t - s) - (s - q)$, or $s = q$, which is not consistent with the assumption that q, s, and t are all different numbers. Therefore, the case for which $t > s$ and $s < q$ is not consistent with the given information and the assumption $t - q = |t - s| + |s - q|$.
Case 3: $t < s$ and $s > q$. In this case, $t - s < 0$ and $s - q > 0$, and so $t - q = |t - s| + |s - q|$ is equivalent to $t - q = -(t - s) + (s - q)$, or $t = s$, which is not consistent with the assumption that q, s, and t are all different numbers. Therefore, the case for which $t < s$ and $s > q$ is not consistent with the given information and the assumption $t - q = |t - s| + |s - q|$.
Case 4: $t < s$ and $s < q$. In this case, $t - s < 0$ and $s - q < 0$, and so $t - q = |t - s| + |s - q|$ is equivalent to $t - q = -(t - s) - (s - q)$, or $t = q$, which is not consistent with the assumption that q, s, and t are all different numbers. Therefore, the case for which $t < s$ and $s < q$ is not consistent with the given information and the assumption $t - q = |t - s| + |s - q|$. The only case that is consistent with the given information and the assumption $t - q = |t - s| + |s - q|$ is Case 1. Therefore, it follows that $t > s$ and $s > q$, and this implies $q < s < t$; SUFFICIENT.

(2) Given that $t > q$, it is possible that $q < s < t$ is true (for example, when s is between t and q) and it is possible that $q < s < t$ is false (for example, when s is greater than t); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS11538

392. What is the median number of employees assigned per project for the projects at Company Z?

(1) 25 percent of the projects at Company Z have 4 or more employees assigned to each project.

(2) 35 percent of the projects at Company Z have 2 or fewer employees assigned to each project.

Arithmetic Statistics

(1) Although 25 percent of the projects have 4 or more employees, there is essentially no information about the middle values of the numbers of employees per project. For example, if there were a total of 100 projects, then the median could be 2 (75 projects that have exactly 2 employees each and 25 projects that have exactly 4 employees each) or the median could be 3 (75 projects that have exactly 3 employees each and 25 projects that have exactly 4 employees each); NOT sufficient.

(2) Although 35 percent of the projects have 2 or fewer employees, there is essentially no information about the middle values of the numbers of employees per project. For example, if there were a total of 100 projects, then the median could be 3 (35 projects

that have exactly 2 employees each and 65 projects that have exactly 3 employees each) or the median could be 4 (35 projects that have exactly 2 employees each and 65 projects that have exactly 4 employees each); NOT sufficient.

Given both (1) and (2), 100 – (25 + 35) percent = 40 percent of the projects have exactly 3 employees. Therefore, when the numbers of employees per project are listed from least to greatest, 35 percent of the numbers are 2 or less and (35 + 40) percent = 75 percent are 3 or less, and hence the median is 3.

The correct answer is C;
both statements together are sufficient.

DS04409

393. Last year, a certain company began manufacturing product X and sold every unit of product X that it produced. Last year the company's total expenses for manufacturing product X were equal to $100,000 plus 5 percent of the company's total revenue from all units of product X sold. If the company made a profit on product X last year, did the company sell more than 21,000 units of product X last year?

(1) The company's total revenue from the sale of product X last year was greater than $110,000.

(2) For each unit of product X sold last year, the company's revenue was $5.

Algebra Applied problems

For a company that made a profit last year from selling product X and had total expenses for product X of $100,000 + 0.05$R$, where R is the total revenue for selling product X, determine whether the company sold more than 21,000 units of product X last year.

Note that since the company made a profit, revenue – cost, which is given by $R - (\$100{,}000 + 0.05R) = 0.95R - \$100{,}000$, must be positive.

(1) It is given that $R > \$110{,}000$. It is possible to vary the unit price and the number of units sold so that $R > \$110{,}000$ and *more than* 21,000 units were sold, and also so that $R > \$110{,}000$ and *less than* 21,000 units were sold. For example, if 25,000 units were sold for $10 per unit, then $R = 25{,}000(\$10) = \$250{,}000 > \$110{,}000$ and $25{,}000 > 21{,}000$. On the other hand, if 20,000 units were sold for $10 per unit, then $R = 20{,}000(\$10) = \$200{,}000 > \$110{,}000$ and $20{,}000 < 21{,}000$; NOT sufficient.

(2) It is given that the company's revenue for each unit of product X was $5. If the company manufactured and sold x units of product X, then its revenue was $\$5x$. Because the company made a profit, $0.95(\$5x) - \$100{,}000 > 0$, and so

$$\begin{aligned} 0.95(\$5x) - \$100{,}000 &> 0 \\ \$4.75x - \$100{,}000 &> 0 \\ \$4.75x &> \$100{,}000 \\ x &> 21{,}052; \text{ SUFFICIENT.} \end{aligned}$$

To avoid long division in the last step, note that $4.75(21{,}000) = 99{,}750$, and thus from $4.75x > 100{,}000$, it follows that $x > 21{,}000$.

The correct answer is B;
statement 2 alone is sufficient.

DS01641

394. Beginning in January of last year, Carl made deposits of $120 into his account on the 15th of each month for several consecutive months and then made withdrawals of $50 from the account on the 15th of each of the remaining months of last year. There were no other transactions in the account last year. If the closing balance of Carl's account for May of last year was $2,600, what was the range of the monthly closing balances of Carl's account last year?

(1) Last year the closing balance of Carl's account for April was less than $2,625.

(2) Last year the closing balance of Carl's account for June was less than $2,675.

Arithmetic Statistics

(1) If Carl began making $50 withdrawals on or before May 15, his account balance on April 16 would be at least $50 greater than it was on the last day of May. Thus, his account balance on April 16 would be at least $2,600 + $50 = $2,650, which is contrary to the information given in (1). Therefore, Carl did not begin making $50 withdrawals until June 15 or later. These observations can be used to give at least two possible ranges. Carl could have had an account balance of

$2,000 on January 1, made $120 deposits in each of the first 11 months of the year, and then made a $50 withdrawal on December 15, which gives a range of monthly closing balances of (120)(10). Also, Carl could have had an account balance of $2,000 on January 1, made $120 deposits in each of the first 10 months of the year, and then made $50 withdrawals on November 15 and on December 15, which gives a range of monthly closing balances of (120)(9); NOT sufficient.

(2) On June 1, Carl's account balance was the same as its closing balance was for May, namely $2,600. Depending on whether Carl made a $120 deposit or a $50 withdrawal on June 15, Carl's account balance on June 16 was either $2,720 or $2,550. It follows from the information given in (2) that Carl's balance on June 16 was $2,550. Therefore, Carl began making $50 withdrawals on or before June 15. These observations can be used to give at least two possible ranges. Carl could have had an account balance of $2,680 on January 1, made one $120 deposit on January 15, and then made a $50 withdrawal in each of the remaining 11 months of the year (this gives a closing balance of $2,600 for May), which gives a range of monthly closing balances of (50)(11). Also, Carl could have had an account balance of $2,510 on January 1, made $120 deposits on January 15 and on February 15, and then made a $50 withdrawal in each of the remaining 10 months of the year (this gives a closing balance of $2,600 for May), which gives a range of monthly closing balances of (50)(10); NOT sufficient.

Given both (1) and (2), it follows from the remarks above that Carl began making $50 withdrawals on June 15. Therefore, the changes to Carl's account balance for each month of last year are known. Since the closing balance for May is given, it follows that the closing balances for each month of last year are known, and hence the range of these 12 known values can be determined.

The correct answer is C;
both statements together are sufficient.

DS16368

395. Are all of the numbers in a certain list of 15 numbers equal?

(1) The sum of all the numbers in the list is 60.

(2) The sum of any 3 numbers in the list is 12.

Arithmetic Properties of numbers

(1) If there are 15 occurrences of the number 4 in the list, then the sum of the numbers in the list is 60 and all the numbers in the list are equal. If there are 13 occurrences of the number 4 in the list, 1 occurrence of the number 3 in the list, and 1 occurrence of the number 5 in the list, then the sum of the numbers in the list is 60 and not all the numbers in the list are equal; NOT sufficient.

(2) Given that the sum of any 3 numbers in the list is 12, arrange the numbers in the list in numerical order, from least to greatest: $a_1 \le a_2 \le a_3 \le \ldots \le a_{15}$.

If $a_1 < 4$, then $a_1 + a_2 + a_3 < 4 + a_2 + a_3$. Therefore, from (2), $12 < 4 + a_2 + a_3$, or $8 < a_2 + a_3$, and so at least one of the values a_2 and a_3 must be greater than 4. Because $a_2 \le a_3$, it follows that $a_3 > 4$. Since the numbers are arranged from least to greatest, it follows that $a_4 > 4$ and $a_5 > 4$. But then, $a_3 + a_4 + a_5 > 4 + 4 + 4 = 12$, contrary to (2), and so $a_1 < 4$ is not true. Therefore, $a_1 \ge 4$. Since a_1 is the least of the 15 numbers, $a_n \ge 4$ for $n = 1, 2, 3, \ldots, 15$.

If $a_{15} > 4$, then $a_{13} + a_{14} + a_{15} > a_{13} + a_{14} + 4$. Therefore, from (2), $12 > a_{13} + a_{14} + 4$, or $8 > a_{13} + a_{14}$, and so at least one of the values a_{13} and a_{14} must be less than 4. Because $a_{13} \le a_{14}$, it follows that $a_{13} < 4$. Since the numbers are arranged from least to greatest, it follows that $a_{11} < 4$ and $a_{12} < 4$. But then $a_{11} + a_{12} + a_{13} < 4 + 4 + 4 = 12$, contrary to (2). Therefore, $a_{15} \le 4$. Since a_{15} is the greatest of the 15 numbers, $a_n \le 4$ for $n = 1, 2, 3, \ldots, 15$.

It has been shown that, for $n = 1, 2, 3, \ldots, 15$, each of $a_n \ge 4$ and $a_n \le 4$ is true. Therefore, $a_n = 4$ for $n = 1, 2, 3, \ldots, 15$; SUFFICIENT.

The correct answer is B;
statement 2 alone is sufficient.

DS16565

396. If the average (arithmetic mean) of six numbers is 75, how many of the numbers are equal to 75 ?

(1) None of the six numbers is less than 75.

(2) None of the six numbers is greater than 75.

Arithmetic Statistics

If the average of six numbers is 75, then $\frac{1}{6}$ of the sum of the numbers is 75. Therefore, the sum of the numbers is (6)(75).

(1) If one of the numbers is greater than 75, then we can write that number as $75 + x$ for some positive number x. Consequently, the sum of the 6 numbers must be at least $(5)(75) + (75 + x) = (6)(75) + x$, which is greater than (6)(75), contrary to the fact that the sum is equal to (6)(75). Hence, none of the numbers can be greater than 75. Since none of the numbers can be less than 75 (given information) and none of the numbers can be greater than 75, it follows that each of the numbers is equal to 75; SUFFICIENT.

(2) If one of the numbers is less than 75, then we can write that number as $75 - x$ for some positive number x. Consequently, the sum of the 6 numbers must be at most $(5)(75) + (75 - x) = (6)(75) - x$, which is less than (6)(75), contrary to the fact that the sum is equal to (6)(75). Hence, none of the numbers can be less than 75. Since none of the numbers can be less than 75 and none of the numbers can be greater than 75 (given information), it follows that each of the numbers is equal to 75; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS16188

397. What amount did Jean earn from the commission on her sales in the first half of 1988 ?

(1) In 1988 Jean's commission was 5 percent of the total amount of her sales.

(2) The amount of Jean's sales in the second half of 1988 averaged $10,000 per month more than in the first half.

Arithmetic Applied problems

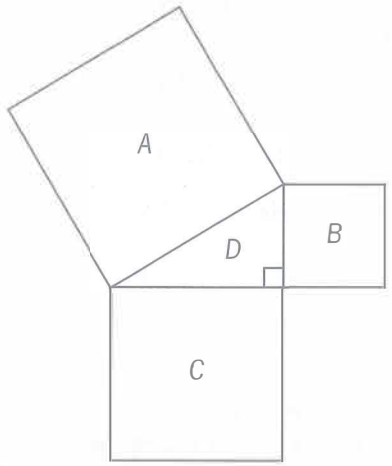
Let A be the amount of Jean's sales in the first half of 1988. Determine the value of A.

(1) If the amount of Jean's sales in the first half of 1988 was $10,000, then her commission in the first half of 1988 would have been (5%)($10,000) = $500. On the other hand, if the amount of Jean's sales in the first half of 1988 was $100,000, then her commission in the first half of 1988 would have been (5%)($100,000) = $5,000; NOT sufficient.

(2) No information is given that relates the amount of Jean's sales to the amount of Jean's commission; NOT sufficient.

Given (1) and (2), from (1) the amount of Jean's commission in the first half of 1988 is (5%)A. From (2) the amount of Jean's sales in the second half of 1988 is A + $60,000. Both statements together do not give information to determine the value of A.

The correct answer is E;
both statements together are still not sufficient.



DS16572

398. In the figure above, if the area of triangular region D is 4, what is the length of a side of square region A ?

(1) The area of square region B is 9.

(2) The area of square region C is $\frac{64}{9}$.

Geometry Area

The area of the triangular region D can be represented by $\frac{1}{2}bh$, where b is the base of the triangle (and is equal to the length of a side of

the square region C) and h is the height of the triangle (and is equal to the length of a side of the square region B). The area of any square is equal to the length of a side squared. The Pythagorean theorem is used to find the length of a side of a right triangle, when the length of the other 2 sides of the triangle are known and is represented by $a^2 + b^2 = c^2$, where a and b are the lengths of the 2 perpendicular sides of the triangle and c is the length of the hypotenuse.

Although completed calculations are provided in what follows, keep in mind that completed calculations are not needed to solve this problem.

(1) If the area of B is 9, then the length of each side is 3. Therefore, $b = 3$. Then, b can be determined, since the area of the triangle is, by substitution, $4 = \frac{1}{2}(3b)$ or $8 = 3b$ or $\frac{8}{3} = b$. Once b is known, the Pythagorean theorem can be used: $\left(\frac{8}{3}\right)^2 + 3^2 = c^2$ or $\frac{64}{9} + 9 = c^2$ or $\frac{145}{9} = c^2$. The length of a side of A is thus $\sqrt{\frac{145}{9}}$; SUFFICIENT.

(2) If the area of C is $\frac{64}{9}$, then the length of each side is $\frac{8}{3}$. Therefore, $b = \frac{8}{3}$. The area of a triangle is $A = \frac{1}{2}bh$ so $4 = \frac{1}{2}\left(\frac{8}{3}h\right)$, $8 = \frac{8}{3}h$, and $3 = h$. Once h is known, the Pythagorean theorem can be used as above; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.

DS16168

399. If n is a positive integer and $k = 5.1 \times 10^n$, what is the value of k?

(1) $6{,}000 < k < 500{,}000$

(2) $k^2 = 2.601 \times 10^9$

Arithmetic Properties of numbers

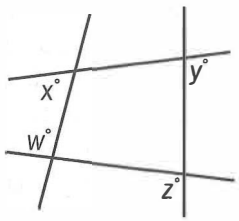
Given that $k = 5.1 \times 10^n$, where n is a positive integer, then the value of k must follow the pattern shown in the following table:

n	k
1	51
2	510
3	5,100
4	51,000
5	510,000
6	5,100,000
.	.
.	.
.	.

(1) Given that $6{,}000 < k < 500{,}000$, then k must have the value 51,000, and so $n = 4$; SUFFICIENT.

(2) Given that $k^2 = 2.601 \times 10^9$, then $k = \sqrt{2.601 \times 10^9} = \sqrt{2{,}601 \times 10^6} = \sqrt{2{,}601} \times \sqrt{10^6}$ $= 51 \times 10^3 = 51{,}000$, and so $n = 4$; SUFFICIENT.

The correct answer is D;
each statement alone is sufficient.



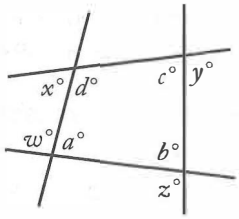
DS06875

400. What is the value of $x + y$ in the figure above?

(1) $w = 95$

(2) $z = 125$

Geometry Angles



In the figure above, a, b, c, and d are the degree measures of the interior angles of the

quadrilateral formed by the four lines and $a+b+c+d=360$. Then

$w+x+y+z$

$=(180-a)+(180-d)+(180-c)+(180-b)$

$=720-(a+b+c+d)$

$=720-360$

$=360.$

Determine the value of $x+y$.

(1) Given that $w=95$, then $95+x+y+z=360$ and $x+y+z=265$. If $z=65$, for example, then $x+y=200$. On the other hand, if $z=100$, then $x+y=165$; NOT sufficient.

(2) Given that $z=125$, then $w+x+y+125=360$ and $w+x+y=235$. If $w=35$, for example, then $x+y=200$. On the other hand, if $w=100$, then $x+y=135$; NOT sufficient.

Taking (1) and (2) together, $95+x+y+125=360$, and so $x+y=140$.

The correct answer is C;
both statements together are sufficient.

DS16370

401. If n and k are positive integers, is $\sqrt{n+k}>2\sqrt{n}$?

(1) $k>3n$

(2) $n+k>3n$

Algebra Inequalities

Determine if $\sqrt{n+k}>2\sqrt{n}$. Since each side is positive, squaring each side preserves the inequality, so $\sqrt{n+k}>2\sqrt{n}$ is equivalent to $(\sqrt{n+k})(\sqrt{n+k})>(2\sqrt{n})(2\sqrt{n})$, which in turn is equivalent to $n+k>4n$, or to $k>3n$.

(1) Given that $k>3n$, then $\sqrt{n+k}>2\sqrt{n}$; SUFFICIENT.

(2) Given that $n+k>3n$, then $k>2n$. However, it is possible for $k>2n$ to be true and $k>3n$ to be false (for example, $k=3$ and $n=1$) and it is possible for $k>2n$ to be true and $k>3n$ to be true (for example, $k=4$ and $n=1$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS16589

402. In a certain business, production index p is directly proportional to efficiency index e, which is in turn directly proportional to investment index i. What is p if $i=70$?

(1) $e=0.5$ whenever $i=60$.

(2) $p=2.0$ whenever $i=50$.

Arithmetic Proportions

(1) This gives only values for e and i, and, while p is directly proportional to e, the nature of this proportion is unknown. Therefore, p cannot be determined; NOT sufficient.

(2) Since p is directly proportional to e, which is directly proportional to i, then p is directly proportional to i. Therefore, the following proportion can be set up: $\frac{p}{i}=\frac{2.0}{50}$. If $i=70$, then $\frac{p}{70}=\frac{2.0}{50}$. Through cross multiplying, this equation yields $50p=140$, or $p=2.8$; SUFFICIENT.

The preceding approach is one method that can be used. Another approach is as follows: It is given that $p=Ke=K(Li)=(KL)i$, where K and L are the proportionality constants, and the value of $70KL$ is to be determined. Statement (1) allows us to determine the value of L, but gives nothing about K, and thus (1) is not sufficient. Statement (2) allows us to determine the value of KL, and thus (2) is sufficient.

The correct answer is B;
statement 2 alone is sufficient.

DS16085

403. If n is a positive integer, what is the tens digit of n ?

(1) The hundreds digit of $10n$ is 6.

(2) The tens digit of $n+1$ is 7.

Arithmetic Properties of numbers

(1) Given that the hundreds digit of $10n$ is 6, the tens digit of n is 6, since the hundreds digit of $10n$ is always equal to the tens digit of n; SUFFICIENT.

(2) Given that the tens digit of $n+1$ is 7, it is possible that the tens digit of n is 7 (for example, $n = 70$) and it is possible that the tens digit of n is 6 (for example, $n = 69$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.

DS16204

404. What is the value of $\frac{2t+t-x}{t-x}$?

(1) $\frac{2t}{t-x} = 3$

(2) $t - x = 5$

Algebra Simplifying algebraic expressions

Determine the value of $\frac{2t+t-x}{t-x}$.

(1) Since $\frac{2t}{t-x} = 3$ and $\frac{2t+t-x}{t-x} = \frac{2t}{t-x} + \frac{t-x}{t-x} = \frac{2t}{t-x} + 1$, it follows that $\frac{2t+t-x}{t-x} = 3+1$;

SUFFICIENT.

(2) Given that $t - x = 5$, it follows that $\frac{2t+t-x}{t-x} = \frac{2t+5}{5} = \frac{2}{5}t + 1$, which can vary when the value of t varies. For example, $\frac{2}{5}t + 1 = 3$ if $t = 5$ (choose $x = 0$ to have $t - x = 5$) and $\frac{2}{5}t + 1 = 5$ if $t = 10$ (choose $x = 5$ to have $t - x = 5$); NOT sufficient.

The correct answer is A;
statement 1 alone is sufficient.